Cheapflation Cycles

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Abstract

We document a new source of fluctuations in inflation inequality. When the cost of upstream inputs rises, varieties within a product category tend to have similar absolute price increases. However, the same absolute price increase constitutes a larger percentage change for low-price products, resulting in excess inflation at the low end ("cheapflation"). Since low-income households tend to buy lower-priced varieties, the inflation rates they face are disproportionately sensitive to upstream costs. Using data on food-at-home purchases, we show that this mechanism generates cycles in inflation inequality and excess volatility in inflation for low-income households relative to high-income households. This channel parsimoniously accounts for observed fluctuations in inflation inequality over time, including surges in cheapflation and inflation inequality during both the Great Recession and the 2021–2023 inflation surge. Income-group price indices constructed from public inflation statistics mask these within-category differences in inflation and thus understate the excess cost-sensitivity and volatility of inflation for low-income households. We provide suggestive evidence that this mechanism applies to categories beyond food at home. The same mechanism also leads to systematic differences in inflation across cities and import price inflation across countries in response to nationwide and global cost shocks.

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1 Introduction

Inflation can differ markedly across households due to differences in the composition of households' expenditures. A large body of research has documented heterogeneity in the inflation rates faced by households across the income distribution due to differences in spending shares across product categories. For example, low-income households devote a greater share of their budgets to necessities like food, energy, and shelter, which leads to differences in inflation for low- and high-income households when the relative prices of these goods fluctuate (e.g., Hobijn and Lagakos 2005; Jaravel 2024; Orchard 2025).

In this paper, we document a new and quantitatively important source of fluctuations in inflation inequality that operates *within* product categories. We show that when firms pass through upstream cost shocks in levels, an increase in costs leads to systematically higher inflation rates for lower-priced varieties within narrow product categories. Since low-income households tend to buy lower-priced varieties, this pricing behavior leads to predictable and cyclical swings in inflation inequality.

Our mechanism builds on the premise that firms tend to pass upstream cost shocks to prices on a dollars-and-cents basis, which Sangani (2026) calls *pass-through in levels*. This means that upstream cost shocks tend to induce similar absolute price changes across product varieties within narrow categories. For example, when the commodity price of coffee beans rises, both premium and budget coffee brands raise prices by similar amounts in cents-per-ounce. However, these identical dollar increases constitute larger percentage price changes for cheaper products. As a result, low-priced varieties systematically exhibit higher inflation rates than high-priced varieties when input costs rise.

Because low-income households disproportionately purchase low-priced varieties, pass-through in levels generates differences in inflation across the income distribution even holding households' spending shares across product categories fixed. Moreover, the extent of inflation inequality fluctuates over time with upstream costs. When the price of upstream inputs rises relative to the overall price level, inflation rates faced by low-income households increase more than those faced by high-income households; when commodity prices fall, the reverse occurs.

Using scanner data for food-at-home purchases, we show that this channel can account for the bulk of variation in inflation inequality over the eighteen-year period from 2006 to 2023. We derive a simple formula that, under pass-through in levels, predicts the gap in product-category inflation rates between low- and high-income households using three statistics: the relative prices of the varieties they buy, the average inflation in the product category, and wage inflation. Within-category inflation differences predicted by this formula alone account for 70 percent of the variation in the gap between the top and bottom income quintiles' food-at-home inflation rates over time. Further accounting for differences in households' expenditure shares across product categories increases the share of variation explained to 73 percent. This formula also parsimoniously accounts for the extent to which inflation inequality was elevated during specific subperiods, such as the Great Recession (2008–2011) or the post-pandemic inflation (2021–2023).

The success of pass-through in levels in accounting for these fluctuations is surprising given that many mechanisms have been proposed to explain relative price movements during specific episodes in our sample. The causes of the 2021–2023 rise in inflation inequality in particular have been hotly debated, with commentators attributing the excess inflation of low-priced varieties during this period—a phenomenon termed "cheapflation" by Cavallo and Kryvtsov (2024)—to the distribution of fiscal stimulus or to price-gouging by retailers. We find that the extent of cheapflation and inflation inequality from 2021–2023 was in line with (in fact, slightly below) what pass-through in levels predicts, leaving limited scope for other channels. Moreover, we show that the cheapflation and elevated inflation inequality were by no means special to the post-pandemic period. The same patterns accompany each instance in the data where input costs rose relative to the aggregate price level, such as during the food commodity booms of 2008 and 2011, though the degree to which input costs rose during 2021–2023 made these phenomena especially dramatic.

An important feature of the mechanism we document is that it operates within narrowly defined product categories. As a result, it is largely obscured in publicly available inflation statistics, which even at their most disaggregated level report inflation rates pooled across varieties within a product category. To take an example, one of the 273 entry-level items (ELIs) tracked by the Bureau of Labor Statistics (BLS) is "roasted and instant coffee." Despite the narrow category definition, scanner data reveals large fluctuations in the inflation gap between low- and high-income households within roasted coffee alone: the gap reached 8 percentage points when coffee commodity prices spiked in 2011 and fell to −3 percent when coffee commodity prices fell in 2012–2013. These fluctuations occur entirely within a single ELI and are therefore invisible in the BLS series.

Thus, even income-group inflation measures built from the most disaggregated price indices publicly available substantially understate fluctuations in inflation inequality that we observe in product-level household purchase data. In scanner data, the pass-through of upstream food manufacturing costs to food-at-home prices for the lowest income quintile is 10 percent higher than that of the highest income quintile. Analogous calculations in income-group-specific price indices constructed with the most granular BLS data available find a differential of only 2 percent. Likewise, the variance of food-at-home inflation rates for the lowest income quintile in scanner data is 21 percent higher than that of the highest income quintile; parallel calculations with BLS data find an 8 percent gap.

While prior work shows that aggregation understates secular differences in inflation across households (Jaravel 2019), our channel implies that the aggregation bias is especially severe when input costs rise sharply—precisely when policymakers may be most concerned about the distributional burden of cost-of-living increases. The post-pandemic inflation is a clear example. From 2021–2023, we find that aggregation to the BLS's entry-level items understates the differential growth in food prices for the lowest-income quintile relative to the highest-income quintile by a factor of seven.[1] About two-thirds of the aggregation bias during this period is due to pass-through in levels of rising input costs, while one-third is due to the secular aggregation bias documented by Jaravel (2019).

Our main results use Laspeyres price indices, which most closely resemble the methodology used by the BLS to construct inflation rates. These results are robust to using alternative price indices that account for households' abilities to substitute across goods and for nonhomothetic preferences. Accounting for substitution with Törnqvist price indices tends to reduce the estimated level and volatility of inflation, and does so roughly proportionately across income groups. In contrast, accounting for nonhomotheticities using the algorithm developed by Jaravel and Lashkari (2024) tends to *increase* the volatility of inflation. This is because when food-at-home prices rise faster than income, households tend to trade down to lower-priced varieties (Jaimovich, Rebelo, and Wong 2019; Argente and Lee 2021), which are exactly those varieties that experience the highest inflation rates when food-at-home input costs rise. When both substitution and nonhomotheticities are jointly accounted for, our estimates of the differences in inflation between low- and high-income households remain largely unchanged.

While our most detailed evidence comes from food-at-home products, we provide suggestive evidence that our channel extends to other categories of consumption. The lack of disaggregated inflation and household purchase data typically makes it challenging to generalize findings from scanner data on fast-moving consumer goods to other product categories. We address this challenge in three ways. First, we use microdata underlying the U.K. consumer price index to detect cheapflation cycles in goods and services beyond food at home. Second, we use BLS price indices from across U.S. cities to detect similar

[1] Our findings echo concerns raised by U.K. anti-poverty advocate Jack Monroe, who argued that the U.K.'s national inflation index "grossly underestimates the real cost of inflation" for low-income households, citing large price increases for several low-price grocery products. Her campaign prompted the U.K. Office for National Statistics to launch a study of inflation for low-price grocery items. However, due to limited product coverage, the study was unable to detect elevated inflation for low-price items.

fluctuations in relative inflation rates faced by low- and high-income cities. Third, we explore the automobile market, where detailed data on household purchases and model-level prices are available from the Consumer Expenditure Survey and industry catalogues. In each case, we find fluctuations in cheapflation and inflation inequality consistent with our main results, suggesting that our conclusions on cycles in inflation inequality may generalize to the broader consumption bundle.

Finally, we show that the same mechanism leads to inflation heterogeneity across any units with different expenditure patterns, not just income groups. For example, pass-through in levels implies that cities that consume more expensive varieties or that have larger distribution margins experience muted responses to input cost shocks. Similarly, countries that import higher-priced varieties will see less import price inflation when global input costs rise. We provide evidence for both predictions. The mechanism thus predicts systematic cross-sectional variation in exposure to common input price changes.

Related literature. This paper relates to a literature that explores differences in inflation across households, surveyed by Jaravel (2021). The vast majority of this literature considers inflation differences that arise due to differences in household spending shares across product categories.[2] We show that aggregation to product categories masks considerable fluctuations in inflation inequality due to differences in spending patterns within narrowly defined categories.

Our focus on cost-driven fluctuations in inflation inequality also differs from Kaplan and Schulhofer-Wohl (2017) and Jaravel (2019), who use scanner data to measure *secular* differences in inflation rates across income groups. We document that differences in the varieties bought by households, combined with pass-through in levels, lead to *fluctuations* in inflation inequality in response to input cost shocks. Given the lack of prior evidence, previous work has typically assumed that the differences in purchased varieties do not lead to such fluctuations. For example, Del Canto et al. (2025) note,

> "Prior work has found that households at different income levels experience different trend inflation in consumption prices, and that this difference is driven

[2]See, e.g., Michael (1979), Hagemann (1982), Garner, Johnson, and Kokoski (1996), Hobijn and Lagakos (2005), Hobijn, Mayer, Stennis, and Topa (2009), Cravino, Lan, and Levchenko (2020), Hottman and Monarch (2020), Klick and Stockburger (2021, 2024), Hochmuth, Pettersson, and Christoffer (2022), Chakrabarti, Garcia, and Pinkovskiy (2023), Ferreira, Leiva, Nuño, Ortiz, Rodrigo, and Vazquez (2023), Pallotti, Paz-Pardo, Slacalek, Tristani, and Violante (2023), Cavallo (2024), Jaravel (2024), Lan, Li, and Li (2024), Olivi, Sterk, and Xhani (2024), Orchard (2025), Del Canto, Grigsby, Qian, and Walsh (2025), Lauper and Mangiante (2025), Lokshin, Sajaia, and Torre (2025), and Neyer and Stempel (2025). Each of these papers considers differences in inflation across households due to heterogeneous spending shares across BLS entry-level items (ELIs) or more aggregated product categories.

> by differences within fine product groups. [.... However], there is no a priori reason to think inflation rates of finer product categories should be differentially responsive to short-run shocks."

In contrast, we show that input cost shocks lead to systematic fluctuations in inflation inequality within narrow product categories. Monetary or productivity shocks that move input prices will thus affect within-category differences in inflation across income groups.

Our finding that pass-through in levels can parsimoniously account for fluctuations in inflation inequality over time relates to prior work that studies cheapflation and inflation inequality during specific periods, such as the Great Recession (Li 2019; Argente and Lee 2021; Becker 2026), the Covid-19 pandemic in 2020 (Jaravel and O'Connell 2020; Weber, Gorodnichenko, and Coibion 2023), and the period of inflation from 2021–2023 (Cavallo and Kryvtsov 2024; Chen, Levell, and O'Connell 2024). Those previous studies do not show that pass-through in levels accounts for the patterns they document. Pass-through in levels likewise offers an explanation for the disproportionate effects of large aggregate cost shocks, such as currency devaluations or tariffs, on low-income households that purchase lower-priced varieties (see e.g., Cravino and Levchenko 2017; Gouvea 2017; Cavallo, Llamas, and Vazquez 2025).

Finally, we argue that fluctuations in cheapflation and inflation inequality arise due to pass-through in levels, relating to a large literature on the theoretical and empirical determinants of pass-through (see e.g., Bulow and Pfleiderer 1983; Campa and Goldberg 2005; Burstein, Eichenbaum, and Rebelo 2006; Nakamura and Zerom 2010; Weyl and Fabinger 2013; Burstein and Gopinath 2014; Mrázová and Neary 2017; Amiti, Itskhoki, and Konings 2019; and Miravete, Seim, and Thurk 2023, 2025). Sangani (2026) makes the case for pass-through in levels across a broad array of industries, surveys previous estimates of pass-through in levels and logs, and derives conditions on demand that lead to pass-through in levels. That paper does not explore its implications for cheapflation and inflation inequality, as we do in this paper.

Layout. Section 2 describes the data sources we use. In Section 3, we use coffee as a laboratory to study pass-through and its effect on cheapflation and inflation inequality. Section 4 broadens the analysis to the entire food-at-home basket. Section 5 analyzes other categories beyond food at home. Section 6 applies our mechanism to heterogeneity in inflation across cities and import price inflation across countries, and Section 7 concludes.

2 Data Sources and Sample Construction

Our main analysis uses two datasets: NielsenIQ Retail Scanner data, which we use to explore how inflation and pass-through vary across the price distribution, and NielsenIQ Household Panelist data, which we use to construct inflation rates faced by different income groups.

2.1 Retail Scanner Data

The NielsenIQ Retail Scanner dataset records weekly sales and quantities sold of individual products (referred to hereafter as universal product codes, or UPCs) at over 30,000 retail stores across the United States. The data cover food-at-home products and other nondurables such as personal care products, household cleaning supplies, and general merchandise. We use data from 2006 to 2023, which cover over $5.4 trillion in cumulative sales. An advantage of the Retail Scanner dataset is that it includes all sales of tracked products in participating retail chains, allowing us to measure product-level inflation rates with much less sampling error than would be possible in panelist data.

For each universal product code i in each retail chain r, we calculate the average price in quarter t as $p_{irt} = \text{Sales}_{irt}/\text{Units}_{irt}$. Our choice to aggregate at the level of UPC by retail chain follows from DellaVigna and Gentzkow (2019), who show that stores within a retail chain tend to exhibit little variation in posted prices for a UPC at each point in time.[3] We use these quarterly prices to construct year-over-year inflation rates for each UPC i in each retail chain r:

$$\pi_{irt} = \frac{p_{irt}}{p_{irt-4}} - 1.$$

We primarily use these year-over-year inflation rates to avoid seasonality effects that may bias inflation measured over shorter horizons.

Appendix Table A1 reports that products with available inflation rates account for 91 percent of sales in the Retail Scanner data in each year. The remaining 9 percent consists of sales of products that are discontinued or whose attributes—such as package size or branding—have changed over the year.[4]

[3]Since we measure prices using a quantity-weighted average within retail chain and quarter, variation in inflation across products can reflect variation in both posted prices and customer shopping behavior. Quantitatively, changes in posted prices appear to account for most of this variation: Appendix C.1 shows that aggregating store–week price observations using an unweighted average yields similar results.

[4]NielsenIQ uses UPC versions to delineate such attribute changes. We are careful to treat different UPC versions as distinct products to avoid conflating quality changes with inflation for continuing varieties.

Construction of unit price groups. To analyze inflation dynamics across the price distribution, we stratify products within narrowly defined categories by unit price. Products in the NielsenIQ data are classified into about 1,200 highly disaggregated "product modules," such as refrigerated yogurt, bar soap, olive oil, paper towels, and antacids. For each of these product categories, we use product package sizes provided by NielsenIQ to calculate unit prices for products in the category (e.g., dollars per ounce of coffee). In each quarter, we calculate the average unit price for each UPC i at each retail chain r using sales and quantities sold over the prior four quarters, to ensure that these unit prices reflect persistent price differences.[5] Then, we order products in each quarter by unit price and split products into N bins with equal sales. The first bin corresponds to UPC-retailer pairs in each product module with the lowest unit prices, while the Nth bin contains the products with the highest unit prices. We primarily use unit price deciles (i.e., $N = 10$), though for some analyses we aggregate to terciles or quintiles ($N = 3$ or 5).

To calculate inflation rates for each unit price group, we aggregate product-level inflation rates using initial sales shares, $\pi^n_{t+4} = \sum_{ir\in n} \lambda_{irt}\pi_{irt+4}$, where λ_{irt} is the sales of UPC i at retailer r in quarter t as a share of the sales of all continuing varieties in group n. Analogous inflation rates for unit price groups within a specific product module m are denoted by π^n_{mt+4}. Likewise, we construct the change in price level for unit price group n as $\Delta p^n_{t+4} = \sum_{ir\in n} \tilde{\lambda}_{irt}\Delta p_{irt+4}$, where $\Delta p_{irt} = p_{irt} - p_{irt-4}$ and $\tilde{\lambda}_{irt}$ is the quantity share (e.g., in ounces) of UPC i at retailer r in group n in quarter t.

2.2 Household Panelist Data

To analyze differences in inflation across income groups, we supplement the Retail Scanner data with data on purchases by a nationally representative panel of households. The NielsenIQ Homescan Consumer Panel includes about 60,000 participating households each year. Households record the date and location of shopping trips, scan all purchased items in NielsenIQ-tracked categories, and report promotions and discounts. NielsenIQ offers participating households a variety of incentives to accurately report data and drops households from the program that do not meet minimum reporting standards.

[5] A concern with using products' initial prices is that subsequent differences in inflation across the price distribution could in part reflect mean reversion. Our use of average unit prices over the prior four quarters is designed to mitigate this concern. Note however that while mean reversion would mechanically imply higher inflation rates for lower-priced products, it would not generally lead to a systematic *covariance* between products' initial prices and future input cost changes. Thus, mean reversion should not generally bias estimates of pass-through across the price distribution. We verify in Appendix C.2 that sorting products into groups using longer-run averages of products' prices does not meaningfully change our results.

Construction of income quintiles. To analyze inflation across the income distribution, we group households each year into income quintiles. We order households by reported income using the sixteen income bins provided by NielsenIQ and further sort households within each income bin by total expenditures divided by the square-root of household size.[6] We then split households into five equally-sized groups using projection weights provided by NielsenIQ. Appendix Figure A1 shows how equivalized expenditures are used to break ties when households in the same income bin are split across income quintiles. For robustness, we repeat our analyses splitting households into income deciles.

We construct inflation rates for each income group by calculating each group's expenditures on each UPC in each quarter and merging in UPC-level inflation rates from the Retail Scanner dataset. Our baseline analysis on inflation rates by income quintile aggregates at the level of UPC rather than UPC-retail chain, because matching by UPC allows us to cover a larger share of household expenditures. Appendix Table A2 reports that over 60 percent of expenditures reported in the Homescan data can be matched by UPC to the Retail Scanner data, compared to 25 percent when matching by both UPC and retail chain. This difference owes to the fact that not all retail chains that households buy from participate in the NielsenIQ Retail Scanner program.[7] We trim the top and bottom 0.5 percent of inflation observations to ensure our results are not driven by extreme outliers.

Our baseline measures of inflation for each income group are measured using Laspeyres indices, $\pi^g_{t+4} = \sum_i \omega^g_{it}\pi_{it+4}$, where ω^g_{it} is the expenditure share of income group g on product i in quarter t and π_{it+4} is the year-over-year inflation in the quantity-weighted average price of UPC i from quarter t to quarter $t+4$. We likewise denote the income-group inflation rate within a specific product module m by π^g_{mt+4}. These Laspeyres indices most closely resemble the way the Bureau of Labor Statistics (BLS) constructs its consumer price index for urban consumers (CPI-U). Of course, the Laspeyres price index does not account for substitution across products in response to price changes or how nonhomotheticities cause households' baskets to change endogenously with real income. Section 4.4 explores how inflation across income groups compares under alternative price indices that account for these forces.

[6] The income inequality literature commonly uses the square-root of household size to equivalize incomes and expenditures across households. See e.g., the BLS's methodology for calculating the distribution of personal consumption expenditures (`https://www.bls.gov/cex/pce-ce-distributions.htm`) and Handbury (2021) and Klick and Stockburger (2024) for applications to price indices across income groups.

[7] In Appendix C.3, we show that inflation dynamics for UPCs and retailers purchased by panelists but not matched to the NielsenIQ data are similar to those in the merged dataset. Moreover, the unit prices of unmatched products exhibit a stronger covariance with buyer income than matched products, suggesting our results on inflation inequality in the merged sample are conservative. Appendix Figure A2 also shows that the match rates for products and retail chains over time are largely similar across income groups.

3 Coffee: A Case Study

We begin by using roasted coffee as a laboratory to explore how pass-through in levels generates heterogeneity in inflation rates across products and households. Coffee provides an ideal empirical setting for this analysis. The primary input in producing coffee—green Arabica coffee beans—is traded on global commodity markets, and its price exhibits large swings due to weather shocks in coffee-growing regions and exchange rate movements. These cost shocks offer ample variation for identifying the pass-through of cost changes to retail prices. The retail coffee market also exhibits substantial cross-sectional variation in prices, with premium brands like Starbucks and Peet's charging much higher unit prices than brands like Folgers and Maxwell House. This variation allows us to examine how a common cost shock translates into different inflation rates across the price distribution.

Our analysis builds on Nakamura and Zerom (2010), who estimate the aggregate pass-through of commodity costs to coffee retail prices from 2000–2005, and Sangani (2026), who documents pass-through in levels for several food products including coffee. We extend their analyses by considering how the pass-through of input cost shocks varies across the price distribution and its implications for cheapflation and inflation inequality. Section 3.1 documents that coffee commodity changes are passed through cent-for-cent into retail prices across the price distribution (consistent with Sangani 2026), and Section 3.2 shows that this pattern generates cycles in cheapflation and inflation inequality within coffee.

3.1 Pass-Through in Levels and Cheapflation

We pair NielsenIQ data on retail coffee prices with commodity price data from the International Coffee Organization (ICO). Specifically, we use retail prices of products in the "ground and whole bean coffee" product module, and we use ICO's price of mild Arabicas at the dock in New York as our measure of commodity input prices.[8] Following Nakamura and Zerom (2010), we adjust the commodity price to account for the fact that green coffee beans lose 19 percent of their weight during the roasting process.

Figure 1 plots inflation in coffee commodity prices alongside inflation for retail coffee products split into three unit price groups. Coffee commodity prices exhibit large swings over our sample period from 2006 to 2023. In three instances—in 2011, 2014, and 2021—

[8] Different coffee quality grades and origins typically trade at "differentials" (in cents/ounce) to the benchmark commodity contract. Using commodity prices for various coffee varieties from Datastream, Appendix Figure A3 shows that these series indeed comove closely with the ICO commodity price measure. Appendix Table A3 finds that changes in the benchmark ICO commodity price are passed through one-for-one to the commodity prices for these different varieties, with no evidence of systematically different pass-through rates across cheaper or more expensive varieties.

coffee commodity prices rose by over 60 percent within the span of one year. These swings in commodity prices appear to be incorporated into retail prices with a lag of about two quarters, as shown by the solid lines indicating inflation for retail coffee products.

The top panel of Figure 1 shows substantial heterogeneity in the response of retail price inflation to commodity prices across unit price groups. In each of the three episodes of rising commodity prices, inflation was substantially higher for products with lower unit prices. For example, in 2011, retail prices for products in the lowest tercile of unit prices rose by 40 percent, compared to 12 percent for products in the top tercile.

These differences across unit price groups disappear in the bottom panel of Figure 1, which instead shows price changes in levels (i.e., cents per ounce). For the largest commodity price increases in 2011 and 2021, unit prices rose by almost exactly the same amount across unit price terciles. The identical increases in price levels across products appear as larger inflation rates for products with lower unit prices, because the same price increase represents a larger percentage change for low-priced products.

In their study of online food prices from 2021–2023, Cavallo and Kryvtsov (2024) term the pattern of higher inflation for lower-priced varieties "cheapflation." Figure 1 shows that cheapflation is not specific to the post-pandemic period: indeed, we find systematic appearances of cheapflation each time green coffee bean prices rise. Moreover, Figure 1 suggests a simple candidate explanation for these cheapflation cycles. If firms raise their prices by the same absolute amount in response to commodity cost increases, these identical absolute price changes generate larger percentage price increases for lower-priced varieties, and thus cheapflation.

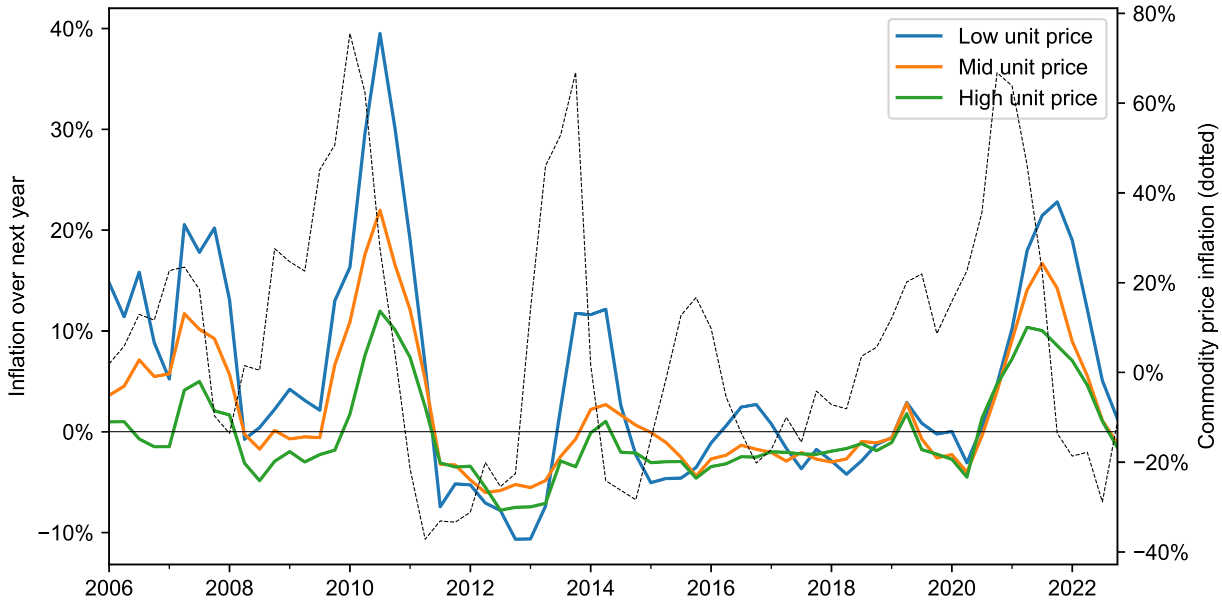
To formalize this intuition, we estimate the long-run pass-through of commodity cost changes to retail prices using the distributed lag specification,

$$\Delta p_{irt} = a_{ir} + \sum_{k=0}^{K} b_k \Delta c_{t-k} + \sum_{k=1}^{4} d_k q_t + \varepsilon_{irt}. \tag{1}$$

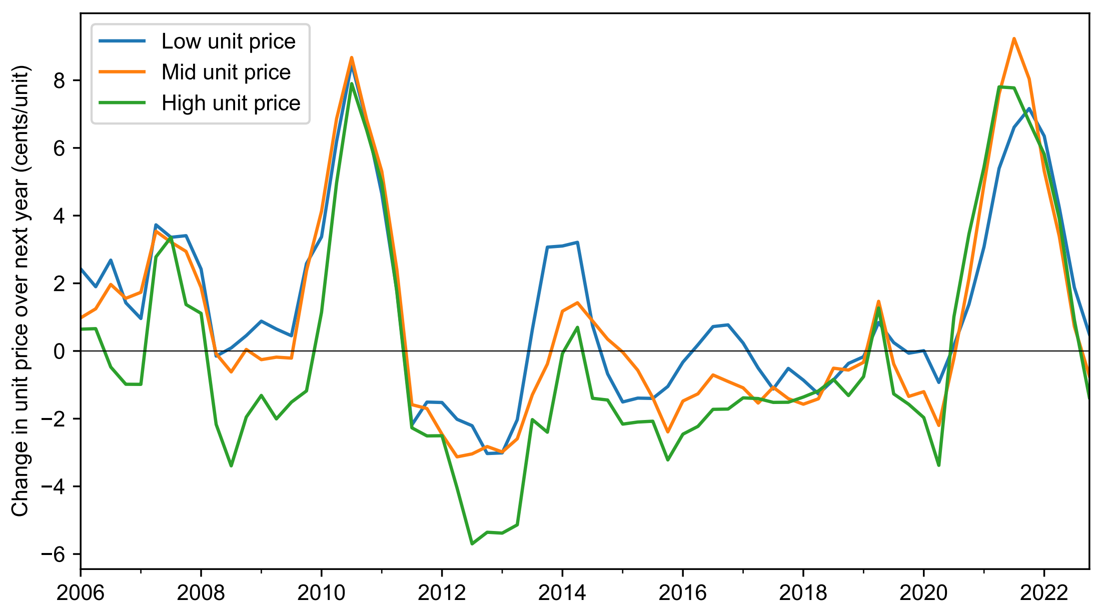
where Δp_{irt} is the change in the price of UPC i at retailer r from quarter $t-1$ to t, Δc_t is the change in the commodity price, a_{ir} are fixed effects that absorb product-retailer trends, and q_t are quarter-of-year dummies that absorb seasonal variation in prices. The coefficients b_k capture the change in retail prices associated with a change in commodity input costs k quarters ago. Accordingly, the long-run pass-through is given by the sum of the coefficients, $\sum_{k=0}^{K} b_k$. For our baseline results, we choose a horizon of $K = 6$ quarters to match Nakamura and Zerom (2010).[9]

[9]Industry analysts report that coffee commodity prices take about nine months to be passed through to retail prices, due to roasting times and roaster inventories (Gutmann and Sangani 2026).

Figure 1: Coffee price changes in percentages and levels, by unit price tercile.



(a) Price changes in percentages (inflation).



(b) Price changes in levels.

Note: Panel (a) plots inflation over the next year for three unit price groups of coffee products, constructed using the procedure described in Section 2.1. The dotted line in panel (a) plots inflation in coffee bean commodity prices over the next year. Panel (b) plots the quantity-weighted average change in unit prices over the next year for coffee products in each unit price group, as described in Section 2.1.

This specification is standard in the literature on pass-through, though the canonical specification uses price and cost changes measured in logs rather than in levels (see e.g., Campa and Goldberg 2005). Using (1) to compare pass-through along the price distribution is advantageous to measuring pass-through rates at, say, six-month or one-year horizons, because pass-through rates at shorter horizons may reflect a combination of desired long-run pass-through and short-run price rigidities. In a calibrated model of the coffee market, Nakamura and Zerom (2010) find that while price rigidities (menu costs in their model) contribute to delays in price adjustment, they have a negligible effect on the long-run pass-through estimated at six quarters.

The left panel of Figure 2 plots the long-run pass-through in levels, estimated for the full sample of coffee products as well as separately for each unit price quintile. The pass-through in levels for the entire sample is 1.01 (standard error 0.11). In other words, increases in green coffee bean prices are eventually passed through cent-for-cent to retail prices. Furthermore, pass-through in levels appears largely uniform across unit price groups, and we cannot reject complete pass-through in levels for any unit price group.[10]

The right panel of Figure 2 plots the long-run pass-through in logs, estimated with the analogous specification,

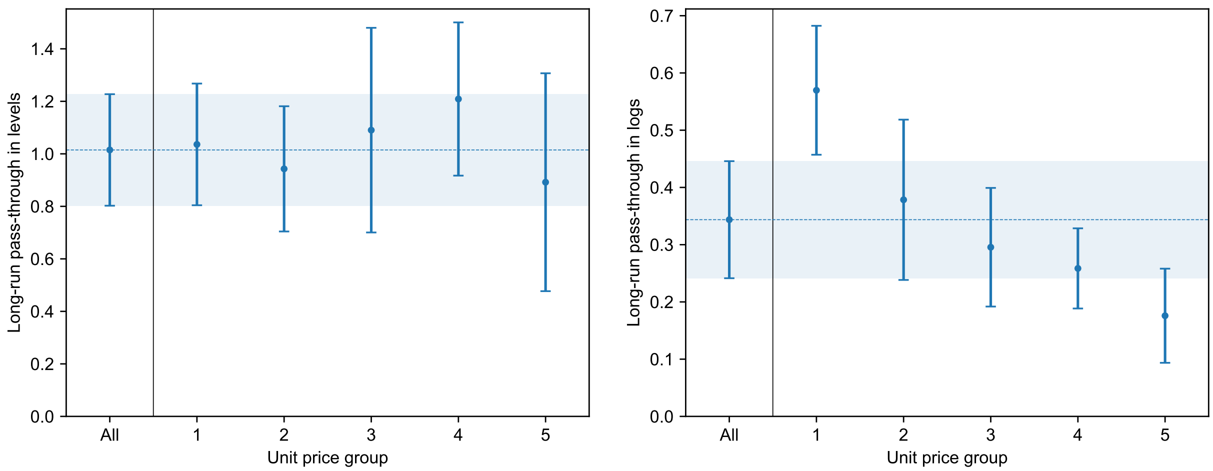
$$\Delta \log p_{irt} = \alpha_{ir} + \sum_{k=0}^{K} \beta_k \Delta \log c_{t-k} + \sum_{k=1}^{4} \delta_k q_t + \varepsilon_{irt}. \tag{2}$$

Since pass-through is complete in levels, pass-through when measured in logs is mechanically incomplete: the log pass-through for the entire sample is 0.34 (standard error: 0.05). Complete pass-through in levels generates heterogeneous sensitivity in inflation rates to commodity inflation across unit price groups. For the lowest unit price quintile of products, the log pass-through of commodity inflation to retail prices is 0.57, compared to 0.18 for the highest unit price quintile.

Thus, we find that coffee products across unit price groups exhibit complete pass-through in levels of changes in commodity input costs. Complete pass-through in levels implies that the inflation rates of low-priced varieties are more sensitive to input cost inflation, because the same absolute price increase (or decrease) represents a larger proportional change in price for lower-priced varieties. Pass-through in levels thus systematically leads to cheapflation when input costs rise.

[10]While Nakamura and Zerom (2010) document that the aggregate pass-through in levels of commodity prices to wholesale and retail coffee prices is close to one, they do not show that pass-through in levels emerges uniformly across the price distribution. In fact, their model instead predicts that pass-through in levels varies systematically with unit price, as we show in Appendix Figure D1.

Figure 2: Pass-through of coffee commodity costs to retail prices by unit price group.



(a) Pass-through in levels. **(b)** Pass-through in logs.

Note: Panels (a) and (b) show estimates of the long-run pass-through in levels $\sum_{k=0}^{K} b_k$ and logs $\sum_{k=0}^{K} \beta_k$, estimated with specifications (1) and (2) respectively. Error bars indicate 95 percent confidence intervals constructed using the delta method, with standard errors two-way clustered by brand and quarter. The dotted line and shaded region indicate the point estimate and 95 percent confidence interval for pass-through estimated in the full sample.

Robustness. One may be concerned that our estimates of the pass-through of commodity costs to prices are biased by reverse causality, for example due to demand shocks that feed back from retail prices to commodity prices, or may wonder whether changes to the specifications in (1) and (2) alter our results. We evaluate robustness to these concerns in Appendix Table A4. We continue to find uniform pass-through in levels across unit price groups and declining log pass-through with unit price when we use exchange rates for Brazil and Colombia and weather shocks to coffee-growing areas in Brazil and Colombia to isolate exogenous changes in commodity prices. Absorbing product-specific seasonality effects or changing the horizon over which we measure the long-run pass-through also does not meaningfully alter our results.

We also test the robustness of our results to measurement choices made in Section 2. Our baseline measures of inflation are constructed using quantity-weighted average prices of each UPC across store–weeks within retail chain and quarter. Appendix Figure C1 finds similar pass-throughs across the price distribution using inflation rates instead constructed from unweighted average prices of UPCs across store–week observations within retail chain and quarter. Our baseline results also sort varieties into unit price groups using their average unit price over the prior year. Appendix Figure C4 finds that sorting varieties into groups using their average prices over a longer horizon—the prior two, three, or five years—has little effect on measured pass-through, allaying the concern that our estimates are biased by mean reversion.

3.2 Within-Category Cycles in Inflation Inequality

If households across the income distribution purchase different varieties, differences in the sensitivity of varieties' inflation rates to commodity shocks will lead to differences in inflation across the income distribution. We estimate how the average unit price of varieties purchased by each income group differs using the specification,

$$\log p^{g}_{\text{coffee},t} = \sum_{q=1}^{5} \beta_q 1\{g = q\} + \phi_t + \varepsilon_{gt}, \tag{3}$$

where $\log p^{g}_{\text{coffee},t}$ is the log of the average unit price of coffee products purchased by households in income quintile g in quarter t and ϕ_t are time fixed effects that absorb changes in unit prices over time. We omit the dummy for the first quintile, so that the coefficients β_q estimate the difference in log unit prices paid by each income quintile relative to the lowest income group.

Panel A of Table 1 shows that high-income households purchase higher-priced varieties: coffee products bought by the top income quintile are on average 24 percent more expensive than those purchased by households in the lowest income quintile. These patterns are consistent with previous work that documents differences in the relative prices of products purchased by income groups within narrow product categories (e.g., Aguiar and Hurst 2007; Faber 2014; Handbury 2021; Sangani 2022).

These differences in the price of varieties purchased by income groups, combined with pass-through in levels, generate differences in the sensitivity of inflation rates faced by each income group to input cost fluctuations. We estimate these differences in inflation sensitivity using the specification,

$$\pi^{g}_{\text{coffee},t} = \sum_{q=1}^{5} \beta_q \left(\pi_{\text{coffee},t} \times 1\{g = q\}\right) + \alpha_g + \phi_t + \varepsilon_{gt}. \tag{4}$$

Here, $\pi^{g}_{\text{coffee},t}$ is the year-over-year coffee inflation rate for income quintile g from quarter $t-4$ to quarter t as defined in Section 2.2, and $\pi_{\text{coffee},t}$ is the overall inflation rate for coffee products in the Retail Scanner data over the same period. The group fixed effects α_g absorb differences in trend inflation across income groups, and time fixed effects ϕ_t absorb average

Table 1: From cheapflation to inflation inequality in coffee.

Panel A. Prices paid		*Panel B. Inflation sensitivity*		
	Log unit price		*Coffee inflation for income group*	
	(1)		(OLS)	(IV)
Income quintile 2	0.036**	Coffee inflation × Income quintile 2	−0.039**	−0.055**
	(0.000)		(0.002)	(0.004)
Income quintile 3	0.093**	Coffee inflation × Income quintile 3	−0.096**	−0.118**
	(0.001)		(0.004)	(0.006)
Income quintile 4	0.157**	Coffee inflation × Income quintile 4	−0.155**	−0.198**
	(0.005)		(0.007)	(0.026)
Income quintile 5	0.236**	Coffee inflation × Income quintile 5	−0.270**	−0.324**
	(0.006)		(0.017)	(0.038)
Time FEs	Yes	Time FEs	Yes	Yes
N	360	N	340	340
R^2	0.99	R^2	1.00	1.00
Within R^2	0.94	Within R^2	0.76	0.19

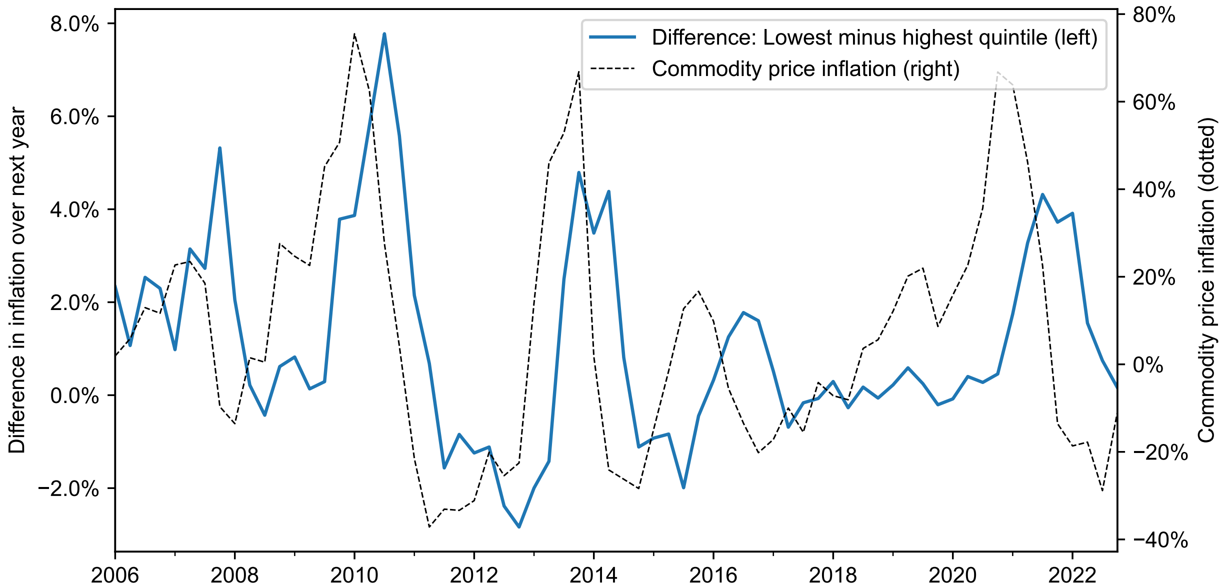
Note: Panel A reports results from specification (3), and Panel B reports results from specification (4) (for ease of display, Panel B does not display the group fixed effects α_g). The IV column in Panel B uses contemporaneous coffee bean commodity inflation rates and two lags to instrument for coffee inflation rates. Regressions weighted by expenditures. Standard errors two-way clustered by year and income group.

coffee inflation over time. We again omit dummies for the first income quintile, so that β_q measures differences in the sensitivity of inflation rates for each income quintile to overall coffee inflation relative to the lowest income group. As noted in Section 2.2, inflation rates for each income group are constructed using Laspeyres price indices, so differences in inflation across income groups are not influenced by how households' expenditure shares subsequently respond to price changes.

As predicted by pass-through in levels, the sensitivity of coffee inflation rates for each income quintile to overall coffee inflation declines systematically with income. Inflation rates for the highest income quintile are 27 percent less sensitive to overall fluctuations in coffee prices than for the lowest income quintile. We find similar results when we isolate input cost-driven fluctuations in coffee prices, using current and lagged inflation rates for coffee bean commodity prices to instrument for overall coffee inflation.

These results imply fluctuations in "inflation inequality"—i.e., the gap in inflation experienced by low- and high-income households—as commodity costs move around. Figure 3 plots the gap between the coffee inflation rate for households in the lowest income quintile and the highest income quintile. Indeed, there are large swings in the

Figure 3: Within-category inflation inequality in coffee.



Note: The solid line plots the difference in coffee inflation rates over the next year for the first and fifth income quintiles, constructed using the procedure described in Section 2.2. The dotted line plots inflation in coffee bean commodity prices over the next year.

extent of the within-category inflation inequality, with spikes in 2011, 2014, and 2022 following increases in coffee commodity costs. The inflation gap is positive on average, consistent with the secular drivers of inflation inequality documented by Jaravel (2019), but also features cyclical variation predicted by pass-through in levels—even becoming negative in periods of commodity price deflation, such as in 2012–2013.

Comparison to BLS data. The within-category fluctuations in inflation inequality shown in Figure 3 are masked when prices are aggregated to the level of publicly released CPI data. To organize price collection, the Bureau of Labor Statistics (BLS) classifies goods and services into 273 entry-level items (ELIs), which are grouped into 243 basic items and further aggregated into 211 item strata. The most disaggregated CPI data released publicly is at the item stratum level (typically reported across 32 geographic areas). A large literature studies differences in inflation across the income distribution by matching inflation rates on item strata to household expenditure shares across items (see the list of studies in Footnote 2).

Such datasets built on ELI- or stratum-level prices can miss substantial fluctuations in inflation inequality that stem from differences in purchasing patterns within product categories. Coffee, for instance, is assigned to item stratum SEFP01, which includes a single ELI (FP011) that covers both roasted and instant coffee. Over our sample period, the maximum year-over-year inflation rate in SEFP01 was 21 percent (from 2010Q3–2011Q3), and the minimum was −7 percent (from 2013Q1–2014Q1). In the disaggregated data, however, the inflation gap between the first and fifth income quintiles *within* roasted coffee alone ranges between 7.8pp (2010Q3–2011Q3) and −2.8pp (2012Q4–2013Q4).[11] In other words, assuming category-level inflation rates apply to households in each income group understates the degree of inflation inequality when costs rise and overstates the degree of inflation inequality when costs fall.

4 Cheapflation & Inflation Inequality Across Food at Home

We now consider fluctuations in cheapflation and inflation inequality across the entire food-at-home basket. The same patterns that we documented in coffee extend to food-at-home products: pass-through in levels generates systematic fluctuations in cheapflation and inflation inequality in response to changes in the relative price of upstream inputs.

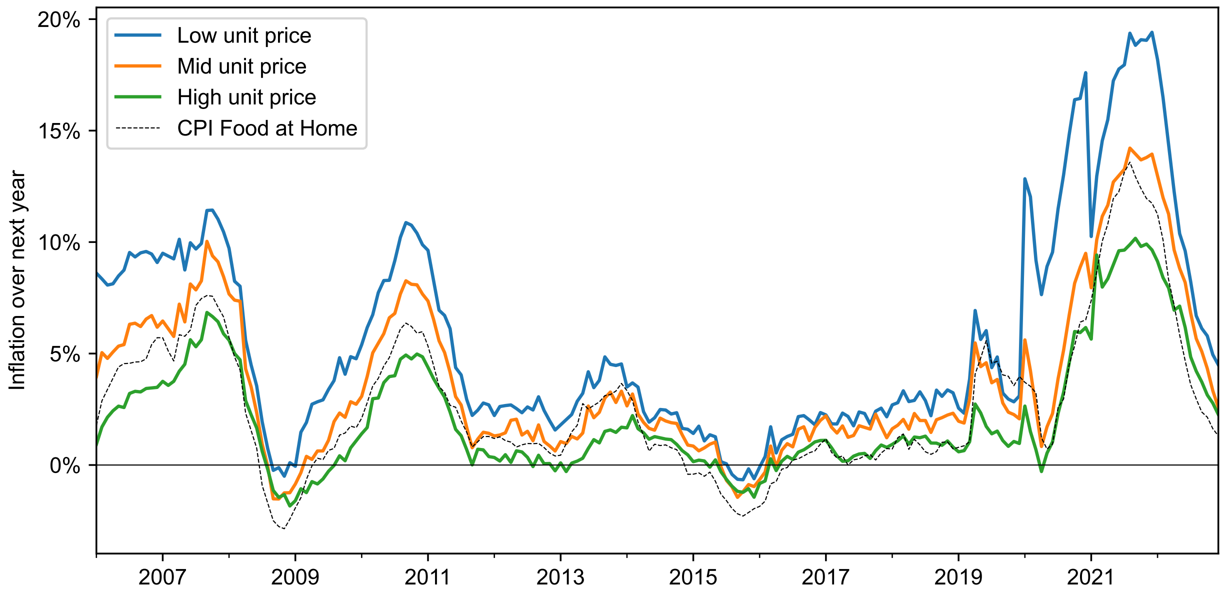
4.1 Pass-Through in Levels and Cheapflation

We use the Retail Scanner data to construct a food-at-home price index, using products in six departments: dry grocery, frozen foods, dairy, deli, packaged meat, and fresh produce. Kaplan and Schulhofer-Wohl (2017) and Beraja, Hurst, and Ospina (2019) show that price indices constructed from NielsenIQ scanner data track BLS statistics closely, and Appendix Figure A4 confirms that this is the case for our food-at-home price index: inflation rates for our food-at-home price index have a correlation with the BLS's food-at-home consumer price index of 0.96.

The advantage of the Retail Scanner data is that we can then disaggregate inflation in the overall food-at-home price index by unit price group to study how inflation varies across the price distribution. Figure 4 plots inflation rates from 2006 to 2023 for three unit price groups. Recall that varieties in each product module are split into groups with equal sales, ensuring that each unit price group has an identical distribution of sales across product modules. Thus, differences in inflation across unit price groups reflect differences in inflation rates for low- and high-priced varieties in each product module, rather than differences in composition across modules.

[11] BLS documentation for ELI FP011 indicates that it includes roasted, instant, and freeze-dried coffee. The NielsenIQ "ground and whole bean coffee" module covers a subset of these products (instant and freeze-dried coffee are categorized separately by NielsenIQ).

Figure 4: Cheapflation: Food-at-home inflation by unit price tercile from 2006–2023.

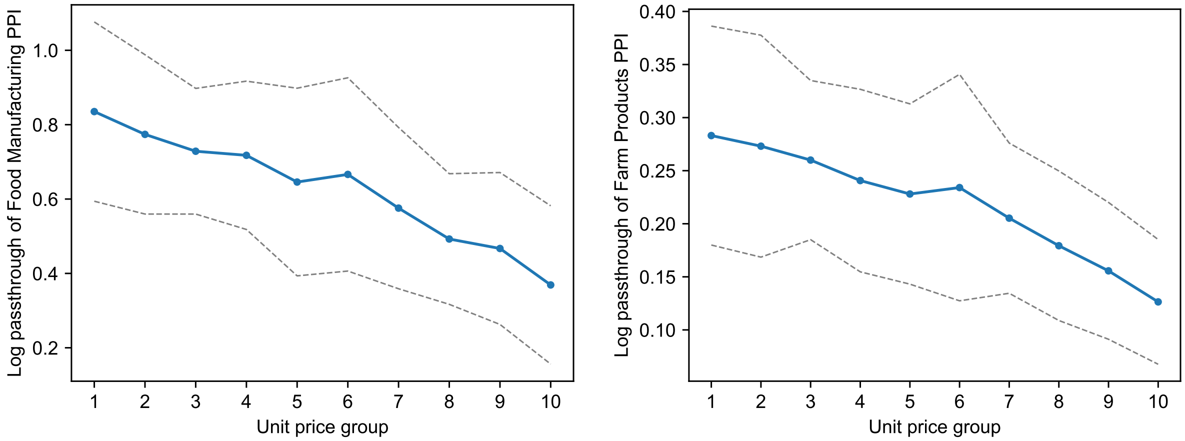


Note: The figure shows inflation over the next year for three unit price groups of food-at-home products, constructed using the procedure described in Section 2.1.

Figure 4 reveals substantial differences in inflation rates for low-priced food-at-home products compared to high-priced food-at-home products. Low-priced varieties have higher inflation rates than high-priced varieties over the whole sample, consistent with the secular trends in inflation inequality documented by Jaravel (2019). But the gap in inflation rates also widens during periods when overall food-at-home inflation is high—such as during the food commodity price booms of 2008, 2011, and 2021—and narrows during periods when aggregate food-at-home inflation rates are low. Viewed in context of the entire time series, the "cheapflation" during the 2021–2023 period is not an isolated incident, but rather is part of a systematic pattern of widening inflation rates between low- and high-priced varieties during times of high food-at-home inflation.

To test whether these differences in inflation by unit price group reflect differential sensitivity to upstream costs, we estimate the pass-through of two upstream producer price indices (PPI) for Food Manufacturing and Farm Products to food-at-home price indices for each unit price group. These two producer price indices reflect changes in input costs of processed and raw food, respectively: Appendix Figure A5 shows that inflation in these price indices tends to lead food-at-home consumer inflation. We estimate log pass-through of these PPI to food-at-home price indices using a distributed lag specification

Figure 5: Lower-price varieties have higher log pass-through of upstream price indices.



(a) Food Manufacturing PPI. **(b)** Farm Products PPI.

Note: Estimates of long-run pass-through of upstream producer price indices by unit price decile from specification (5), estimated over the period 2006–2020. Dotted lines indicate 95 percent confidence intervals, constructed using the delta method with Newey-West standard errors.

analogous to the one that we used to measure pass-through in coffee,[12]

$$\Delta \log p_t^n = \alpha_n + \sum_{k=0}^{K} \beta_k^n \Delta \log \mathrm{PPI}_{t-k} + \sum_{k=1}^{4} \delta_k^n q_t + \varepsilon_{nt}. \tag{5}$$

Here, $\Delta \log p_t^n$ is the change in the food-at-home price index for unit price decile n from quarter $t-1$ to t and $\Delta \log \mathrm{PPI}_t$ is the change in the producer price index over the same period. The intercepts α_n absorb secular differences in inflation rates across unit price groups, and quarter-of-year dummies q_t absorb seasonal effects. We choose a horizon of $K = 6$ quarters, matching our specification for coffee products. For our baseline results, we estimate these pass-through regressions over the period 2006–2020 to avoid the influence of the post-pandemic inflation, where changes in upstream producer price indices are correlated with changes in the aggregate price level.[13]

[12]Of course, unlike for coffee, these PPI reflect an average of price changes for several inputs to food at home. Appendix B shows formally that differences in PPI pass-through across unit price groups can be interpreted as differences in the average pass-through of each input cost to each variety (appropriately averaged across inputs and varieties) and/or differences in the covariance between the average pass-through of an input cost to a unit price group and that input's sensitivity to changes in the PPI.

[13]Including the period from 2021–2023 results in higher estimates of PPI pass-through since PPI increases over that period were accompanied by inflation in the aggregate price level, which raises retailers' other wage and distribution costs. We can control for concomitant changes in the aggregate price level by augmenting (5) to include the pass-through of wage changes. Appendix Figure A6 shows that doing so yields similar estimates of PPI pass-through whether or not we include the period from 2021–2023.

Figure 5 shows estimated long-run log pass-through, $\sum_{k=0}^{K} \beta_k^n$, of PPI changes to food-at-home prices by unit price decile. For both upstream cost indices, log pass-through systematically declines with unit price. For Food Manufacturing, log pass-through declines from 0.84 for the lowest price decile to 0.37 for the highest price decile, while for Farm Products, pass-through falls from 0.28 for the lowest decile to 0.13 for the highest price decile. In both cases, the pass-through of upstream cost movements to prices for products in the highest unit price decile is less than half that of the lowest decile.

Robustness. As for our analysis of coffee products, we test whether our results on pass-through for food-at-home products are sensitive to the measurement choices described in Section 2. Appendix Figure C2 shows that pass-through of the Food Manufacturing and Farm Products PPIs to retail prices is similar if we construct inflation rates using unweighted, rather than quantity-weighted, average prices of UPCs within each retail chain and quarter. Appendix Figure C5 also reports similar results using different measures of products' initial prices to sort them into unit price deciles, indicating that pass-through differences across unit price groups are not due to mean reversion. Finally, to check that our estimates do not conflate short-run price rigidities with long-run pass-through, Appendix Figure A7 shows that these pass-through estimates are stable as we vary the long-run pass-through horizon between four and eight quarters.

A simple formula for inflation differences under pass-through in levels. While the decline in log pass-through across unit price groups is consistent with pass-through in levels, it does not tell us whether pass-through in levels quantitatively accounts for the *extent* of pass-through differences across unit price groups. We can test whether pass-through in levels is quantitatively consistent with differences in inflation across unit price groups by deriving an expression that relates varieties' initial price differences to predicted differences in subsequent inflation.

To do so, we assume a simple pricing equation for each variety that yields pass-through in levels of input cost changes. Suppose each product category contains a continuum of varieties indexed by $i \in [0, 1]$, and that the unit price of variety i at time t is

$$p_{it} = m_t + \beta_i w_t, \tag{6}$$

where m_t denotes a common material input cost, w_t is the wage rate, and β_i are variety-specific weights on labor. Note that this pricing equation assumes that a unit of each variety contains the same quantity of the common material input (e.g., each ounce of roasted coffee across brands contains the same amount of coffee beans).[14]

Equation (6) can be the outcome of several different microfoundations. First, (6) obtains if varieties are produced by perfectly competitive firms, and each variety i is produced with a Leontief production technology in the material input and labor with varying production weights β_i on labor. Differences in log pass-through of changes in the material cost m_t across varieties then reflect differences in the material cost share. In this vein, Cravino and Levchenko (2017) explain differences in inflation across households after the 1994 Mexican peso devaluation with competitive retailers that combine physical goods and distribution services in different proportions.

Second, (6) can result even under imperfect competition if firms' prices are constrained by limit pricing. Suppose potential entrants—or consumers themselves—can buy the material input at cost m_t and convert it into a perfect-substitute product with labor cost $\beta_i w_t$. An incumbent then cannot sustain a price above $m_t + \beta_i w_t$, so prices amount to a margin over value added rather than a proportional markup over total cost (see Modigliani 1958 and Bils and Chang 2000 for formal treatments of this sort of limit pricing). Even if firm demand does not completely fall to zero above $m_t + \beta_i w_t$, (6) can be sustained by kinks in firms' residual demand curves at a fixed margin over the material cost, due for example to strategic interactions between oligopolistic firms (Maskin and Tirole 1988).

Third, the pricing function in (6) can obtain even when firms face smooth, downward-sloping residual demand curves if each variety is produced by a single firm with marginal cost m_t, and the quantity demanded of each variety is given by the generalized logit demand system,

$$q_{it} = \frac{\exp\left(\delta_i - \beta_i^{-1} p_{it}/w_t\right)}{\int_0^1 \exp\left(\delta_k - \beta_k^{-1} p_{kt}/w_t\right) dk},$$

where δ_i are taste shifters and β_i are price sensitivity parameters that vary across varieties. This demand system is one special case of a broader class of "shift invariant" demand systems characterized by Sangani (2026) that yield complete pass-through in levels of common cost shocks.

Regardless of the exact microfoundation, (6) has the desirable properties that material cost changes are passed through to prices completely in levels (i.e., holding w_t fixed,

[14]As we show in Table 2 below, (6) predicts inflation differences across varieties that quantitatively match the data. At first glance, the assumption of a common material input across varieties may appear at odds with evidence that higher-quality varieties often use more expensive inputs (see e.g., Faber 2014). But these two facts are not necessarily inconsistent. For example, high- and low-quality varieties may use different suppliers that themselves combine a common input with different amounts of labor. In practice, different quality grades of a commodity input are also often priced at fixed premia or discounts relative to a benchmark contract, as we discussed for coffee markets in Footnote 8.

$\Delta p_{it} = \Delta m_t$) while prices are neutral to changes in the aggregate price level (i.e., scaling material costs m_t and wages w_t by a common factor leads to proportional changes in each price p_{it}).

Given (6), we can characterize how a perturbation to materials and labor prices, denoted $\pi^m = d\log m$ and $\pi^w = d\log w$, shape inflation rates across groups of varieties. Let λ_i denote the initial sales share of variety i, and let $p = (\int_0^1 \lambda_i/p_i di)^{-1}$ denote the (unit-weighted) average unit price across varieties in the product category.

Proposition 1 (Differential inflation under pass-through in levels)**.** *Consider a group g of varieties defined by variety expenditure shares λ_i^g. Let p^g denote the initial (unit-weighted) average unit price of varieties in group g and $\pi^g = \int_0^1 \lambda_i^g \pi_i di$ denote the inflation rate on a Laspeyres price index for group g. Under* (6)*, the inflation rate π^g to first order is*

$$\pi^g \approx \pi^{all} + \left(\frac{p}{p^g} - 1\right)\left(\pi^{all} - \pi^w\right),$$

where $\pi^{all} = \int_0^1 \lambda_i \pi_i di$ is the inflation rate on a Laspeyres price index for the category.

Proof. See Appendix B. □

Proposition 1 shows that the difference between the inflation rate for a group of varieties g and the overall category inflation rate depends on the average unit price of varieties in group g relative to all varieties, p/p^g, and the gap between the category inflation rate and wage inflation. Intuitively, when $\pi^{\text{all}} > \pi^w$, the price of materials is increasing faster than the wage. Pass-through in levels then implies that lower-priced varieties will have larger percentage price increases, leading to higher inflation rates $\pi^g > \pi^{\text{all}}$ for groups of varieties where the average unit price $p^g < p$.

Thus, if prices follow (6), differences in inflation across subgroups of varieties can be predicted using three sufficient statistics: the ratio of the average unit price of products in the subgroup to all varieties p^g/p, the inflation rate for all varieties π^{all}, and the wage inflation rate π^w. Likewise, the extent of cheapflation—i.e., the gap in inflation between a group of low-priced products L and high-priced products H—can be estimated using

$$\underbrace{\pi^L - \pi^H}_{\text{“Cheapflation”}} \approx \underbrace{\left(p/p^L - p/p^H\right)}_{\text{“Price gap”}} \underbrace{\left(\pi^{\text{all}} - \pi^w\right)}_{\text{“Excess inflation”}} . \tag{7}$$

We term the first bracketed quantity the “price gap” because it captures the difference in relative prices between the two groups of products. The second bracketed quantity is the excess inflation for the product category relative to wages.

Table 2: Explaining cheapflation with pass-through in levels.

	Inflation of lowest-price decile − highest-price decile					
	(1)	(2)	(3)	(4)	(5)	(6)
Excess module inflation × Price gap	1.182**		1.187**		1.194**	
	(0.028)		(0.026)		(0.019)	
Module inflation × Price gap		1.187**		1.188**		1.195**
		(0.025)		(0.025)		(0.019)
Wage inflation × Price gap		-2.286**		-1.544**		-2.201**
		(0.454)		(0.480)		(0.568)
Post-2020			-0.069**	-0.055*		
			(0.018)	(0.028)		
Price gap	-0.004	0.045**	0.005	0.019	0.071**	0.108**
	(0.012)	(0.018)	(0.011)	(0.013)	(0.013)	(0.031)
Time FEs					Yes	Yes
Product Module FEs					Yes	Yes
N	42137	42137	42137	42137	42137	42137
R^2	0.92	0.92	0.92	0.92	0.93	0.93

Note: The table reports results from specifications (8) and (9). We measure wage inflation as the year-over-year log change in the BLS employment cost index for private industry workers. Inflation rates are measured from quarter t to quarter $t+4$, and the price gap in quarter t is measured as defined in (7) and is winsorized at the 1 percent level. Post-2020 is a binary indicator for years after 2020. Regressions weighted by sales. Driscoll-Kraay standard errors with four lags. * indicates significance at 10%, ** at 5%.

Are fluctuations in cheapflation due to pass-through in levels? We use (7) to evaluate whether pass-through in levels can quantitatively account for the differences in pass-through that we see across unit price groups. Based on (7), we estimate the specification,

$$\text{Cheapflation}_{mt+4} = \beta\left(\text{PriceGap}_{mt} \times \text{ExcessModuleInflation}_{mt+4}\right) + \delta\text{PriceGap}_{mt} + \varepsilon_{mt}, \quad (8)$$

where $\text{Cheapflation}_{mt+4} = \pi^1_{mt+4} - \pi^{10}_{mt+4}$ is the difference between the inflation rate for the lowest unit price decile and highest unit price decile of varieties in product module m from quarter t to quarter $t+4$, PriceGap_{mt} is the quantity defined in (7) calculated using average unit prices in quarter t, and $\text{ExcessModuleInflation}_{mt+4}$ is the overall inflation rate of product module m in excess of wage inflation. We measure wage inflation as the log change in the BLS employment cost index. In addition to the interaction term, we include PriceGap_{mt} separately in the regression to allow for secular differences in inflation rates across high- and low-priced varieties in each product module.

The first-order approximation in (7) predicts that under pass-through in levels, $\beta \approx 1$.

Column 1 of Table 2 reports that differences in inflation across the low- and high-priced products in the data line up closely with these predictions: we estimate $\hat{\beta} \approx 1.2$.[15] Thus, the differential response of low- and high-priced varieties to overall price fluctuations is quantitatively consistent with pass-through in levels.

In column 2, we report the results from a specification where we split excess inflation in each product module into the module inflation rate and the wage inflation rate:

$$\begin{aligned}\text{Cheapflation}_{mt+4} = \beta^{\text{module}} \left(\text{PriceGap}_{mt} \times \text{ModuleInflation}_{mt+4}\right) \\ + \beta^{\text{wage}} \left(\text{PriceGap}_{mt} \times \text{WageInflation}_{t+4}\right) + \delta \text{PriceGap}_{mt} + \varepsilon_{mt}, \quad (9)\end{aligned}$$

Equation (7) predicts that $\beta^{\text{module}} \approx 1$ and $\beta^{\text{wage}} \approx -1$: higher inflation rates for a module lead to more cheapflation, but higher wage inflation narrows the inflation differences between low- and high-priced varieties. These predictions are broadly borne out in column 2: we find that the estimated coefficient $\hat{\beta}^{\text{module}} \approx 1.2$, while $\hat{\beta}^{\text{wage}} \approx -2.3$.[16] The fact that wage inflation enters as an offsetting force supports our conjecture that cheapflation is a result of changes in input prices *relative* to the aggregate price level, rather than a consequence of generalized inflation.

How much of the cheapflation during the 2021–2023 inflation surge can be accounted for by (7)? In columns 3–4, we include an indicator for the post-2020 period to evaluate whether there was excess cheapflation during this period, over and above what would be predicted by pass-through in levels. In fact, we find the opposite—after accounting for pass-through in levels, the extent of cheapflation during the period from 2021 to 2023 was slightly less than expected given the rise in food-at-home prices relative to wages.

We extend this logic to evaluate whether other time period- or product category-specific factors are important to explain patterns of cheapflation in columns 5–6. Studies of specific episodes such as the Great Recession or the post-pandemic inflation often attribute relative price changes across low- and high-priced varieties to how business cycle conditions or the distribution of fiscal stimulus affect price sensitivity for different income groups (e.g., Li 2019; Mongey and Waugh 2025; Becker 2026). Under this view, the correlation between overall inflation rates and cheapflation in Figure 4 is due to coincident changes in households' incomes or asset positions. Columns 5 and 6 test for

[15] Appendix Table A6 shows that similar results obtain if we instead split products in each module into five unit price groups and measure cheapflation as the gap between the first and fifth unit price quintiles.

[16] The estimate of $\hat{\beta}^{\text{wage}}$ is somewhat sensitive to the measure of wage inflation. For example, Appendix Table A5 instead uses the average hourly earnings of production and nonsupervisory workers to measure wage inflation and estimates $\hat{\beta}^{\text{wage}} \approx -1.6$ (the estimate is not significantly different from -1). We use the employment cost index as our baseline, since Almuzara, Audoly, and Melcangi (2025) show that changes in workforce composition have large effects on the average hourly earnings measure after 2020.

these competing explanations by including time fixed effects, which capture the effect of any period-specific economic conditions on the extent of cheapflation and thus isolate how cross-sectional variation in product-category inflation rates relates to within-category cheapflation. The columns also include product category fixed effects, to test whether our results are affected by a small set of categories. The resulting estimates of the coefficients on the interaction terms do not change substantially, indicating that our estimates are not driven by a single period or product category. Moreover, the share of variation in cheapflation explained by the independent variables rises marginally from 0.92 to 0.93, suggesting that there is limited variation in cheapflation due to differences across product categories or time periods that is not explained by pass-through in levels.

4.2 Cycles in Inflation Inequality

As we saw for coffee products, fluctuations in cheapflation due to the pass-through of upstream costs lead to fluctuations in inflation inequality if income groups purchase differently priced varieties. Table 3 panel A shows that high-income households indeed buy more expensive varieties across the food-at-home basket: varieties purchased by the highest income quintile have 8.2 percent higher unit prices than those purchased by the lowest income quintile on average.

We test for whether these differences in the prices of varieties purchased lead to differences in inflation sensitivity across income groups using the specification,

$$\pi_{mt}^{g} = \sum_{q=1}^{5} \beta_q \left(\pi_{mt} \times 1\{g = q\}\right) + \alpha_g + \phi_{mt} + \varepsilon_{gmt}, \tag{10}$$

where π_{mt}^{g} is the year-over-year inflation rate for product module m for income quintile g from quarter $t-4$ to quarter t as defined in Section 2.2, π_{mt} is the overall inflation rate for product module m in the Retail Scanner data over the same window, and α_g and ϕ_{mt} are income quintile and module-quarter fixed effects.

Table 3 panel B shows that the sensitivity of inflation rates experienced by each income quintile to overall product category inflation rates declines with income. When the average inflation rate for a product category is 10 percent, inflation for the highest income quintile is 0.5pp less than inflation for the lowest income quintile on average.

These differences in inflation sensitivity lead to meaningful fluctuations in the level of food-at-home inflation inequality over time. Figure 6 plots the difference in food-at-home inflation rates experienced by low- and high-income households in our data from 2006 to 2023. The gap is positive on average, but also varies with the overall rate of food-at-home

Table 3: From cheapflation to inflation inequality in food at home.

Panel A. Prices paid		*Panel B. Inflation sensitivity*	
	Log unit price (1)		*Module inflation for income group* (2)
Income quintile 2	0.012**	Avg. module inflation × Income quintile 2	0.000
	(0.003)		(0.003)
Income quintile 3	0.027**	Avg. module inflation × Income quintile 3	−0.008*
	(0.004)		(0.004)
Income quintile 4	0.049**	Avg. module inflation × Income quintile 4	−0.025**
	(0.006)		(0.006)
Income quintile 5	0.082**	Avg. module inflation × Income quintile 5	−0.052**
	(0.008)		(0.010)
Module-Time FEs	Yes	Module-Time FEs	Yes
N	213971	N	201667
R^2	1.00	R^2	0.99
Within R^2	0.19	Within R^2	0.08

Note: Panel A reports results from a regression of log unit prices on income quintile dummies and product module-time fixed effects. Panel B reports results from specification (10) (for ease of display, Panel B does not display the group fixed effects α_g). Avg. module inflation rates are winsorized at the 1 percent level. Regressions weighted by expenditures. Standard errors two-way clustered by year and product module.

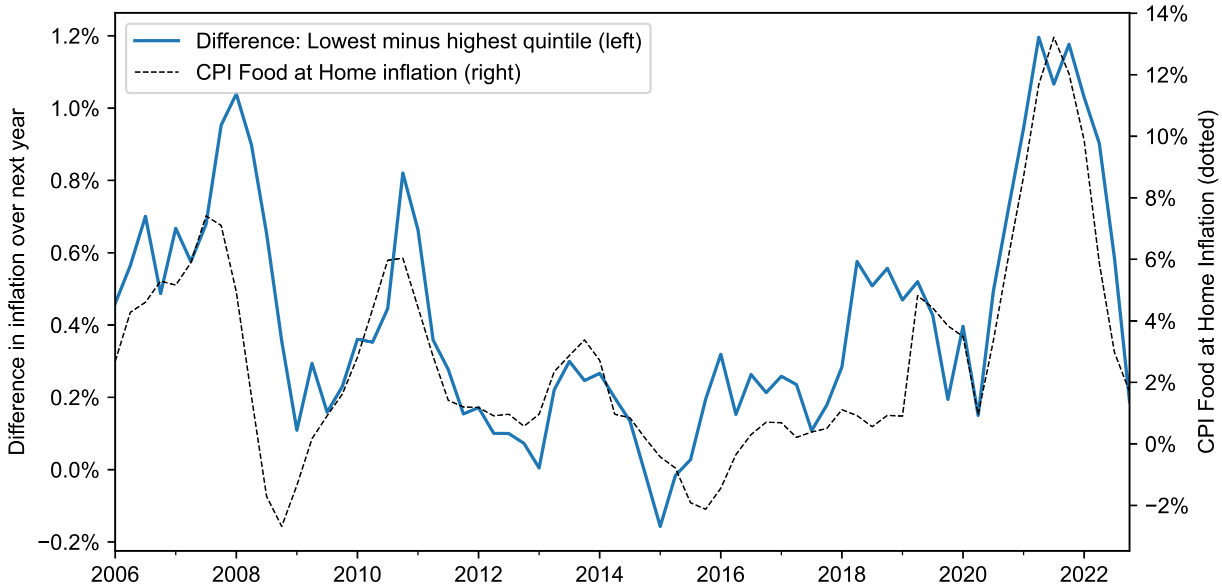
inflation, widening when food-at-home inflation is high in 2008, 2011, and 2021, and narrowing when food-at-home inflation is low in 2009 and 2015. Falling food prices can even lead the gap to become negative, such as in 2015.

How much variation can pass-through in levels explain? We construct a measure of the predicted gap in inflation rates between the first and fifth income quintiles from quarter t to quarter $t+4$ using

$$\text{PredictedInflationGap}_{t+4} = \underbrace{\sum_m \left(\omega^1_{mt} - \omega^5_{mt}\right)\pi^1_{mt+4}}_{\substack{\text{Due to differences} \\ \text{in spending shares}}} + \underbrace{\sum_m \omega^5_{mt}\widehat{\Delta\pi_{mt+4}}}_{\substack{\text{Due to pass-} \\ \text{through in levels}}}, \tag{11}$$

where ω^1_{mt} and ω^5_{mt} are the first and fifth income quintiles' expenditure shares on product module m in quarter t, π^1_{mt+4} is the inflation rate for module m from quarter t to quarter $t+4$ for the first income quintile, and the predicted gap in inflation rates within each product

Figure 6: Gap in food-at-home inflation rates between lowest and highest income quintile.



Note: The solid line plots the difference in food-at-home inflation rates over the next year for the first and fifth income quintiles, constructed using the procedure described in Section 2.2. The dotted line plots inflation in the BLS's consumer price index for food at home.

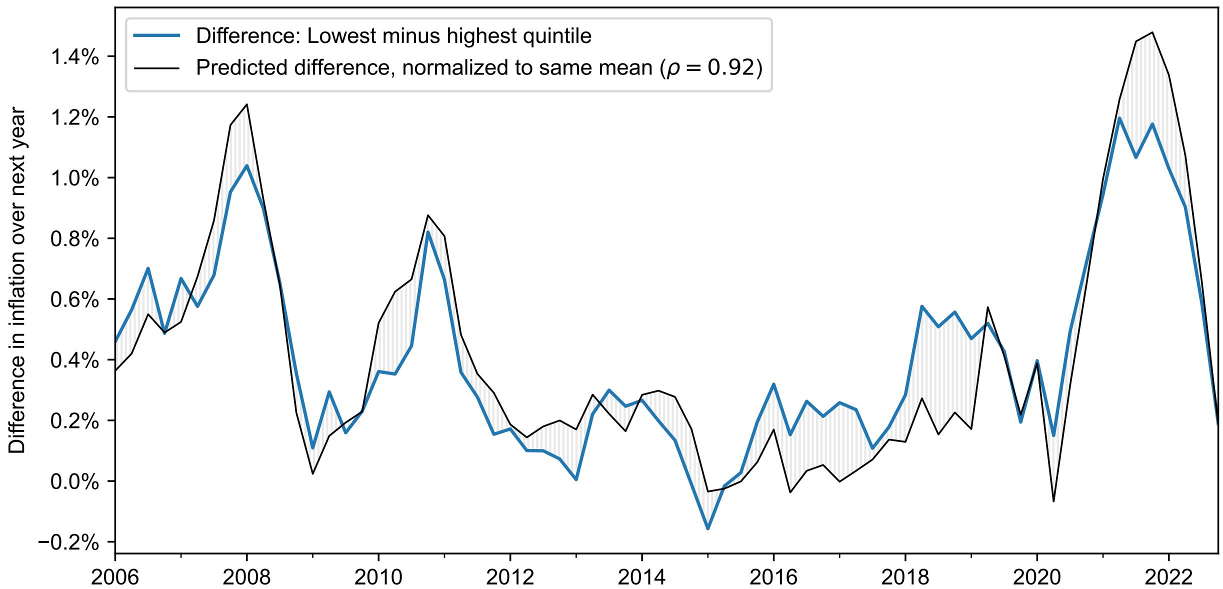
module m, $\widehat{\Delta\pi_{mt+4}}$, is

$$\widehat{\Delta\pi_{mt+4}} \equiv \left(p_{mt}/p^1_{mt} - p_{mt}/p^5_{mt}\right)\left(\pi^{\text{all}}_{mt+4} - \pi^{\text{w}}_{t+4}\right).$$

This expression for the within-module gap in inflation rates across income quintiles uses the expression derived in (7). Note that our predictions for the inflation gap within each module, $\widehat{\Delta\pi_{mt+4}}$, and the overall inflation gap, $\text{PredictedInflationGap}_{t+4}$, use only information about differences in income groups' spending shares and unit prices in quarter t—alongside the inflation rates π^{all}_{mt+4}, π^1_{mt+4}, and π^{w}_{t+4}—to predict gaps in inflation rates over the next year. Figure 7 plots the predicted inflation gap from (11) alongside the actual inflation gap over the sample; the correlation between the two is 0.92.

Table 4 quantifies how much of the variation in inflation inequality over time can be accounted for by these predictions. Over the full sample from 2006 to 2023, the variance of the gap in food-at-home inflation rates between the bottom and top quintiles is 0.100 squared percentage points. The variance of the inflation gap after accounting for predicted differences in within-module inflation rates—i.e., the second term in (11)—is 0.030. In other words, 70 percent of the variance in inflation inequality over time can be explained with within-module inflation differences predicted by pass-through in levels.

Figure 7: Predicted gap in food-at-home inflation rates between lowest and highest income quintile, alongside observed inflation gap.



Note: The figure plots the gap in food-at-home inflation rates for the lowest and highest income quintile as shown in Figure 6 against the predicted inflation gap from Equation (11). The predicted inflation gap is shifted upward to account for secular differences in inflation over time (i.e., it is shifted upward by 0.258% so that both the predicted and observed inflation gaps have the same mean over the sample).

Once we further account for differences in expenditure shares across product modules, the residual gap in inflation rates across the first and fifth income quintiles has a variance of 0.027 squared percentage points. That is, the predicted inflation gap in (11) accounts for 73 percent of the variance in inflation inequality over the sample.[17]

The predicted inflation gap can also account for heightened inflation inequality during specific periods, such as the Great Recession and the post-pandemic inflation. Argente and Lee (2021) and Becker (2026) document larger gaps in inflation rates across income groups during the Great Recession. The second column of Table 4 shows that inflation inequality from 2008–2011 is indeed 0.122 percentage points higher than the sample average. However, the gap disappears (and in fact becomes negative) after accounting for pass-through in levels and differences in expenditure shares across income groups. Indeed, Figure 7 suggests that the elevated inflation inequality from 2008–2011 was largely the result of two surges in food commodity prices in 2008 and 2011. Equation (11) like-

[17]As we found for cheapflation, other time-period specific factors, such as business cycle conditions, have limited explanatory power for fluctuations in inflation inequality after accounting for pass-through in levels. A regression of the inflation gap on the predicted gap from (11) has an R^2 of 0.84. Further including the unemployment gap and the GDP gap as explanatory variables marginally increases the R^2 to 0.85.

Table 4: Explaining variation in inflation inequality over time.

	Variance of inflation gap (pp^2)	*Excess inflation inequality relative to full sample average* (percentage points)	
	All Years	2008–2011	2021–2023
Actual	0.100	0.122	0.323
After accounting for pass-through in levels	0.030	−0.016	0.088
After accounting for pass-through in levels and module expenditure shares	0.027	−0.060	−0.072

Note: The inflation gap is the difference between food-at-home inflation rates for the first and fifth income quintile, as plotted in Figure 6. The "Excess inflation inequality" columns report the average inflation gap over the prior year for all years in the reported range, inclusive of the start and end year. "Accounting for pass-through in levels" reports statistics after subtracting the second term in (11), and "Accounting for pass-through in levels and module expenditure shares" reports statistics after subtracting the predicted inflation gap from (11).

wise predicts the entirety of the surge in inflation inequality during the post-pandemic inflation from 2021–2023, as shown in the third column of Table 4. If anything, inflation inequality was slightly lower than expected, echoing our results on lower-than-expected cheapflation post-2020 in Table 2.

4.3 Aggregation Bias in BLS Statistics

Consumer price inflation statistics published by the Bureau of Labor Statistics are built up from inflation measured across 273 product categories referred to as entry-level items (ELIs). While Kaplan and Schulhofer-Wohl (2017) and Jaravel (2019) document that aggregation to these categories understates differences in trend inflation across income groups, our analysis suggests that aggregation also masks differences in inflation fluctuations across income groups.

To evaluate how this aggregation affects the measurement of cyclical fluctuations in inflation inequality, we compare our food-at-home price indices constructed with scanner data to food-at-home inflation indices created with publicly available BLS data. For the latter, we build on the distributional consumer price indices (D-CPIs) developed by Jaravel (2024). Jaravel (2024) creates these distributional consumer price indices by matching BLS price index data to income-group expenditure shares from the Consumer Expenditure Survey at the finest level of disaggregation available in public data. We adapt his code to produce distributional consumer price indices for food-at-home products across income quintiles, which we can compare on equal footing to the food-at-home inflation indices that we construct in the scanner data.

Pass-through of upstream costs and inflation volatility over 2006–2020. The left panel of Figure 8 plots differences in the long-run, log pass-through of the Food Manufacturing PPI to food-at-home prices across income groups, which we estimate using specification (5). The pass-through of upstream costs to prices is 9.8 percent higher for the lowest-income quintile than the highest income quintile. This difference is primarily due to differences in the varieties that households buy within narrow product categories: Figure 8 shows that the pass-through of the Food Manufacturing PPI to food-at-home indices constructed with ELI-aggregated price data differs across income groups by only 1.7 percent. In other words, aggregation to ELIs masks 83 percent of the disproportionate sensitivity of food-at-home inflation rates for low-income households to upstream costs.[18]

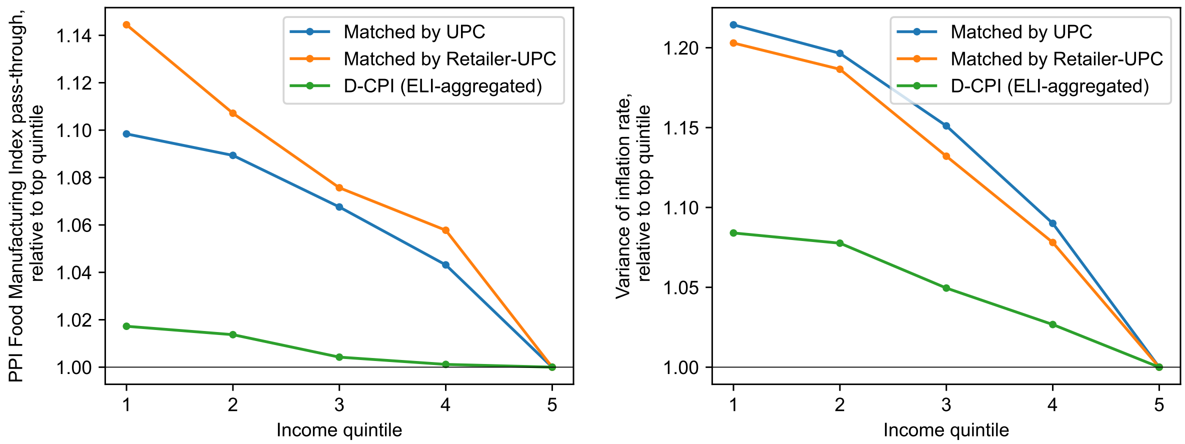
Similar results obtain for the pass-through of the Farm Products PPI to income groups' food-at-home prices (not pictured). The pass-through of Farm Products PPI to prices for the lowest-income quintile is 5.8 percent higher than for the highest-income quintile. Analogous calculations with ELI-aggregated data yield a differential of only 2.8 percent.

These differences in the pass-through of upstream costs to prices across income groups translate into differences in inflation volatility. The right panel of Figure 8 shows that the variance of year-over-year food-at-home inflation rates for the lowest-income quintile is 21.4 percent higher than that of the highest income quintile, a difference that can largely be accounted for by the differences in pass-through of upstream costs across quintiles. In contrast, food-at-home inflation rates constructed using ELI-aggregated price data are only 8.4 percent more volatile for the lowest income quintile than the highest income quintile. Appendix Figure A8 shows that these results on elevated pass-through of upstream costs and inflation volatility for low-income households are robust to instead splitting households into income deciles.

Price growth over 2021–2023. The different sensitivity of inflation to upstream costs across income groups implies that periods when the cost of living rises are likely to be accompanied by especially severe inflation for low-income households, and that this disproportionate burden of inflation is invisible in income-group price indices constructed from publicly available inflation data. The period from 2021–2023, which saw a sharp rise

[18]Disaggregating household purchases further by retail chain and UPC increases the difference in pass-through across income groups to 14.2 percent. However, this difference may be due in part to the lower share of panelists' spending that we can match to inflation data when matching at the retailer-UPC level.

Figure 8: Food-at-home inflation for lower income quintiles is more sensitive to upstream costs and more volatile.



(a) Pass-through of Food Manufacturing PPI.

(b) Variance of inflation rates.

Note: Panel (a) reports the long-run log pass-through of the Food Manufacturing producer price index to the food-at-home price index for each income group, estimated using (5) for the period 2006–2020. Panel (b) reports the variance of year-over-year inflation rates for each income group for 2006–2020. In both panels, estimates for each income group are normalized by the estimate for the top income quintile. The "matched by UPC" and "matched by retailer-UPC" estimates use food-at-home price indices constructed with the Retail Scanner data, disaggregating household purchases and product-level inflation at the UPC and retailer-UPC level respectively. The "D-CPI" estimates use food-at-home distributional price indices constructed from BLS price data by Jaravel (2024).

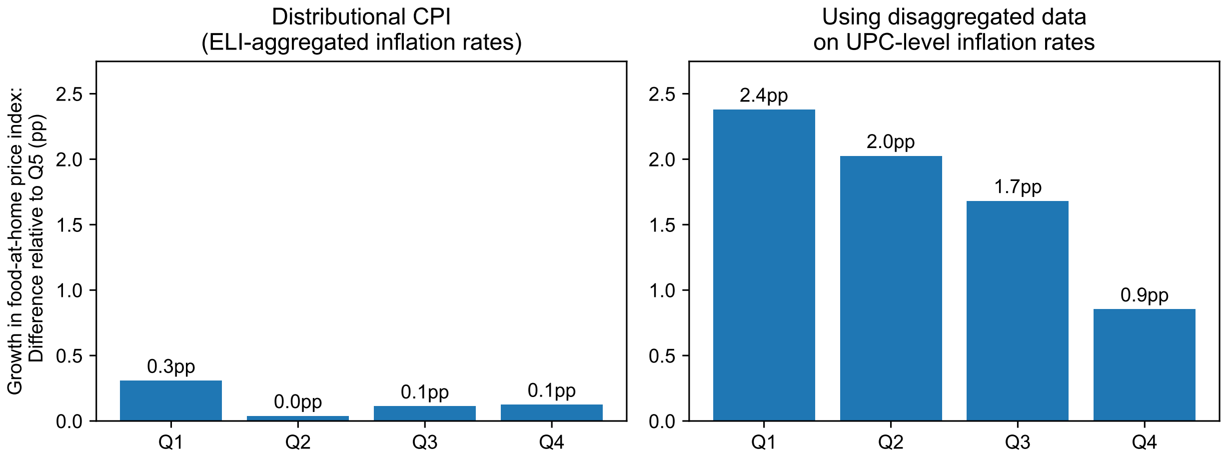
in input costs, provides an illustration. Figure 9 shows that price indices constructed with ELI-aggregated data show small differences in inflation across income groups during this period, with prices for the bottom quintile increasing 0.3pp more than the top quintile.

Differences in price growth measured using the disaggregated data are an order of magnitude larger, with low-income households experiencing 2.4pp higher growth in food-at-home prices than high-income households. Relative to the overall growth in the food-at-home consumer price index over this period, this difference represents a 10 percent larger increase in prices for low-income households than high-income households.

The dramatic differences in the extent of inflation inequality that we find using disaggregated data compared to ELI-aggregated data may partly explain why studies have come to contrasting conclusions about the level of inflation inequality over the post-pandemic period.[19] Differing results may also owe to differences in the breadth of con-

[19]For example, "Inflation Usually Hits Harder for Poor Families. For a Couple of Years, It Didn't" (*Wall Street Journal*) uses ELI-aggregated data from Jaravel (2024) to report that inflation was broadly similar for the bottom and top income deciles from May 2020 to 2022, in contrast with other studies that document

Figure 9: Differences in food-at-home price growth by income quintile, 2020Q4 to 2023Q4.



Note: Each subfigure plots the difference between the growth of the food-at-home price index for the income quintile from 2020Q4 to 2023Q4 and the growth of the food-at-home price index for the highest income quintile over the same period. The left panel uses food-at-home distributional price indices constructed from BLS price data by Jaravel (2024), and the right panel uses food-at-home price indices constructed with the Retail Scanner data.

sumption categories covered and underlying data sources and methods.[20]

Note that the larger inflation gap that we document in disaggregated data from 2021–2023 reflects both *secular* differences in trend inflation that are masked by aggregation (as emphasized by Jaravel 2019) and the *excess* inflation inequality caused by rising input prices and pass-through in levels. In Figure 7, we estimated that a secular inflation gap of 0.26pp per year is needed to reconcile the level of inflation inequality in the data with that predicted by pass-through in levels. Applying this estimate to 2021–2023 suggests that 68 percent of the 2.4pp inflation gap during this period can be attributed to our mechanism operating through rising input costs, with the remaining 32 percent due to secular inflation differences across households.[21] In other words, ELI-aggregated data misses secular differences in inflation inequality, but the aggregation bias is as much as three times as large from 2021–2023 due to rising input costs and pass-through in levels.

Table 5 summarizes the aggregation bias in income-group price indices built from

elevated inflation inequality during this period.

[20] For example, studies that control for household size when constructing income groups tend to find a greater degree of inflation inequality.

[21] Appendix Figure A9 provides an alternative estimate based on regressing the level of inflation inequality on food-at-home inflation from 2006–2023, yielding an estimate of 0.36pp per year. Applying this estimate instead attributes 55 percent of the inflation gap from 2021–2023 to our mechanism.

Table 5: How much do aggregated statistics miss?

Measure	*Difference between lowest and highest income quintile* Disaggregated UPC-level data	BLS-based data	Understatement in BLS-based data
Pass-through of Food Manufacturing PPI to food-at-home inflation, 2006–2020	9.8%	1.7%	83%
Pass-through of Farm Products PPI to food-at-home inflation, 2006–2020	5.8%	2.8%	52%
Variance of food-at-home inflation rates, 2006–2020	21.4%	8.4%	61%
Growth in food-at-home price index, 2020Q4–2023Q4	2.4pp	0.3pp	87%

Note: "UPC-level data" refers to food-at-home price indices we construct with Retail Scanner data, and "BLS-based data" refer to food-at-home price indices constructed from BLS data following Jaravel (2024).

BLS data across several measures, including the pass-through of upstream costs, inflation volatility, and price growth from 2021–2023. In each case, the ELI-aggregated data substantially understates differences in inflation experienced across the income distribution.

4.4 Accounting for Substitution and Nonhomotheticity

Our baseline measures of inflation across income groups use Laspeyres price indices, which weight product-level inflation rates by households' initial expenditure shares. As noted by Boskin, Dulberger, Gordon, Griliches, and Jorgenson (1998), Laspeyres measures of inflation can overstate the true increase in the cost of living because they do not account for substitution across varieties and across product categories in response to relative price changes. They also do not account for how nonhomotheticities—i.e., changes to households' preferred consumption basket in response to real income changes—affect the change in household utility that results from price changes.

We explore how accounting for substitution and nonhomothetic preferences affects our results on the differences in inflation experienced by households across the income distribution. To account for the effect of substitution on cost of living changes, we construct the inflation rate on a Törnqvist price index for each income group, given by

$$\pi_t^{g,\text{Törnqvist}} = \prod_i (1 + \pi_{it})^{\frac{1}{2}(\omega_{it-4}^g + \omega_{it}^g)} - 1, \tag{12}$$

Table 6: Price indices accounting for substitution and nonhomotheticity.

Income quintile	*Panel A. Growth in food-at-home price index from 2020Q4–2023Q4 (pp)* Homothetic Laspeyres (1)	Homothetic Törnqvist (2)	Nonhomothetic (Base=2020Q4) (3)	*Panel B. Variance of food-at-home inflation rates from 2006–2020 (pp^2)* Homothetic Laspeyres (4)	Homothetic Törnqvist (5)	Nonhomothetic (Base=2006Q1) (6)
1	27.27	25.43	27.01	4.69	4.67	4.89
2	26.92	24.90	26.41	4.62	4.59	4.92
3	26.57	24.52	25.99	4.45	4.43	4.78
4	25.74	23.87	25.27	4.21	4.23	4.55
5	24.89	23.13	24.45	3.86	3.90	4.15
Q1/Q5	1.10	1.10	1.10	1.21	1.20	1.18

Note: Columns 1 and 4 are our baseline measures of inflation using Laspeyres price indices. Columns 2 and 5 account for substitution using the Törnqvist inflation measure from (12). Columns 3 and 6 use the second-order nonhomothetic adjustment from Jaravel and Lashkari (2024). For both panels, the base period is chosen to be the first quarter of the sample and is listed in the column heading.

where π_{it} is the year-over-year inflation rate for UPC i from quarter $t-4$ to quarter t, and ω_{it}^{g} is the expenditure share of income group g on UPC i in quarter t. This Törnqvist measure incorporates the effects of substitution on cost of living, because it weights product-level inflation rates using an average of initial and final expenditure shares. As noted by Diewert (1976), the Törnqvist price index is a superlative index, providing a second-order approximation to the true cost-of-living index for any twice differentiable, homothetic expenditure function.

To further account for nonhomotheticities, we use the second-order algorithm developed by Jaravel and Lashkari (2024) to estimate changes in real consumption and price indices for each income group in each period. This algorithm uses cross-sectional variation in inflation across households with different real consumption levels to estimate the elasticity of real consumption to total expenditure.[22] Computing these nonhomothetic price indices requires choosing a base period that subsequent changes in real consumption are measured against. For our baseline results, we choose the first quarter of the sample as the base period. Appendix Table A7 shows that results are similar if we instead measure changes in real consumption relative to the final quarter of the sample.

[22]In particular, we use the algorithm from Jaravel and Lashkari (2024) Appendix A.2.1, which estimates changes in real consumption growth to a second order. We use a fourth-order polynomial (i.e., $K_N = 4$) to fit the relationship between inflation and real consumption in each period. For expenditures of households in each income quintile, we use average income by quintile by year from Census TableA-4b.

Table 6 shows how inflation experiences across income groups change under these alternative price indices, as compared to our baseline Laspeyres measures. Panel A shows the growth in the food-at-home price index across income quintiles over 2021–2023. For the lowest (highest) income quintile, accounting for substitution changes the growth in the food-at-home price index by −1.84pp (−1.76pp), and further accounting for nonhomotheticities increases the food-at-home price index by +1.58pp (+1.32pp). The two forces roughly net out, and under each measure the inflation experienced by the lowest-income quintile is about 10 percent higher than the highest quintile.

Why do nonhomotheticities increase the estimated rise in the cost of living? During 2021–2023, food-at-home prices grew faster than income. Declining real consumption leads households to trade down to lower-priced varieties (Jaimovich et al. 2019; Argente and Lee 2021), which are exactly those products that experienced the highest inflation rates from pass-through in levels. Since income effects tilted households' consumption baskets toward products whose prices were rising the fastest, they amplified the rise in households' costs of living.

This effect of nonhomotheticities on cost of living extends to other cycles in food-at-home inflation rates over the sample. Panel B of Table 6 shows that the volatility of inflation rates experienced by each income quintile from 2006–2020 rises by 5–7 percent after accounting for substitution and nonhomothetic preferences. Pass-through in levels leads to higher inflation rates for precisely those products that households trade down toward in periods where real food-at-home consumption is falling, and less deflation for products that households trade up toward in periods where real food-at-home consumption is rising. While accounting for substitution and nonhomotheticities alters the level of inflation volatility, it does not change the conclusion that low-income households face greater inflation volatility than high-income households.

5 Beyond Food at Home

The analyses so far suggest that aggregation to product categories masks systematic cycles in cheapflation and inflation inequality in food at home. While detailed scanner and panelist data facilitate these analyses, the lack of comparable data for other categories of consumption makes it difficult to assess whether such patterns generalize to other parts of the consumption bundle.

In this section, we provide suggestive evidence of fluctuations in cheapflation and inflation inequality in other consumption categories. First, using price quote data underlying the U.K. CPI and data on inflation across U.S. cities from the BLS's metropolitan area price indices, we find evidence of heightened inflation sensitivity for both lower-priced varieties and lower-income cities in categories beyond food at home. Second, we use microdata on household vehicle purchases and prices to document similar fluctuations in cheapflation and inflation inequality for automobiles. These analyses suggest that our conclusions on fluctuations in inflation inequality within narrow product categories may apply beyond food at home, though we caution that the lack of finely disaggregated data prevents us from conclusively attributing these patterns to pass-through in levels alone.

5.1 Evidence on Cheapflation from U.K. CPI Microdata

We use microdata underlying the United Kingdom's consumer price indices, made available by the U.K. Office for National Statistics (ONS).[23] These data include price quotes that the ONS collects from individual stores; the ONS supplements these data with scanner data to produce the U.K.'s consumer price indices. Price quotes in the data are categorized into about 1,200 items (e.g., "Basmati Rice, 0.5–1kg") spanning nondurables, semi-durables, durables, and services. In the description that follows, we refer to an observation of an item at a specific shop in a specific region as a variety. Price quotes for each variety are collected at a monthly frequency. Appendix Table A8 reports the number of unique items, varieties, and price observations by broad category in our sample.

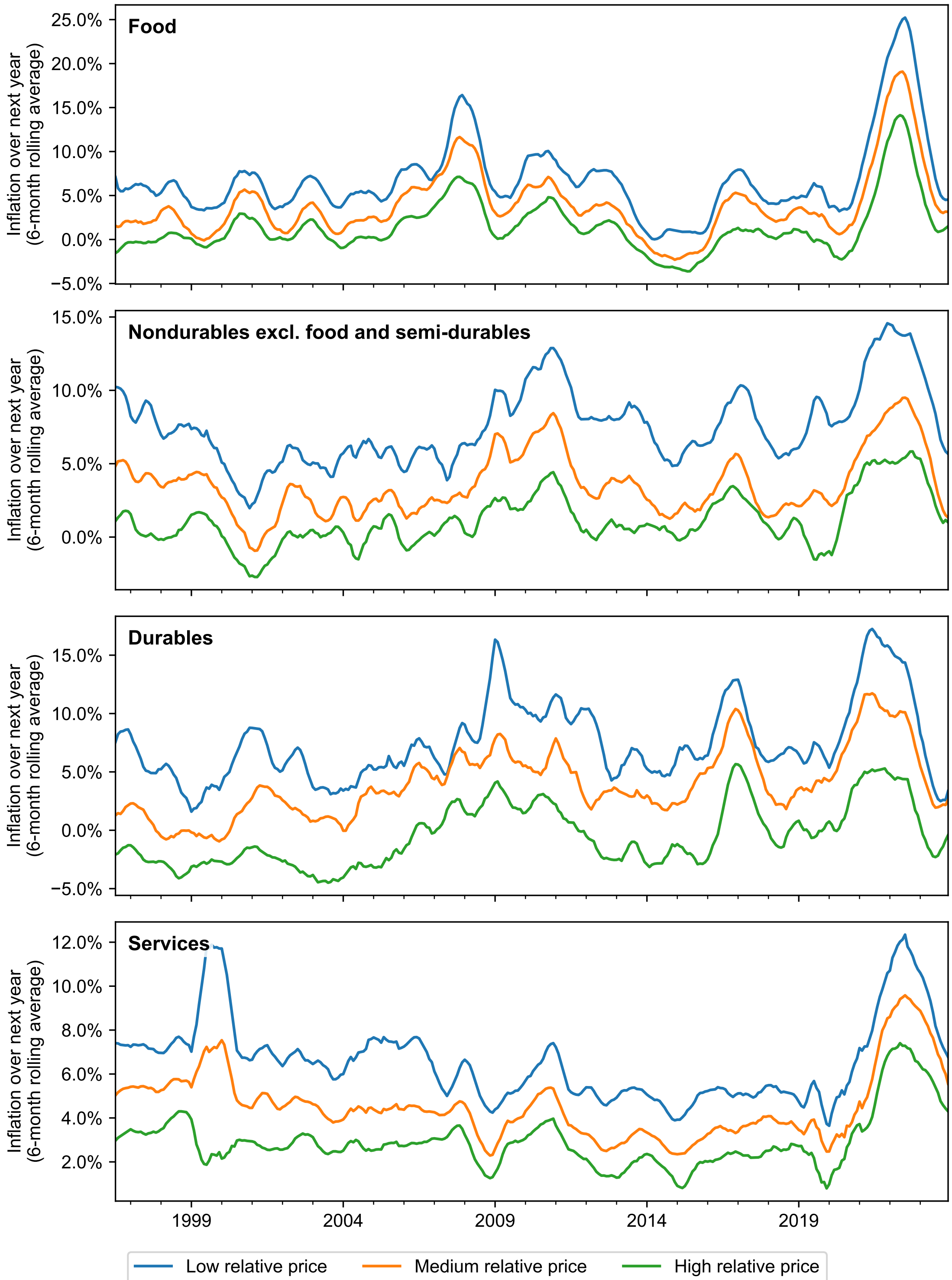
To explore whether inflation for low-priced varieties in the U.K. microdata is more sensitive to the overall inflation rate, we follow a similar procedure to Figures 1 and 4: we split varieties within each item into groups using their relative prices over the prior year and plot the average inflation rate for varieties in each group over the subsequent year. As we saw for U.S. food-at-home products in the NielsenIQ data, the top panel of Figure 10 shows that the gap in inflation between low- and high-priced food varieties tends to widen when overall food inflation is high—such as in 2008 and 2021—and narrows when food inflation is low, such as in 2015.[24] The remaining panels of Figure 10 show similar patterns for other categories beyond food: other nondurable and semi-durable products, durables, and services. In each of these categories, when the overall category inflation rate is high, the gap in inflation between lower- and higher-priced varieties tends to widen, and vice versa.

The idea that cheapflation cycles emerge even in labor-intensive categories like ser-

[23]We build on code by Gado and Schoenle (2026) to clean the publicly available microdata.

[24]These patterns are somewhat less pronounced than in the NielsenIQ data shown in Figure 4. While this may reflect genuine differences between the U.S. and U.K., it likely also owes to the lack of package size information in the U.K. microdata. Because variety prices are a noisy proxy for unit prices (they may also reflect differences in package size), relative prices are measured with error, attenuating measured inflation differences between cheap and expensive varieties.

Figure 10: Cheapflation cycles across goods and services in U.K. CPI microdata.



Note: Varieties within each item are grouped into terciles of relative price. We winsorize year-over-year inflation rates for each variety at the 2.5 percent level. We construct item-level inflation rates for each relative price group by aggregating individual variety inflation rates with stratum weights, and construct aggregate inflation rates by aggregating item-level inflation rates with item weights. The figures show 6-month rolling averages of aggregate inflation rates for each relative price group in each category.

vices may seem surprising, because our pass-through-in-levels mechanism requires that production uses some inputs whose prices do not co-move exactly with labor. Yet, many services do in fact use goods in production: for example, food away from home and accommodation services use food inputs, and repair services use industrial parts. In Appendix Figure A10, we use the BEA input–output tables to estimate the share of production costs spent on goods, services, and labor compensation for subcomponents of U.S. personal consumption expenditures. Even producers of services spend a meaningful share of production expenses (often 10–25 percent) on goods. Fluctuations in the price of these inputs relative to labor can lead to diverging inflation rates for cheaper versus more expensive varieties.

We formally test whether low-priced varieties of an item exhibit more inflation sensitivity using the specification,

$$\begin{aligned}\text{Inflation}_{kit+12} = \beta\left(\text{AvgItemInflation}_{it+12} \times \text{RelativePrice}_{kit}\right) \\ + \gamma\text{RelativePrice}_{kit} + \delta_{it} + \varepsilon_{kit}, \quad (13)\end{aligned}$$

where $\text{Inflation}_{kit+12}$ is the inflation of variety k in item i from month t to month $t+12$, $\text{AvgItemInflation}_{it+12}$ is the average inflation rate across all varieties in item i over the same period, and $\text{RelativePrice}_{kit}$ is the log-deviation of the price of variety k relative to the average price of all varieties in item i in the same month, averaged across the prior 12 months. The coefficient of interest is β, which estimates whether inflation rates for relatively cheap products are more sensitive to subsequent inflation than expensive products. We include a control for variety k's relative price to absorb secular differences in inflation rates across cheap and expensive varieties and item–time fixed effects δ_{it} to absorb variation in average inflation across varieties of an item in each month.

Column 1 of Table 7 reports an estimate of $\beta \approx -0.31$ for the full sample. In words, when prices for an item increase by 10 percent on average, a variety that is 10 percent cheaper than the average tends to exhibit a 0.31pp higher inflation rate than the average. To test whether these patterns are specific to food at home or extend to other categories, column 2 estimates a variant of specification (13) that allows for separate estimates of β and γ for food products, nondurables excluding food, semi-durables, durables, and services. The magnitude of the relationship between relative price and inflation sensitivity is greatest for food products, but we nevertheless find evidence that low-priced varieties are more sensitive to item inflation across all categories.

The presence of cheapflation cycles for labor-intensive categories like services raises the concern that some other mechanism, besides the pass-through of common input costs,

Table 7: Cheapflation cycles beyond food at home: Evidence from U.K. CPI microdata.

	Inflation$_{kit+12}$		
	(1)	(2)	(3)
Relative Price$_{kit}$ × Avg. Item Inflation$_{it+12}$	−0.307**		
	(0.046)		
Relative Price$_{kit}$ × Avg. Item Inflation$_{it+12}$× Food		−0.552**	−0.548**
		(0.047)	(0.056)
Relative Price$_{kit}$ × Avg. Item Inflation$_{it+12}$× Other nondurables		−0.259**	−0.251**
		(0.115)	(0.118)
Relative Price$_{kit}$ × Avg. Item Inflation$_{it+12}$× Semi-durables		−0.231**	−0.229**
		(0.038)	(0.047)
Relative Price$_{kit}$ × Avg. Item Inflation$_{it+12}$× Durables		−0.191**	−0.194**
		(0.077)	(0.071)
Relative Price$_{kit}$ × Avg. Item Inflation$_{it+12}$× Services		−0.177**	−0.147**
		(0.074)	(0.070)
Item-Time FEs	Yes	Yes	Yes
Relative Price$_{kit}$× Time FEs			Yes
N (millions)	21.6	21.6	21.6
R^2	0.12	0.12	0.12

Note: The table reports the results from specification (13). We winsorize both Inflation$_{kit+12}$ and RelativePrice$_{kit}$ at the 2.5 percent level and construct Avg. Item Inflation$_{it+12}$ using winsorized variety-level inflation rates aggregated with stratum weights. Standard errors two-way clustered by COICOP4 and year. * indicates significance at 10%, ** at 5%.

may explain these results. Many other proposed explanations for cheapflation emphasize time-specific forces, such as changes in household price sensitivity or trading down over the business cycle (e.g., Li 2019; Orchard 2025; Becker 2026). Such forces would shift relative inflation rates for low- versus high-priced varieties across all items in the economy, while our mechanism predicts that cheapflation is tied to each item's own inflation rate due to pass-through of item-specific cost shocks. We can thus discriminate between these competing mechanisms by including interactions between varieties' relative prices and time fixed effects, which absorb any time-varying economy-wide tilts in inflation across cheap and expensive varieties. Column 3 estimates (13) augmented with these additional controls. The β coefficients for each group are largely unchanged, suggesting that sensitivity to item-level inflation, rather than economy-wide fluctuations in the extent of cheapflation, is largely responsible for variety-level inflation differences.

Robustness. Appendix Table A9 reports the results of several robustness exercises. As with the NielsenIQ data, a concern is that mean reversion may create a mechanical correlation between initial prices and future price changes. We find similar results if we measure varieties' relative prices using the average price for each variety over the past two or three years. Likewise, one may worry that the excess sensitivity of low-priced varieties to overall item inflation reflects low-priced varieties "catching up" to trend item inflation. Augmenting specification (13) to include interactions of varieties' relative prices with lagged item inflation rates leaves our results largely unchanged. Finally, Appendix Table A9 also reports similar results for a variant of specification (13) with more granular, variety-level fixed effects to absorb all trend inflation differences across varieties.

5.2 Evidence on Inflation Inequality from BLS Area Price Indices

While the BLS aggregates price data for products at the entry-level item (ELI) level, it does so across 32 geographic areas using prices collected in each area. Differences in inflation across geographic areas can thus indicate differences due to the composition of varieties consumed by households across areas with different income levels.

We merge BLS price indices for 27 metropolitan areas from January 1997 to April 2025 with annual measures of per-capita income by metropolitan area from the Bureau of Economic Analysis (CAINC1).[25] Figure 11 plots the average inflation in consumer prices excluding shelter across cities in the top and bottom quartile of per-capita income. Inflation in high-income cities indeed appears to be less volatile, for example reaching a peak of 9.5 percent in 2021 compared to 10.4 percent for low-income cities.

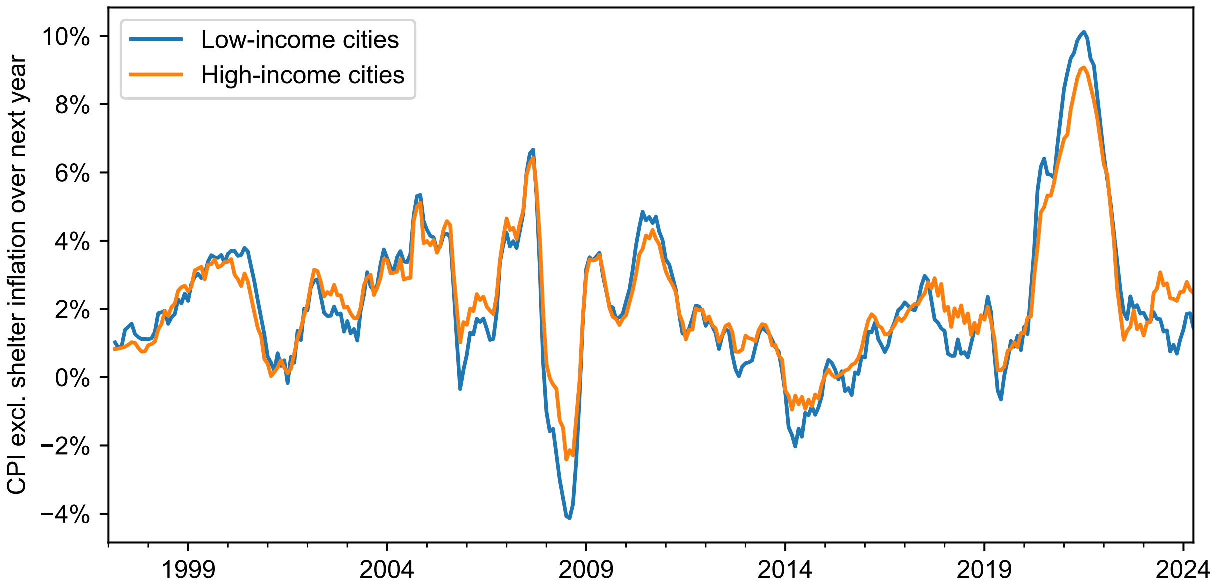
To formally test for differences in inflation sensitivity across metropolitan areas with different income levels, we use the specification,

$$\text{Inflation}_{ct+12} = \beta\left(\text{USInflation}_{t+12} \times \text{LogIncome}_{ct}\right) + \gamma\text{LogIncome}_{ct} + \alpha_c + \delta_t + \varepsilon_{ct}, \quad (14)$$

where Inflation_{ct+12} is the year-over-year inflation rate for metropolitan area c starting in month t, $\text{USInflation}_{t+12}$ is the analogous inflation rate for all urban consumers in a U.S. city average, and LogIncome_{ct} is the log of per-capita income in metropolitan area c in the year of month t. Metropolitan area and time fixed effects α_c and δ_t absorb secular differences in inflation across cities and across time periods. Pass-through in levels predicts that consumer price inflation in higher income areas is less sensitive to national fluctuations in inflation, i.e., $\beta < 0$.

[25]The BLS does not make data available for all 32 geographic areas it uses internally, and the publicly available data for different cities are also available at different frequencies (e.g., monthly, bi-monthly, or semi-annual). We include only cities with either monthly or bi-monthly price series observations. Appendix Table A10 reports the list of metropolitan areas grouped into income quartiles. The list is split for before/after 2018, since the geographic areas used by the BLS were revised in 2018 (see `https://www.bls.gov/cpi/additional-resources/geographic-revision-2018.htm`).

Figure 11: Inflation in consumer prices excluding shelter for low- vs. high-income cities.



Note: High- and low-income cities are cities in the top and bottom quartile of per-capita income. The list of cities changes with the BLS's revision of geographic areas in 2018; see Appendix Table A10 for the list of cities in each quartile. The plotted lines are three-month moving averages of year-over-year inflation in the consumer price index excluding shelter (`SA0L2`), averaged across cities in each group.

Table 8 reports the results from estimating (14) for consumer price inflation overall and for several aggregates that capture various parts of consumer expenditures. Column 1 finds that a 10 percent increase in per-capita income is associated with 9.7 percent lower sensitivity of consumer price inflation in a city to nationwide consumer price inflation. Column 2 excludes shelter, since we think the forces that dictate differences in shelter inflation across cities are likely to differ from our mechanism of pass-through in levels. Narrowing in on inflation in consumer prices excluding shelter, we find that a 10 percent increase in per-capita income is associated with a 4.5 percent lower sensitivity of inflation to nationwide inflation rates.

Similar results obtain for specific categories of consumption. Columns 3, 5, and 6 split the consumer price index excluding shelter into three mutually exclusive categories: food, other goods excluding food, and services excluding shelter. Column 4 also splits out food away from home to explore whether the inflation inequality results we document in food-at-home extend to other food consumption. In each case, we find that higher-income cities exhibit less sensitivity of inflation rates to nationwide inflation.[26]

[26]The coefficient β in (14) combines two forces: the degree to which relative prices of varieties in a city rise with city income, and the degree to which varieties with higher relative prices are less sensitive to nationwide inflation. Thus, a more negative estimate for β can indicate a greater responsiveness of

Table 8: Inflation inequality cycles beyond food at home: Evidence from BLS metropolitan area price indices.

			Inflation in Metropolitan Area			
Series	CPI	CPI Ex. Shelter	All Food	Food Away from Home	Goods Ex. Food	Services Ex. Shelter
BLS Code	`SA0`	`SA0L2`	`SAF1`	`SEFV`	`SACL1`	`SASL2RS`
	(1)	(2)	(3)	(4)	(5)	(6)
U.S. Avg. Inflation × Log Income	−0.970**	−0.452**	−0.258**	−0.589**	−0.227**	−0.271*
	(0.085)	(0.066)	(0.084)	(0.242)	(0.041)	(0.161)
Time FEs	Yes	Yes	Yes	Yes	Yes	Yes
Metropolitan Area FEs	Yes	Yes	Yes	Yes	Yes	Yes
N	3540	3534	3528	3515	3534	3534
R^2	0.85	0.89	0.75	0.42	0.92	0.57

Note: The table reports the results from specification (14). Driscoll–Kraay standard errors with twelve lags in parentheses. * indicates significance at 10%, ** at 5%.

For aggregated consumer price indices, such as overall consumer price inflation or inflation excluding shelter, differences in inflation sensitivity across cities may be due in part to differences in expenditure shares across categories rather than differences in within-category inflation rates. For example, households in higher income cities may spend a lower share of total expenditures on volatile consumption categories such as food and gasoline. To estimate how much of the reduced sensitivity of inflation rates in high-income cities is due to differences in expenditure shares, we combine the BLS's published expenditure weights by metropolitan area with national inflation rates to construct counterfactual city-level inflation rates that would result if all cities faced identical inflation rates within each category and only differed in the composition of expenditures. Appendix Table A11 finds that the difference in the sensitivity of city-level inflation rates to national inflation is half as large when using these counterfactual inflation rates. This exercise suggests that heterogeneity in inflation across cities *within* BLS categories accounts for half of the differences in inflation sensitivity to nationwide inflation across cities.[27]

5.3 Cheapflation and Inflation Inequality in Automobiles

Finally, we use microdata on household vehicle purchases to evaluate whether the same patterns of cheapflation and inflation inequality apply to vehicle purchases. Net outlays

low-priced varieties to nationwide inflation, a steeper increase in relative prices across cities, or both.

[27] We also show in Appendix Table A12 that our findings on different inflation sensitivity across cities are not driven by the 2021–2023 period: if anything, the post-pandemic period saw smaller differences in the sensitivity of city-level inflation rates to nationwide inflation rates across cities with different incomes.

on vehicle purchases represent nearly as large a share of consumption expenditures as food at home, constituting 6.8 percent of consumer expenditures from 2021–2023 compared to 7.8 percent for food at home.

We use data on the prices and characteristics of vehicle makes and models from Ward's Automotive. We combine these data with the average income of a vehicle make's buyers, which we construct from data on household vehicle purchases from the Consumer Expenditure Survey. Appendix E describes the procedures we use to clean the data and reports summary statistics for the merged dataset, which spans the period 2006–2018.

We find evidence of differential inflation sensitivity for both lower-priced car models and for vehicles with lower-income buyers (see Appendix Table E2). Within a vehicle make, the year-over-year growth in manufacturer-suggested retail prices (MSRPs) for a model with a 10 percent lower initial price is 3.9–7.0 percent more sensitive to average price growth. These results are robust to controlling for secular differences in price growth across models and for changes in models' characteristics across years. Across all vehicles, we also find that models with a 10 percent lower-income customer base have inflation rates that are 8.6–9.1 percent more sensitive to overall vehicle inflation. These differences mirror the cycles in cheapflation and inflation inequality we found for food-at-home products.

Aggregation in CPI data masks this heterogeneity in inflation rates across vehicle makes and models. The BLS constructs two price indices for new cars and trucks (ELI code TA011) and used cars and trucks (ELI code TA021), but does not further disaggregate inflation across vehicle makes and models. Thus, as in food at home, income-group price indices constructed from BLS data are likely to miss quantitatively important differences in inflation rates faced by households across the income distribution due to differences in households' purchasing patterns.

6 Unequal Incidence of Cost Shocks in Other Contexts

Our analysis thus far has focused on differences in the incidence of cost shocks across income groups. However, pass-through in levels also implies systematic cross-sectional heterogeneity in inflation across other units—such as cities and countries—in response to common cost shocks.

Inflation across cities. Appendix F.1 shows that pass-through in levels leads to spatial variation in inflation in response to common cost shocks. Cities that consume lower-priced varieties—either because of lower distribution costs or differences in consumer preferences—are more exposed to national cost shocks, resulting in higher local inflation when adverse shocks occur. We document this mechanism using coffee products as a case study, showing that cities with lower-priced coffee varieties experience higher coffee inflation when commodity prices rise.

Import price inflation across countries. Appendix F.2 extends this logic to import price inflation across countries. Pass-through in levels implies that countries that import lower-priced varieties will experience larger increases in import price inflation following a global cost shock. Using trade data on coffee imports, we show that this prediction holds in the data: countries with lower average unit values for imported coffee have greater percentage increases in import prices when global coffee prices rise. Thus, differences in the relative price of imported varieties lead to systematic heterogeneity in countries' exposures to global commodity shocks.

7 Conclusion

Pass-through in levels generates systematic differences in inflation across product varieties in response to fluctuations in the relative price of inputs. This generates "cheapflation" when the relative prices of inputs rise and leads to systematic fluctuations in inflation inequality. This mechanism can parsimoniously account for fluctuations in the extent of cheapflation and inflation inequality over time. In particular, we find that pass-through in levels, along with differences in expenditure shares across categories, can explain elevated inflation inequality during the Great Recession and the post-pandemic period without the need for other explanations such as price gouging by retailers or differential changes in households' price elasticities of demand.

Income-group price indices constructed from publicly available inflation statistics underreport the uneven burden of cost-driven inflation across the income distribution. For various measures of inflation volatility and sensitivity to upstream costs, we find that food-at-home price indices based on aggregated data underestimate the disparities across income groups by more than 50 percent. The aggregation bias is particularly severe when the relative price of inputs increases sharply, such as during 2021–2023. For policymakers, our results suggest that times when the cost of living is rising most rapidly are also attended by heightened inflation inequality, and that publicly available statistics may be a poor guide to the extent of inflation inequality during these times. Our analysis of microdata underlying the U.K. consumer price index, price indices across U.S. cities, and data on vehicle purchases suggests that these lessons are likely to extend to other categories of household consumption.

While this paper focuses on the unequal incidence of cost shocks across income groups, our mechanism also implies differences in cities' and countries' exposures to common cost shocks. We show that cities with lower-priced varieties face higher inflation in response to national cost shocks, and countries that import lower-priced varieties are more exposed to higher import price inflation following global cost shocks. Thus, pass-through in levels can be a source of systematic, cross-sectional variation in inflation in response to shocks.

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Online Appendix for *Cheapflation Cycles*

(Not for publication)

Appendix A Additional Tables and Figures

Table A1: Share of quarterly sales matched to inflation rate over the next year.

Unit price decile	UPC	Retailer-UPC
1	92.9	89.6
2	94.0	90.7
3	94.5	91.0
4	95.1	91.5
5	95.4	91.8
6	95.6	91.8
7	95.6	91.7
8	95.5	91.6
9	95.6	91.5
10	94.9	90.6
All	94.9	91.2

Note: The share of sales matched to inflation data excludes 2023, the last year of the sample.

Table A2: Share of panelist expenditures matched to Retail Scanner data by income group.

Income	Matched to UPC		Matched to retailer-UPC	
quintile	Total	With inflation	Total	With inflation
1	62.4	57.4	24.0	20.0
2	61.9	57.3	24.9	21.1
3	62.1	57.6	25.8	22.0
4	62.2	58.3	27.0	23.6
5	60.9	56.3	28.6	24.6

Note: The share of expenditures matched to inflation data excludes 2023, the last year of the sample.

Table A3: Pass-through of benchmark coffee commodity price to coffee variety prices.

	$\Delta Price_{it}$
Δ Benchmark Commodity Price_t	0.989**
	(0.015)
$\text{Differential}_i \times \Delta$ Benchmark Commodity Price_t	0.003
	(0.003)
N	3215
R^2	0.94

Note: The table reports results from the specification,

$$\Delta\text{Price}_{it} = \alpha + \beta\left(\Delta\text{BenchmarkCommodityPrice}_t\right) + \gamma\left(\text{Differential}_i \times \Delta\text{BenchmarkCommodityPrice}_t\right) + \varepsilon_{it},$$

where ΔPrice_{it} is the change in the price of coffee commodity series i from month $t-12$ to month t, $\Delta\text{BenchmarkCommodityPrice}_t$ is the change in the price of the benchmark coffee commodity series (the ICO price of mild Arabicas at the dock in New York) over the same period, and Differential_i is the average differential for series i reported in Figure A3. We estimate this specification using the same set of coffee commodity price series in Figure A3. Driscoll-Kraay standard errors with twelve lags. * indicates significance at 10%, ** at 5%.

Table A4: Robustness: Difference in pass-through across coffee unit price groups.

Unit price quintile	Baseline	Exchange rates IV	Weather shocks IV	Product-quarter fixed effects	Horizon $K = 4$	Horizon $K = 8$
	Panel A. Pass-through in levels					
1	0.13	0.11	0.25	0.22	0.24	−0.38
2	−0.44	−0.44	−0.17	−0.64	−0.40	0.18
3	0.33	0.56	0.24	−0.02	−0.16	0.06
4	1.05	0.91	−0.32	1.43	1.14	1.28
5	−0.52	−0.37	−0.30	−0.36	−0.52	−0.18
	Panel B. Pass-through in logs					
1	2.91**	2.35**	1.71*	2.82**	2.59**	2.27**
2	0.39	0.37	0.14	0.25	0.14	1.05
3	−0.65	−0.37	−0.53	−0.89	−0.89	−0.77
4	−1.35	−1.21	−2.03**	−1.02	−0.76	−0.90
5	−2.51**	−2.43**	−2.06**	−2.29**	−2.23**	−2.32**

Note: Each cell reports $(\rho_i - \rho_{\text{all}})/(\sigma_i^2 + \sigma_{\text{all}}^2)^{1/2}$, where ρ_i is the estimated long-run pass-through of commodity costs to retail prices for unit price group i, ρ_{all} is the estimated pass-through for the entire sample, and σ_i and σ_{all} are the standard errors of the respective pass-through estimates. The column "Exchange rates IV" uses current and lagged Brazil and Colombia exchange rates (FRED series `CCUSMA02BRM618N` and `COLCCUSMA02STM`) to instrument for commodity cost changes. The column "Weather shocks IV" uses current and lagged minimum and maximum temperatures in coffee-growing regions in Brazil (21.55°S, 45.34°W) and Colombia (4.81°N, 75.70°W) to instrument for commodity cost changes. * (**) indicates that the absolute value of the statistic is greater than 1.65 (1.96).

Table A5: Cheapflation within product modules: Robustness using average hourly earnings of production and nonsupervisory employees.

	Inflation of lowest-price decile − highest-price decile					
	(1)	(2)	(3)	(4)	(5)	(6)
Excess module inflation × Price gap	1.183**		1.188**		1.194**	
	(0.027)		(0.025)		(0.019)	
Module inflation × Price gap		1.186**		1.188**		1.194**
		(0.025)		(0.025)		(0.019)
Wage inflation × Price gap		-1.657**		-1.045**		-1.529**
		(0.337)		(0.299)		(0.460)
Post-2020			-0.067**	-0.073**		
			(0.018)	(0.024)		
Price gap	0.006	0.030**	0.014	0.008	0.079**	0.094**
	(0.012)	(0.012)	(0.011)	(0.007)	(0.014)	(0.024)
Time FEs					Yes	Yes
Product Module FEs					Yes	Yes
N	42137	42137	42137	42137	42137	42137
R^2	0.92	0.92	0.92	0.92	0.93	0.93

Note: The table reports results from specifications (8) and (9). We measure wage inflation as the year-over-year log change in the average hourly earnings of production and nonsupervisory employees. All inflation rates are measured over the next year from quarter t to quarter $t + 4$. The price gap in quarter t is measured as defined in (7) and is winsorized at the 1 percent level. Post-2020 is an indicator equal to one if the observation year is strictly greater than 2020. Regressions weighted by sales. Driscoll-Kraay standard errors with four lags. * indicates significance at 10%, ** at 5%.

Table A6: Cheapflation within product modules: Robustness using unit price quintiles.

	Inflation of lowest-price quintile − highest-price quintile					
	(1)	(2)	(3)	(4)	(5)	(6)
Excess module inflation × Price gap	1.174**		1.180**		1.188**	
	(0.028)		(0.025)		(0.019)	
Module inflation × Price gap		1.180**		1.181**		1.188**
		(0.025)		(0.025)		(0.019)
Wage inflation × Price gap		-2.206**		-1.460**		-1.797**
		(0.411)		(0.397)		(0.437)
Post-2020			-0.050**	-0.041**		
			(0.011)	(0.015)		
Price gap	-0.006	0.038**	0.003	0.013	0.045**	0.068**
	(0.010)	(0.017)	(0.007)	(0.012)	(0.008)	(0.020)
Time FEs					Yes	Yes
Product Module FEs					Yes	Yes
N	42405	42405	42405	42405	42405	42405
R^2	0.92	0.93	0.93	0.93	0.94	0.94

Note: The table reports results from specifications (8) and (9). We measure wage inflation as the year-over-year log change in the BLS employment cost index for private industry workers. All inflation rates are measured over the next year from quarter t to quarter $t+4$. The price gap in quarter t is measured as defined in (7) and is winsorized at the 1 percent level. Post-2020 is an indicator equal to one if the observation year is strictly greater than 2020. Regressions weighted by sales. Driscoll-Kraay standard errors with four lags. * indicates significance at 10%, ** at 5%.

Table A7: Nonhomothetic price indices: Robustness to choice of base period.

Income quintile	*Panel A. Growth in food-at-home price index from 2020Q4–2023Q4 (pp)* Baseline (1)	Nonhomothetic (b=2020Q4) (2)	Nonhomothetic (b=2023Q4) (3)	*Panel B. Variance of food-at-home inflation rates from 2006–2020 (pp^2)* Baseline (4)	Nonhomothetic (b=2006Q1) (5)	Nonhomothetic (b=2020Q4) (6)
1	27.27	27.01	27.02	4.69	4.89	5.04
2	26.92	26.41	26.35	4.62	4.92	4.84
3	26.57	25.99	25.91	4.45	4.78	4.62
4	25.74	25.27	25.18	4.21	4.55	4.41
5	24.89	24.45	24.42	3.86	4.15	4.09
Q1/Q5	1.10	1.10	1.11	1.21	1.18	1.23

Note: Columns 1 and 4 are our baseline measures of inflation using Laspeyres price indices. Columns 2, 3, 5, and 6 use the second-order nonhomothetic adjustment from Jaravel and Lashkari (2024). In columns 2 and 5, the base period (b) is chosen to be the first quarter of the sample, as in Table 6 in the main text. In columns 3 and 6, the base period is instead chosen to be the last quarter of the sample.

Table A8: Items and varieties in the U.K. CPI microdata.

Category	COICOP-4	Description	Items	Varieties	Obs.
Food	10101	Cereals and cereal products	52	16,492	787,136
Food	10102	Live animals, and meat and other parts of slaughtered land animals	62	26,046	969,430
Food	10103	Fish and other seafood	11	3,861	195,559
Food	10104	Milk, other dairy products and eggs	35	10,182	500,582
Food	10105	Oils and fats	10	3,488	181,603
Food	10106	Fruits and nuts	40	24,027	750,378
Food	10107	Vegetables, tubers, plantains, cooking bananas and pulses	56	29,299	1,173,738
Food	10108	Sugar, confectionery and desserts	26	11,011	430,855
Food	10109	Ready-made food and other food products	21	6,290	290,232
Food	10201	Fruit and vegetable juices	11	3,459	214,331
Food	10202	Coffee and coffee substitutes	27	9,306	507,551
ND	20101	Spirits and liquors	11	4,044	213,141
ND	20102	Wine	20	6,236	362,864
ND	20103	Beer	15	3,288	183,820
ND	40301	Security equipment and materials for dwelling maintenance and repair	18	8,203	322,967
ND	40503	Liquid fuels	1	410	29,351
ND	40504	Solid fuels	4	1,262	64,474
ND	50601	Non-durable household goods	20	6,932	329,432
ND	60101	Medicines	6	2,704	159,595
ND	60102	Medical products	12	3,818	156,045
ND	70202	Fuels and lubricants for personal transport equipment	4	4,032	112,544
ND	90301	Garden products, plants and flowers	22	8,098	280,321
ND	90303	Gardens, plants, and flowers	12	4,870	147,941
ND	90304	Pets and related products	19	6,524	277,112
ND	90503	Miscellaneous printed matter	8	6,703	260,134
SD	30102	Garments	144	82,487	2,855,681
SD	30103	Other articles of clothing and clothing accessories	11	5,599	172,059
SD	30200	Footwear	34	21,107	646,075
SD	50200	Household textiles	12	7,422	282,949
SD	50400	Glassware, tableware and household utensils	17	8,222	292,464
SD	70201	Parts and accessories for personal transport equipment	9	4,752	168,699
SD	90201	Games, toys and hobby-related articles	4	871	30,277
SD	90502	Audiovisual media	1	246	5,924
SD	120302	Other personal effects	14	6,358	189,051
D	50101	Furniture, furnishings and loose carpets	39	23,080	769,968
D	50301	Major electric and other household appliances	25	10,787	396,151
D	50500	Tools and equipment for house and garden	19	7,593	281,754
D	70102	Motorcycles	5	2,594	87,875
D	90101	Photographic and cinematographic equipment and optical instruments	36	11,356	319,016
D	90102	Major recreational durables	6	1,497	41,551
D	90103	Information processing equipment	2	399	4,444
D	90104	Recording media	23	5,916	170,954
D	90302	Pets and pet products	17	5,415	152,593
D	90501	Musical instruments	10	4,671	189,481
D	120301	Jewellery, clocks, and watches	15	9,821	324,884
S	20200	Alcohol production services	19	5,965	258,513
S	30104	Cleaning, repair, tailoring and hire of clothing	3	1,649	61,982
S	40302	Services for the maintenance, repair and security of the dwelling	10	4,422	210,116
S	50102	Repair, installation and hire of furniture, furnishings and loose carpets	15	7,906	232,244
S	50303	Repair, installation and hire of household appliances	2	622	28,996
S	50602	Domestic services and household services	4	2,340	103,287
S	60201	Preventive care services	3	1,062	69,114
S	60202	Outpatient dental services	1	334	30,966
S	60300	Inpatient care services	3	514	30,274
S	70203	Maintenance and repair of personal transport equipment	14	4,275	129,153
S	70204	Other services related to personal transport equipment	6	5,571	297,096
S	70302	Passenger transport by road	4	1,298	65,201
S	80100	Information and communication equipment	1	136	2,083
S	80200	Software, excluding games	4	1,722	67,364
S	90105	Repair of audio-visual, photographic and information processing equipment	2	847	31,587
S	90401	Hire and repair of photographic and cinematographic equipment and optical instruments	14	4,082	188,231
S	90402	Hire, maintenance and repair of major recreational durables	6	2,261	67,199
S	100000	Education services	1	278	4,344
S	110101	Restaurants, cafés and the like	65	54,531	1,923,989
S	110102	Canteens, cafeterias and refectories	10	5,596	224,617
S	110200	Accommodation services	1	287	17,519
S	120101	Life and accident insurance	5	3,718	145,021
S	120102	Insurance connected with health	43	16,887	833,427
S	120400	Social protection	10	3,187	160,968
S	120700	Other services n.e.c.	10	2,687	155,525

Table A9: Cheapflation cycles in U.K. CPI microdata: Robustness.

	Inflation$_{kit+12}$				
Specification:	Baseline (1)	Variety FEs (2)	Controls for Item Inflation Last Year (3)	Relative Price Over Last 2 Years (4)	Relative Price Over Last 3 Years (5)
Relative Price$_{kit}$ $\times$ Avg. Item Inflation$_{it+12}$ $\times$ Food	−0.552**	−0.499**	−0.541**	−0.542**	−0.537**
	(0.047)	(0.046)	(0.051)	(0.050)	(0.055)
Relative Price$_{kit}$ $\times$ Avg. Item Inflation$_{it+12}$ $\times$ Other nondurables	−0.259**	−0.229*	−0.256**	−0.228*	−0.211*
	(0.115)	(0.126)	(0.114)	(0.113)	(0.112)
Relative Price$_{kit}$ $\times$ Avg. Item Inflation$_{it+12}$ $\times$ Semi-durables	−0.231**	−0.231**	−0.228**	−0.212**	−0.203**
	(0.038)	(0.037)	(0.036)	(0.038)	(0.040)
Relative Price$_{kit}$ $\times$ Avg. Item Inflation$_{it+12}$ $\times$ Durables	−0.191**	−0.281**	−0.221**	−0.196**	−0.190**
	(0.077)	(0.062)	(0.070)	(0.078)	(0.080)
Relative Price$_{kit}$ $\times$ Avg. Item Inflation$_{it+12}$ $\times$ Services	−0.177**	−0.136*	−0.159**	−0.149*	−0.136*
	(0.074)	(0.076)	(0.074)	(0.078)	(0.079)
Relative Price$_{kit}$ $\times$ Avg. Item Inflation$_{it}$ $\times$ Food			−0.036		
			(0.024)		
Relative Price$_{kit}$ $\times$ Avg. Item Inflation$_{it}$ $\times$ Other nondurables			−0.081		
			(0.053)		
Relative Price$_{kit}$ $\times$ Avg. Item Inflation$_{it}$ $\times$ Semi-durables			−0.012		
			(0.013)		
Relative Price$_{kit}$ $\times$ Avg. Item Inflation$_{it}$ $\times$ Durables			0.078		
			(0.074)		
Relative Price$_{kit}$ $\times$ Avg. Item Inflation$_{it}$ $\times$ Services			−0.058		
			(0.038)		
Item-Date FEs	Yes	Yes	Yes	Yes	Yes
Variety FEs		Yes			
N (millions)	21.6	21.6	19.9	21.6	21.6
R^2	0.12	0.26	0.12	0.11	0.11

Note: The table reports the results from variants of specification (13). Inflation$_{kit+12}$ and RelativePrice$_{kit}$ are winsorized at the 2.5 percent level. Standard errors two-way clustered by COICOP4 and year. * indicates significance at 10%, ** at 5%.

Table A10: List of BLS metropolitan areas with bi-monthly or monthly data on consumer price index excluding shelter.

Quartile	Code	BLS Area Description	Frequency
Prior to 2018:			
1	A210	Cleveland-Akron, OH	Bi-monthly
1	A421	Los Angeles-Riverside-Orange County, CA	Monthly
1	S23B	Detroit-Warren-Dearborn, MI	Bi-monthly
1	S35C	Atlanta-Sandy Springs-Roswell, GA	Bi-monthly
2	S35B	Miami-Fort Lauderdale-West Palm Beach, FL	Bi-monthly
2	S37A	Dallas-Fort Worth-Arlington, TX	Bi-monthly
2	S37B	Houston-The Woodlands-Sugar Land, TX	Bi-monthly
2	S49A	Los Angeles-Long Beach-Anaheim, CA	Monthly
3	S12B	Philadelphia-Camden-Wilmington, PA-NJ-DE-MD	Bi-monthly
3	S23A	Chicago-Naperville-Elgin, IL-IN-WI	Monthly
3	S35E	Baltimore-Columbia-Towson, MD	Bi-monthly
3	S49D	Seattle-Tacoma-Bellevue WA	Bi-monthly
4	A311	Washington-Baltimore, DC-MD-VA-WV	Bi-monthly
4	S11A	Boston-Cambridge-Newton, MA-NH	Bi-monthly
4	S12A	New York-Newark-Jersey City, NY-NJ-PA	Monthly
4	S35A	Washington-Arlington-Alexandria, DC-VA-MD-WV	Bi-monthly
4	S49B	San Francisco-Oakland-Hayward, CA	Bi-monthly
After 2018:			
1	S23B	Detroit-Warren-Dearborn, MI	Bi-monthly
1	S35C	Atlanta-Sandy Springs-Roswell, GA	Bi-monthly
1	S35D	Tampa-St. Petersburg-Clearwater, FL	Bi-monthly
1	S48A	Phoenix-Mesa-Scottsdale, AZ	Bi-monthly
1	S49C	Riverside-San Bernardino-Ontario, CA	Bi-monthly
1	S49F	Urban Hawaii	Bi-monthly
2	S24B	St. Louis, MO-IL	Bi-monthly
2	S35B	Miami-Fort Lauderdale-West Palm Beach, FL	Bi-monthly
2	S37A	Dallas-Fort Worth-Arlington, TX	Bi-monthly
2	S37B	Houston-The Woodlands-Sugar Land, TX	Bi-monthly
2	S49A	Los Angeles-Long Beach-Anaheim, CA	Monthly
2	S49G	Urban Alaska	Bi-monthly
3	S12B	Philadelphia-Camden-Wilmington, PA-NJ-DE-MD	Bi-monthly
3	S23A	Chicago-Naperville-Elgin, IL-IN-WI	Monthly
3	S24A	Minneapolis-St.Paul-Bloomington, MN-WI	Bi-monthly
3	S35E	Baltimore-Columbia-Towson, MD	Bi-monthly
3	S48B	Denver-Aurora-Lakewood, CO	Bi-monthly
3	S49D	Seattle-Tacoma-Bellevue WA	Bi-monthly
3	S49E	San Diego-Carlsbad, CA	Bi-monthly
4	S11A	Boston-Cambridge-Newton, MA-NH	Bi-monthly
4	S12A	New York-Newark-Jersey City, NY-NJ-PA	Monthly
4	S35A	Washington-Arlington-Alexandria, DC-VA-MD-WV	Bi-monthly
4	S49B	San Francisco-Oakland-Hayward, CA	Bi-monthly

Table A11: Inflation inequality across U.S. metropolitan areas due to consumption weights.

	Inflation in Metropolitan Area			
	CPI Data		Constructed with Area Weights	
	CPI (1)	CPI Ex. Shelter (2)	CPI (3)	CPI Ex. Shelter (4)
U.S. Avg. Inflation × Log Income	−0.970**	−0.452**	−0.208**	−0.224**
	(0.085)	(0.066)	(0.035)	(0.030)
Time FEs	Yes	Yes	Yes	Yes
Metropolitan Area FEs	Yes	Yes	Yes	Yes
N	3540	3534	7452	7452
R^2	0.85	0.89	0.99	0.99

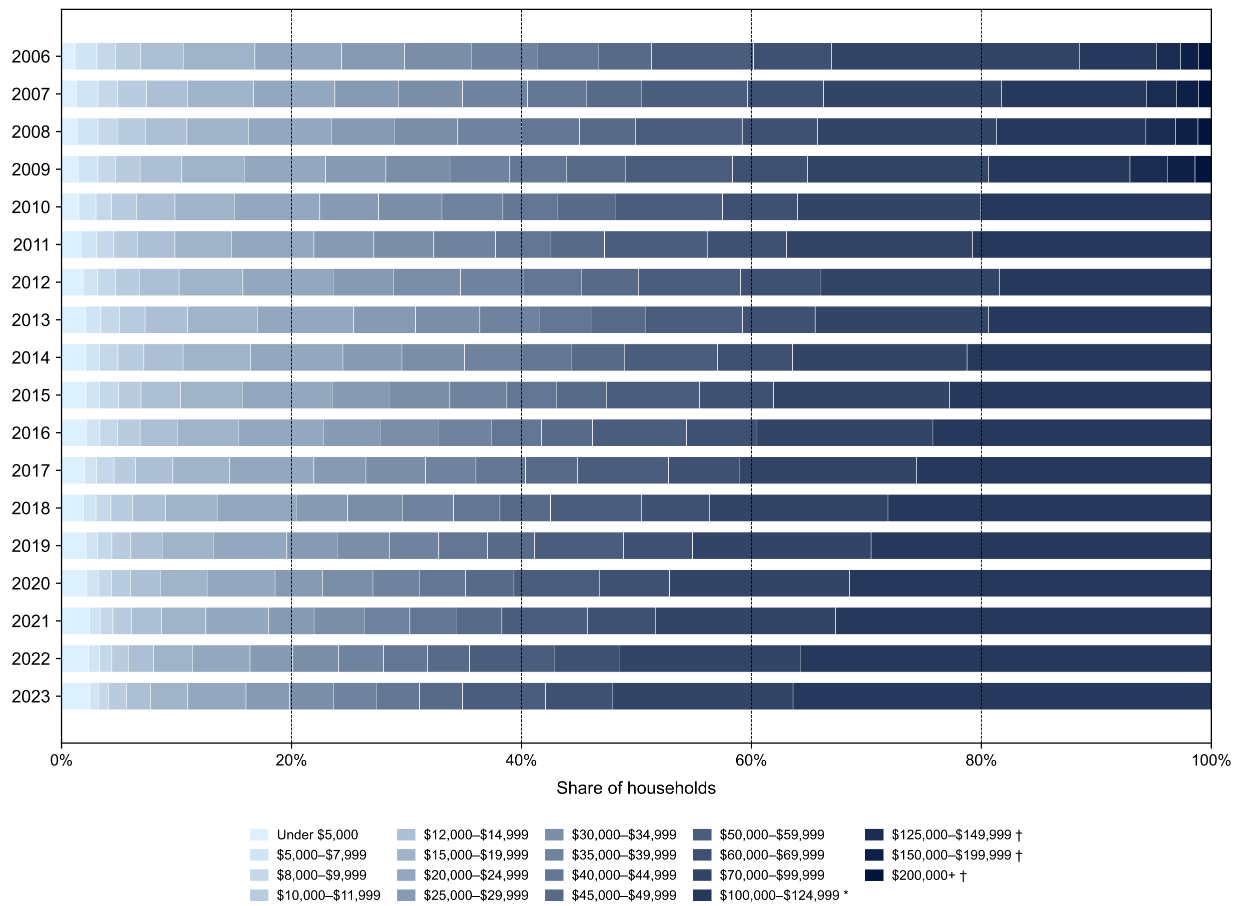
Note: The table reports the results from specification (14). Columns 1–2 use metropolitan area CPI data. Columns 3–4 use counterfactual inflation rates for each metropolitan area using the metropolitan area's relative importance weights across items but using the inflation rate for each item for the U.S. city average. Relative importance weights for each metropolitan area are from the BLS's December 2024 release, which was the most recent release at the time of writing (available at `https://www.bls.gov/cpi/tables/relative-importance`). Driscoll–Kraay standard errors with twelve lags in parentheses. * indicates significance at 10%, ** at 5%.

Table A12: Breaking out post-2020 period for inflation inequality across U.S. metropolitan areas.

	Inflation in Metropolitan Area	
	CPI (1)	CPI Ex. Shelter (2)
U.S. Average Inflation × Log Income	−0.986**	−0.575**
	(0.151)	(0.066)
U.S. Average Inflation × Log Income × Post-2020	−0.046	0.202*
	(0.168)	(0.105)
Time FEs	Yes	Yes
Metropolitan Area FEs	Yes	Yes
N	3540	3534
R^2	0.85	0.89

Note: The table reports the results from specification (14), augmented to include an additional interaction term for U.S. Average Inflation × Log Income × Post-2020. Driscoll–Kraay standard errors with twelve lags in parentheses. * indicates significance at 10%, ** at 5%.

Figure A1: Sorting households into income quintiles.

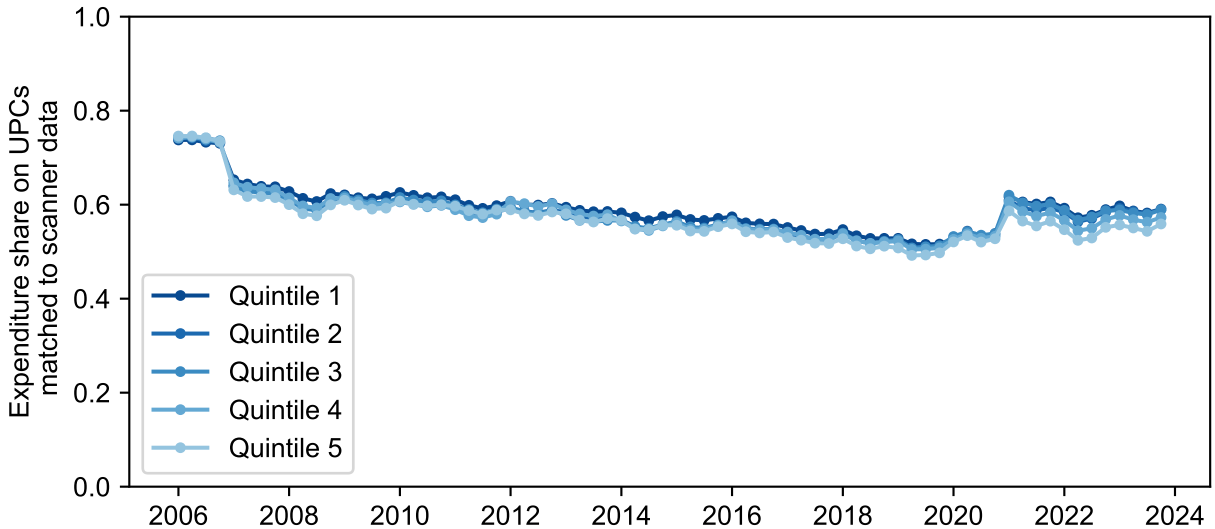


Note: The figure shows the distribution of NielsenIQ Homescan panelists by income bin (using projection weights provided by NielsenIQ). We sort households into income quintiles using these income bins and use total expenditures divided by the square-root of household size as a tiebreaker within income bins.

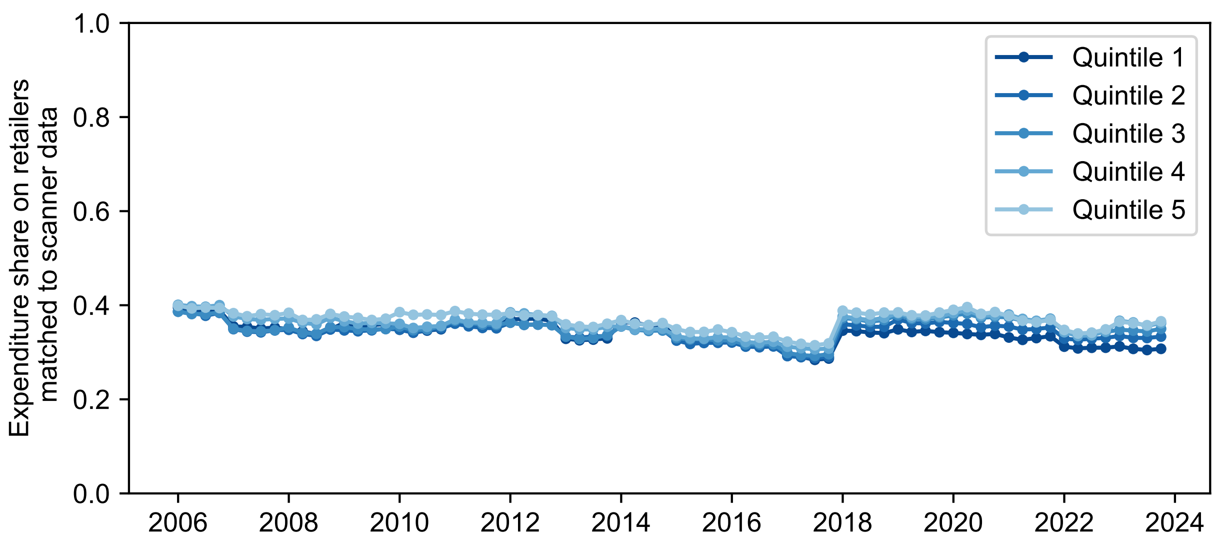
* All households with $100,000+ income for years 2010–2023.

† The split of income groups above $100,000 is only available for 2006–2009.

Figure A2: Homescan panelist expenditures matched to Retail Scanner data by income quintile over time.



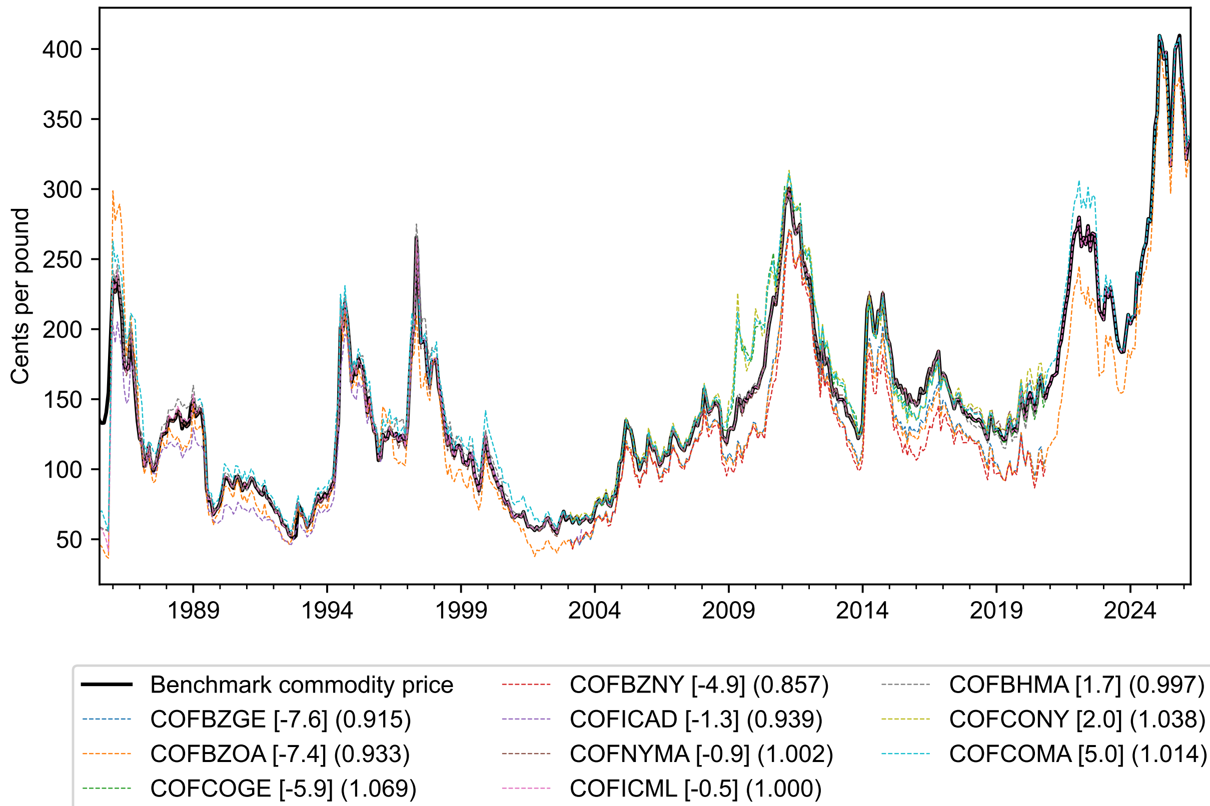
(a) Share of expenditures on UPCs matched to scanner data.



(b) Share of expenditures on retailers matched to scanner data.

Note: A UPC is marked as matched if it appears in the Retail Scanner data in the same year, and a retailer is marked as matched if the retailer code appears in the Retail Scanner data in the same year.

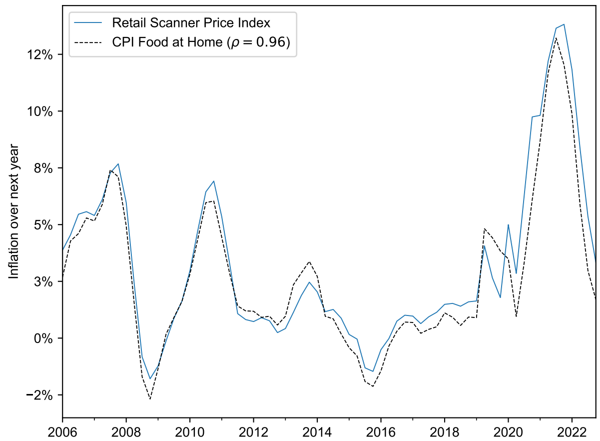
Figure A3: Comparing prices for coffee varieties to benchmark commodity series.



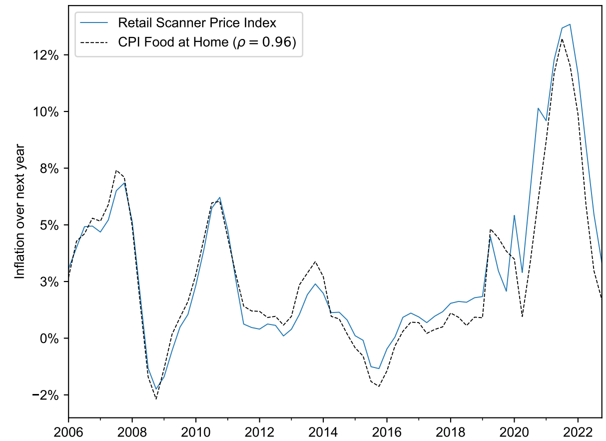
Note: The thick black line (Benchmark commodity price) is the ICO price of mild Arabicas at the dock in New York, which we use as our benchmark measure of coffee commodity input prices. Thin dashed lines show prices of various other coffee commodity series from Datastream. Datastream mnemonics for each series are shown in the legend (the initials after COF typically refer to origin—e.g., CO refers to Colombian origin, BZ to Brazilian origin, ICA to Central American origin, and BH to Brazil/Honduras origin—and the final initials typically refer to the exchange or commodity descriptor—e.g., NY to New York, GE to Hamburg, and MA to Mild Arabica composites). The values in square brackets and parentheses in the legend are series-specific estimates of differentials d_i and pass-throughs b_i from the regression,

$$\text{Price}_{it} = d_i + b_i \text{BenchmarkCommodityPrice}_t + \varepsilon_{it}.$$

Figure A4: Comparing food-at-home inflation in Retail Scanner data to CPI.

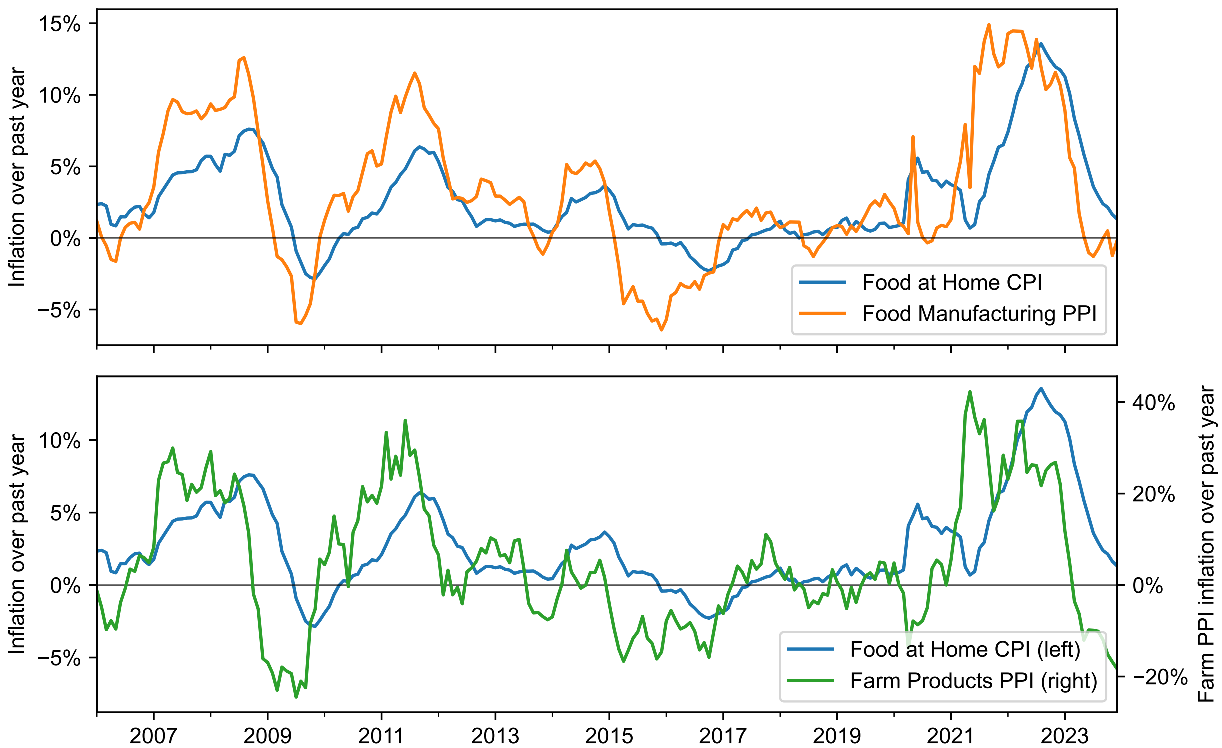


(a) Level of aggregation: Retailer-UPC



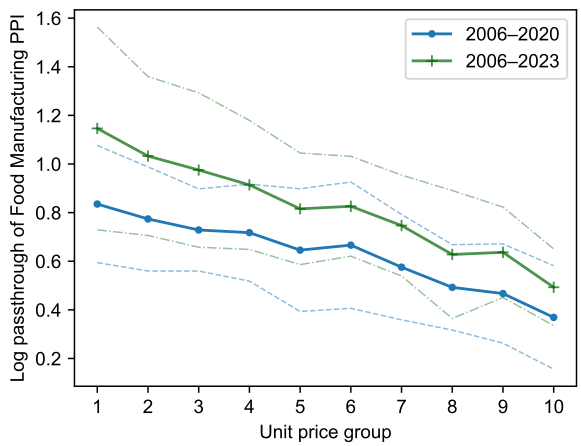
(b) Level of aggregation: UPC

Figure A5: Food at Home CPI, Food Manufacturing PPI, and Farm Products PPI.

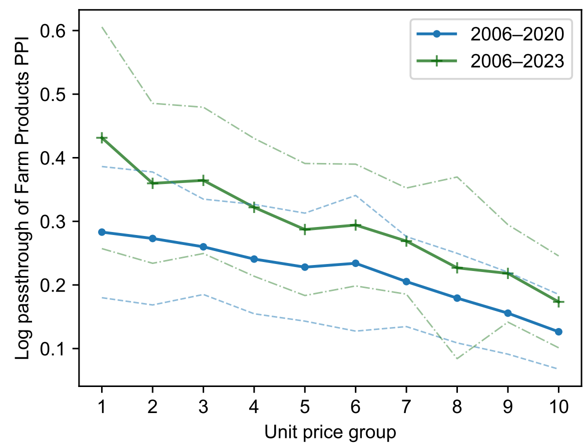


Note: Data pulled from FRED: consumer price index for Food at Home (`CUSR0000SAF11`), producer price index for Food Manufacturing (`PCU311311`), and producer price index for Farm Products (`WPU01`).

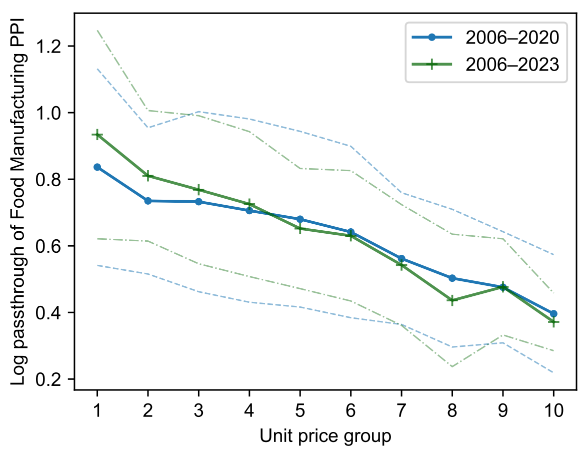
Figure A6: Log pass-through of upstream price indices by unit price group: Comparing estimates from 2006–2020 to estimates from 2006–2023.



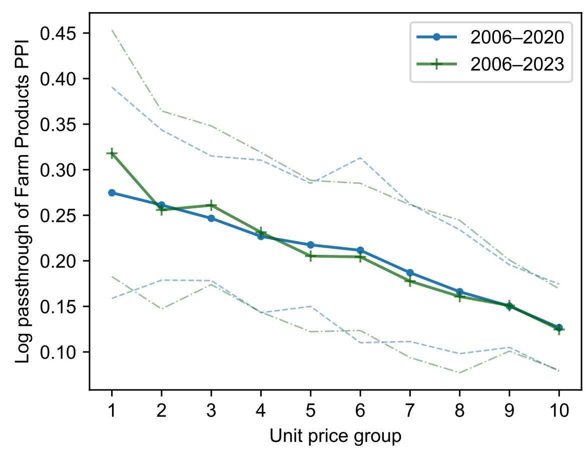
Food Manufacturing PPI:
(a) Baseline specification.



Farm Products PPI:
(b) Baseline specification.



Food Manufacturing PPI:
(c) Specification with wage pass-through.



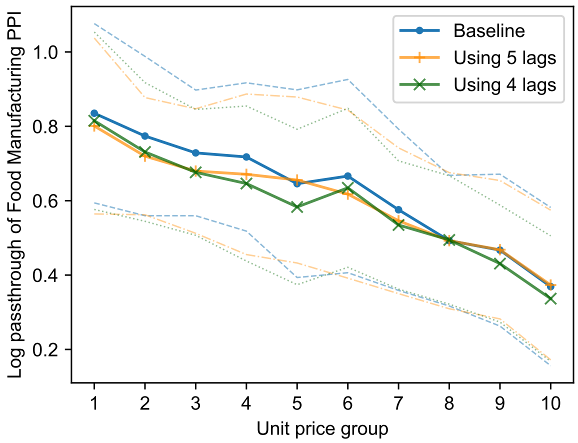
Farm Products PPI:
(d) Specification with wage pass-through.

Note: Panels (a) and (b) report estimates of long-run pass-through of upstream producer price indices by unit price decile using specification (5). Panels (c) and (d) report the long-run pass-through of upstream PPI using a specification augmented with wage pass-through,

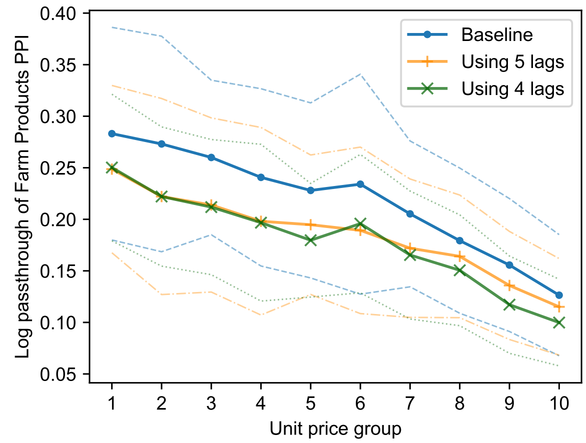
$$\Delta \log p_t^n = \alpha_n + \sum_{k=0}^{K} \beta_k^n \Delta \log \mathrm{PPI}_{t-k} + \sum_{k=0}^{K} \gamma_k^n \Delta \log w_{t-k} + \sum_{k=1}^{4} \delta_k^n q_t + \varepsilon_{nt},$$

where $\Delta \log w_t$ is the change in the log of the employment cost index from quarter $t-1$ to t. Dotted blue lines indicate 95 percent confidence intervals for pass-through estimates using 2006–2020, and dot-dashed green lines indicate 95 percent confidence intervals for pass-through estimates using 2006–2023. Both are constructed using the delta method with Newey-West standard errors.

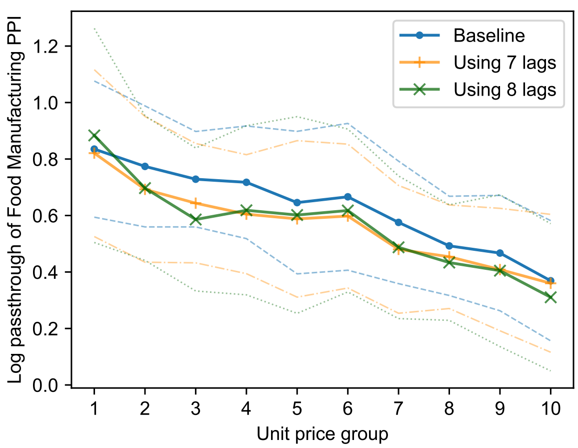
Figure A7: Log pass-through of upstream price indices by unit price group: Varying horizon for long-run pass-through estimates.



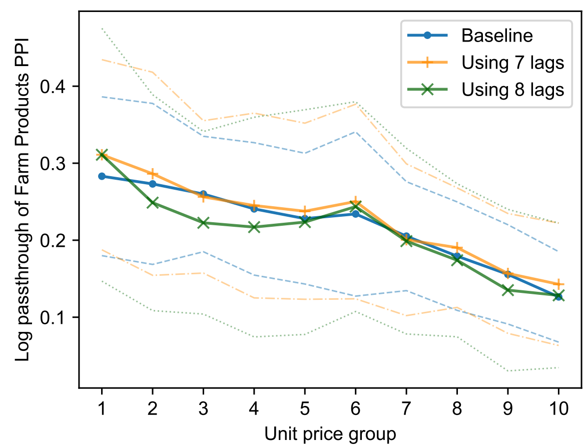
(a) Food Manufacturing PPI: $K \in \{4, 5, 6\}$.



(b) Farm Products PPI: $K \in \{4, 5, 6\}$.



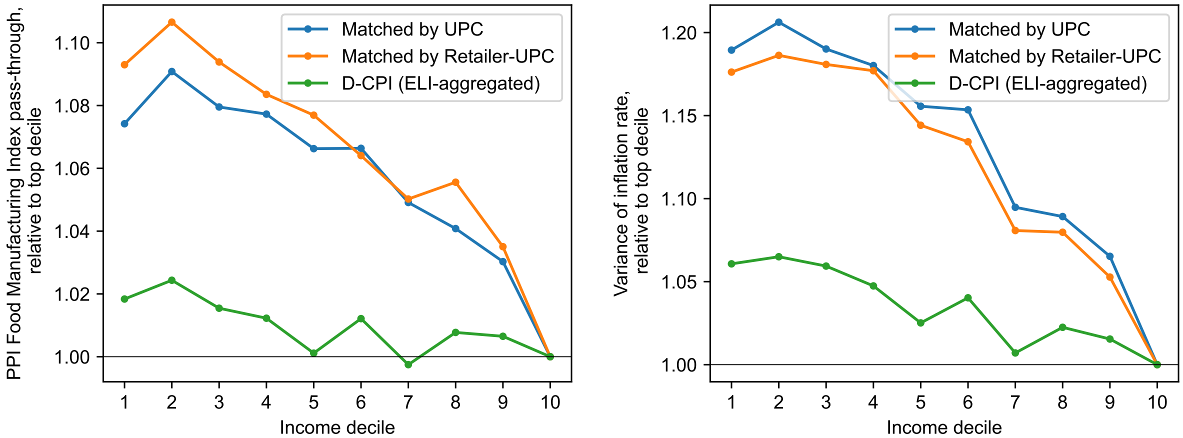
(c) Food Manufacturing PPI: $K \in \{6, 7, 8\}$.



(d) Farm Products PPI: $K \in \{6, 7, 8\}$.

Note: The figures report estimates of the long-run pass-through $\sum_{k=0}^{K} \beta_k^n$ from (5), varying the number of lagged PPI changes (K) in the regression. The baseline estimates use $K = 6$ quarters. Dotted lines indicate 95 percent confidence intervals constructed using the delta method with Newey-West standard errors.

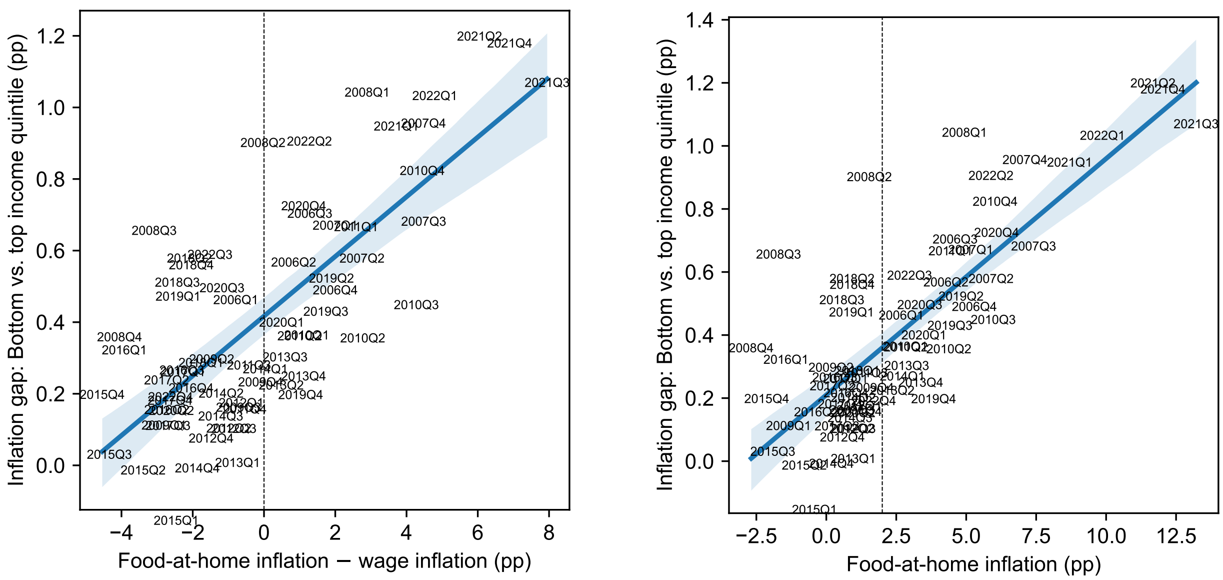
Figure A8: Differences by income decile: Sensitivity of food-at-home inflation to upstream PPI and variance of food-at-home inflation rates.



(a) Pass-through of Food Manufacturing PPI.

(b) Variance of inflation rates.

Figure A9: Estimating secular inflation inequality.

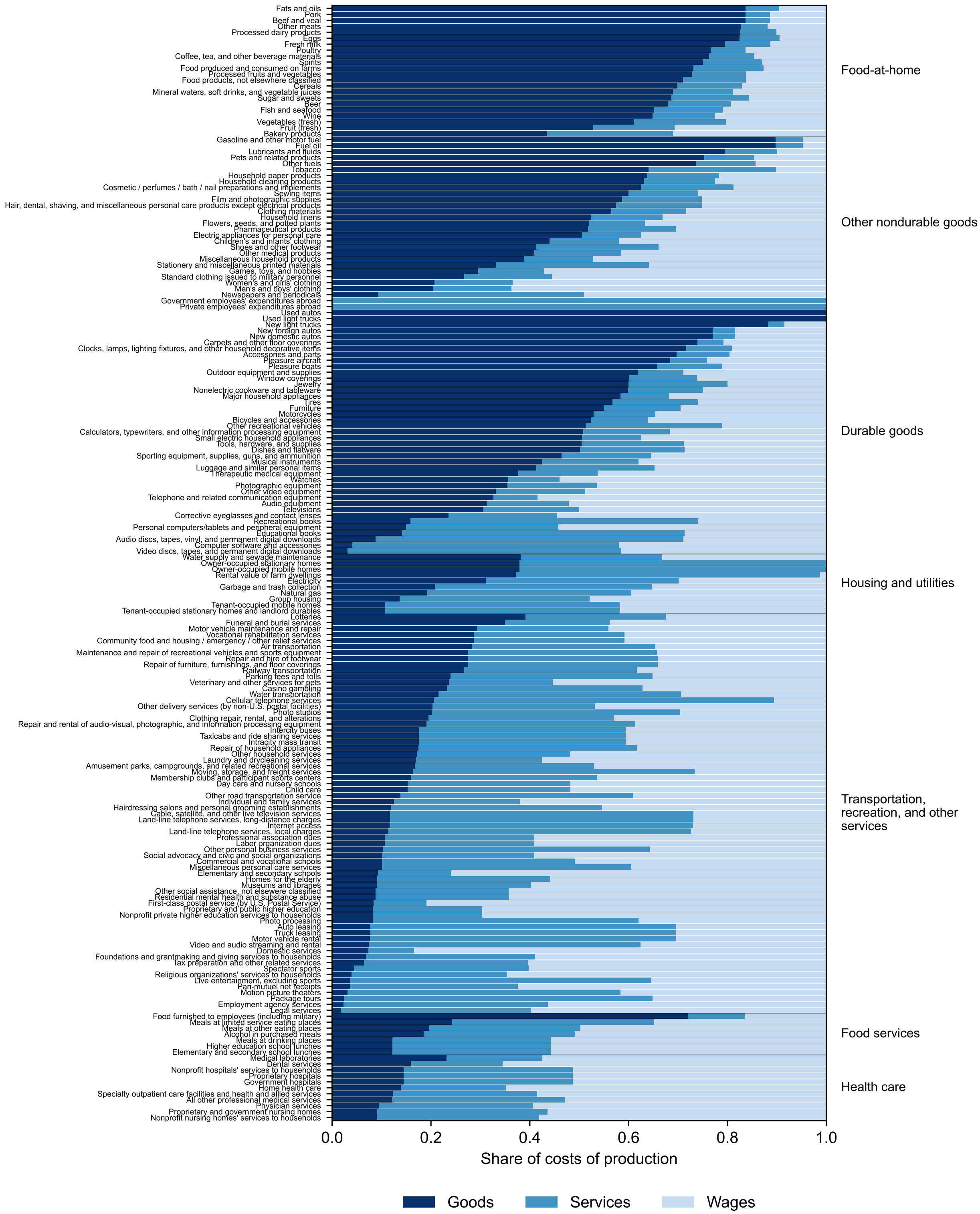


(a) Food-at-home − wage inflation ($R^2 = 0.54$).

(b) Food-at-home inflation ($R^2 = 0.63$).

Note: The *y*-axis in both panels plots the gap in inflation over the next year between the bottom and top income quintiles in quarterly data from 2006–2023. In the left panel, the *x*-axis is food-at-home inflation in excess of wage inflation (measured using the employment cost index) over the next year. In the right panel, the *x*-axis is food-at-home inflation. We estimate the secular inflation gap in the left panel as the inflation gap associated with food-at-home inflation equal to wage inflation, and in the right panel as the inflation gap associated with food-at-home inflation of 2 percent.

Figure A10: Production costs spent on goods, services, and labor by PCE subcomponent.



Note: We match each NIPA line to constituent BEA commodity codes using the 2017 PCE bridge. For each BEA commodity code, we then construct the share of costs spent on goods (NAICS 11, 21–23, 31–33), services, and labor compensation using the BEA 2017 detail input–output table. The chart excludes financial services, net foreign travel, and nonprofits.

Appendix B Proofs

Interpretation of PPI pass-through. Suppose that changes in the PPI reflect a weighted average of changes in individual food materials costs indexed by j,

$$\Delta \log \mathrm{PPI}_t = \sum_j \lambda_j^{\mathrm{PPI}} \Delta \log m_{jt},$$

where λ_j^{PPI} are the weights assigned to each component input cost in the PPI. Suppose changes in the price of each retail variety i in unit price group n can be written as

$$\Delta \log p_{it}^n = \sum_j \rho_{ij}^n \left(\Delta \log m_{jt}\right) + \varepsilon_{it}^n,$$

where ρ_{ij}^n is the log pass-through of changes in input cost m_j to the price of i, and ε_{it}^n are changes in i's retail price orthogonal to changes in any m_j. Then, the change in the Laspeyres price index for all varieties in unit price group n to first order is

$$\begin{aligned}
\Delta \log p_t^n &\approx \sum_i \lambda_i \Delta \log p_{it}^n = \sum_i \lambda_i \left(\sum_j \rho_{ij}^n \left(\Delta \log m_{jt}\right) + \varepsilon_{it}^n \right) \\
&= \bar{\rho}^n \sum_j \lambda_j^{\mathrm{PPI}} \left(\Delta \log m_{jt}\right) + \left[\sum_j \left(\mathbb{E}_\lambda \left[\rho_{ij}^n\right] - \bar{\rho}^n + \bar{\rho}^n \left(1 - \lambda_j^{\mathrm{PPI}}\right) \right) \left(\Delta \log m_{jt}\right) \right] + \mathbb{E}_\lambda \left[\varepsilon_{it}^n\right] \\
&= \bar{\rho}^n \Delta \log \mathrm{PPI}_t - \bar{\rho}^n Cov\left(\lambda_j^{\mathrm{PPI}}, \Delta \log m_{jt}\right) + Cov\left(\mathbb{E}_\lambda \left[\rho_{ij}^n\right], \Delta \log m_{jt}\right) + \mathbb{E}_\lambda \left[\varepsilon_{it}^n\right],
\end{aligned}$$

where $\mathbb{E}_\lambda[x_i] = \sum_i \lambda_i x_i$, $\bar{\rho}^n = \sum_j \mathbb{E}_\lambda[\rho_{ij}^n]$, and $Cov(x_j, y_j) = \sum_j x_j y_j - \sum_j x_j \sum_j y_j$. Dividing through by $\Delta \log \mathrm{PPI}_t$, and noting that ε_{it}^n are assumed orthogonal to changes in any m_j, we find that the pass-through of PPI changes to the Laspeyres price index for group n is:

$$\frac{\Delta \log p_t^n}{\Delta \log \mathrm{PPI}_t} = \underbrace{\bar{\rho}^n \left[1 - Cov\left(\lambda_j^{\mathrm{PPI}}, \frac{\Delta \log m_{jt}}{\Delta \log \mathrm{PPI}_t} \right) \right]}_{\text{Common to all unit price groups}} + Cov\left(\mathbb{E}_\lambda \left[\rho_{ij}^n\right], \frac{\Delta \log m_{jt}}{\Delta \log \mathrm{PPI}_t} \right).$$

Notice that a higher pass-through of the PPI to retail prices for some unit price group n can either reflect a higher $\bar{\rho}^n$ (i.e., higher pass-through of individual input costs, averaged across inputs and varieties) or a higher covariance between average pass-throughs of specific inputs j and their sensitivity to the PPI. □

Proof of Proposition 1. Given a perturbation to materials and labor prices, $d \log m$ and

$d\log w$, the change in the price of variety i to first order is

$$d\log p_i \approx \frac{m}{p_i} d\log m + \frac{\beta_i w}{p_i} d\log w.$$

Thus, the inflation rate on a Laspeyres index with initial expenditure shares λ_i^g is

$$\begin{aligned}
\pi^g \equiv \int_0^1 \lambda_i^g d\log p_i di &\approx \int_0^1 \lambda_i^g \left(\frac{m}{p_i} d\log m + \frac{\beta_i w}{p_i} d\log w \right) di \\
&= m \left[\int_0^1 \frac{\lambda_i^g}{p_i} di \right] d\log m + \left(1 - m \left[\int_0^1 \frac{\lambda_i^g}{p_i} di \right] \right) d\log w \\
&= \frac{m}{p^g} d\log m + \left(1 - \frac{m}{p^g} \right) d\log w,
\end{aligned}$$

where p^g is the λ_i^g-weighted harmonic average of p_i, i.e., $p^g = \left(\int_0^1 \lambda_i^g / p_i di \right)^{-1}$. Using π^{all} and p to denote the Laspeyres inflation rate and average unit price constructed with varieties' sales shares λ_i, we can express $d\log m$ in terms of π^{all} and the change in the wage,

$$d\log m = \frac{p}{m} \pi^{\text{all}} - \left(\frac{p}{m} - 1 \right) d\log w.$$

Substituting in, we get

$$\pi^g = \frac{p}{p^g} \pi^{\text{all}} + \left(1 - \frac{p}{p^g} \right) d\log w,$$

which can be rearranged to yield Proposition 1. $\square$

Appendix C Measurement Robustness

This appendix contains three sets of robustness exercises. Section C.1 considers how our results change if we measure prices for each product in each quarter using an unweighted average of posted prices across store–weeks, rather than the quantity-weighted averages used in the main text. Section C.2 tests the robustness of our results to mean reversion. Section C.3 tests whether UPCs and retail chains observed in the NielsenIQ Consumer Panel data but not the Retail Scanner data exhibit similar inflation dynamics to those matched to the Retail Scanner data.

C.1 Quantity-Weighted Average *vs.* Unweighted Average Prices

In the main text, we measure the inflation rate for UPC i at retail chain r in quarter t as

$$\pi_{irt} = \frac{p_{irt}}{p_{irt-4}} - 1, \qquad \text{where} \qquad p_{irt} = \frac{\text{Sales}_{irt}}{\text{Units}_{irt}}.$$

This price p_{irt} is a quantity-weighted average across all stores and weeks within retail chain r and quarter t. A concern is that variation in these prices—and thus variation in inflation rates—could reflect changes in shopping behavior (i.e., the quantities purchased at each posted price) rather than changes in posted prices themselves.

To assess the sensitivity of our results to this concern, we build an alternative measure of inflation for each product using unweighted averages of prices across stores and weeks,

$$\pi^{\text{unwt}}_{irt} = \frac{p^{\text{unwt}}_{irt}}{p^{\text{unwt}}_{irt-4}} - 1, \qquad \text{where} \qquad p^{\text{unwt}}_{irt} = \frac{1}{N_{irt}} \sum_{s \in \mathcal{S}(r)} \sum_{w \in \mathcal{W}(t)} p_{isw},$$

where p_{isw} is the price of UPC i in store s in week w, $\mathcal{S}(r)$ is the set of stores in retail chain r, $\mathcal{W}(t)$ is the set of weeks in quarter t, and N_{irt} is the number of store–week observations of UPC i in retail chain r and quarter t.

The unweighted-average price measure addresses concerns about shopping behavior affecting measured inflation rates, but introduces other drawbacks. Because NielsenIQ records a price for a UPC at a store only when the product is sold, the unweighted average price is taken over the set of store–weeks with positive sales. This makes the measure sensitive to small changes in sales (e.g., zero vs. one unit). The quantity-weighted average price measure naturally deals with this problem, since a store–week with no sales receives zero weight. It is also less sensitive to the entry or exit of small stores from the sample.

Nevertheless, Table C1 shows that product-level inflation rates constructed using

Table C1: Comparing inflation rates measured using quantity-weighted average prices (baseline) *vs.* unweighted average prices.

Panel A. Correlation	Roasted coffee (1)	All food-at-home (2)
Correlation between inflation measures	0.9988	0.9741
N (millions)	2.1	151.0
Panel B. Regression	*Inflation with quantity-weighted avg. prices*	
	Roasted coffee (1)	All food-at-home (2)
Inflation with unweighted avg. prices	0.985**	0.968**
	(0.007)	(0.039)
N (millions)	2.1	151.0
R^2	1.00	0.95

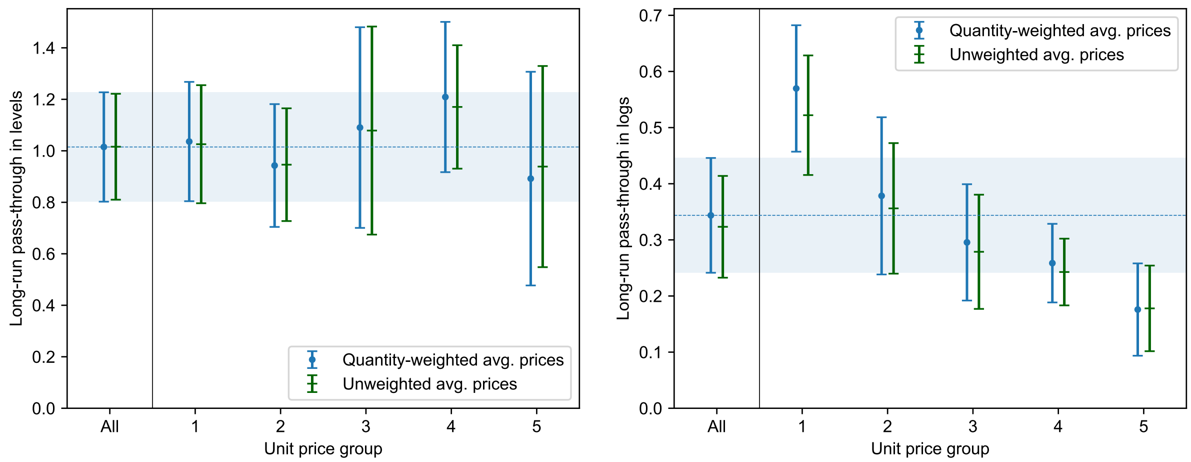
Note: Panel A reports the correlation between product-level inflation rates measured using quantity-weighted average prices *vs.* unweighted average prices each quarter. Panel B reports the results of a regression of inflation rates measured using quantity-weighted average prices on inflation rates measured using unweighted average prices. In Panel B, standard errors are clustered by quarter in column 1 and two-way clustered by product module and quarter in column 2. * indicates significance at 10%, ** at 5%.

quantity-weighted average prices and unweighted average prices are highly correlated. For coffee products studied in Section 3, variety-level inflation rates exhibit a correlation of 0.9988; for the entire set of food-at-home products studied in Section 4, the correlation is 0.9741. Likewise, regressions of our baseline measures of inflation on the unweighted measures yield coefficients and R^2 values close to one. These results suggest that changes in posted prices, rather than changes in quantity weights coming from customer shopping behavior, constitute the dominant share of variation in our baseline measure of inflation.[28]

We can also directly test whether our results on pass-through across the price distribution are affected by the averaging approach. Figure C1 replicates our results on pass-through in levels and logs of coffee commodity prices to retail prices (Figure 2 in the main text) using both the quantity-weighted and unweighted measures of prices. As with our baseline measures, the pass-through in levels of coffee commodity prices to retail prices is statistically indistinguishable from one across all unit price groups, while the pass-through in logs is incomplete and systematically declines with unit price. Likewise, Figure C2 shows that our results on the log pass-through of upstream PPIs to food-at-

[28]These results also suggest that variation in variety-level inflation rates largely reflects variation in changes in regular prices rather than changes in promotional intensity (studied by, e.g., Kryvtsov and Vincent 2021), since customer responses to temporary promotions would create a divergence between the quantity-weighted and unweighted measures.

Figure C1: Pass-through of coffee commodity costs to retail prices in levels and logs.

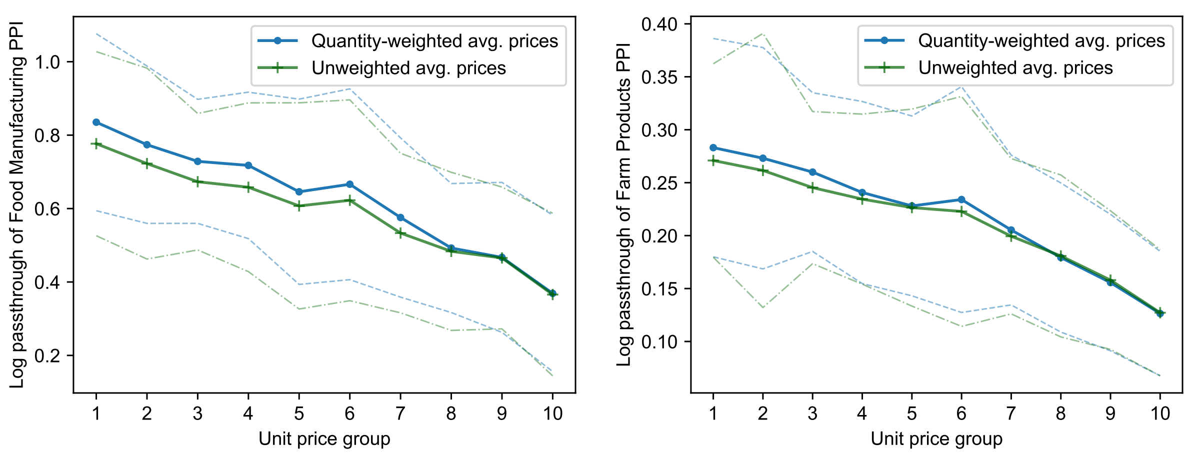


(a) Pass-through in levels. **(b)** Pass-through in logs.

Note: This figure replicates Figure 2, using both quantity-weighted average and unweighted average prices.

home prices (Figure 5 in the main text) are similar whether we use quantity-weighted average prices or unweighted average prices: in both cases, the pass-through of the Food Manufacturing PPI and Farm Products PPI to food-at-home prices declines systematically with products' relative prices.

Figure C2: Log pass-through of upstream price indices with unweighted average prices.



(a) Pass-through: Food Manufacturing PPI **(b)** Pass-through: Farm Products PPI

Note: This figure replicates Figure 5, using both quantity-weighted average and unweighted average prices.

C.2 Constructing Unit Price Groups with Alternative Rankings

Our baseline results sort varieties within a product category into groups using their average unit prices over the prior year. A concern with using a short window to measure varieties' prices is mean reversion: transient deviations in a variety's price relative to trend can create a mechanical correlation between initial price and subsequent price changes.

To explore the degree to which our baseline unit price groups reflect transient deviations in relative prices, Figure C3 compares the unit price groups constructed using average unit prices over the prior year to unit price groups constructed using other sorting variables. Panels (a) and (b) sort varieties using longer-run average unit prices—over the prior three and five years, respectively. Most products are categorized into the same or a neighboring unit price decile. Panel (c) instead sorts products into unit price groups using the average relative unit price of the product over its lifetime, defined as

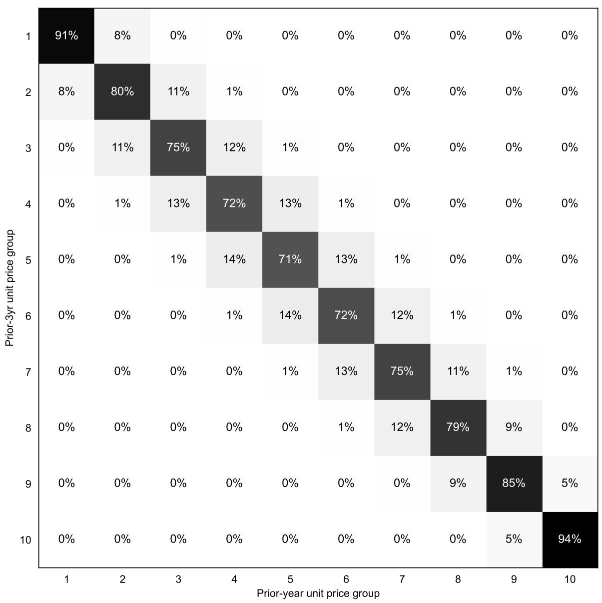
$$\text{Lifetime Average Relative Unit Price}_i = \frac{1}{|\mathcal{T}_i|} \sum_{t \in \mathcal{T}_i} \log(p_{it}/\bar{p}_t),$$

where p_{it} is the unit price of product i in quarter t, $\bar{p}_t$ is the average unit price of products in i's product category in quarter t, and $\mathcal{T}_i$ is the set of quarters in which product i is observed. Most products in panel (c) cluster near the diagonal, suggesting that average unit prices over the prior year correlate strongly with persistent price differences.[29] Finally, panel (d) shows that grouping varieties by their unit prices over the next year leads to similar rankings, suggesting that our unit price rankings are persistent.

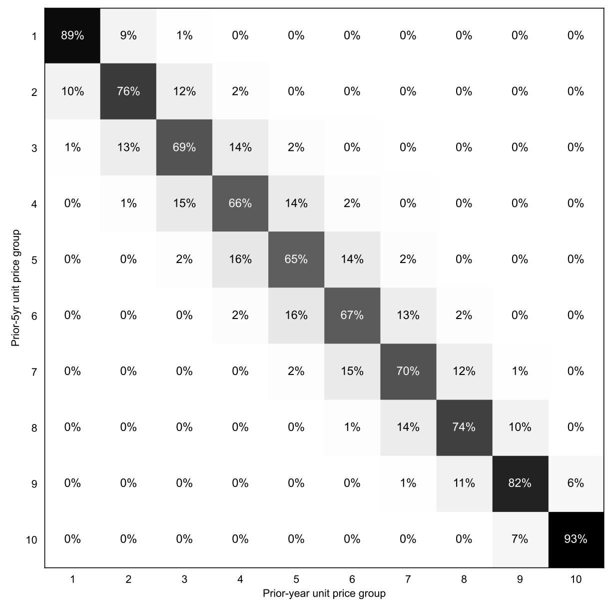
In the main text, we argue that while mean reversion could lead us to overstate secular differences in inflation between low- and high-priced varieties, it should not generally lead to a systematic covariance between varieties' initial prices and future input cost changes. We can directly test this claim by seeing if our results on pass-through change when we use alternative measures to sort varieties into unit price groups. Figure C4 shows the pass-through of coffee commodity prices to retail prices in levels and logs when we group products into unit price quintiles using both shorter-horizon measures of relative prices—only the current quarter price—and longer-horizon measures. The results are

[29] Although the lifetime average relative unit price of a product by construction removes transient deviations in relative prices, it has two disadvantages. First, it incorporates information about a product's future price, risking a mechanical correlation between its relative price at t and subsequent price changes. Second, it captures more than the persistent component of the product's relative price at t: because new product cohorts tend to enter at higher prices (Jaravel 2019; Argente, Lee, and Moreira 2024; Liu and Moreira 2026), the lifetime average relative price of a product also reflects a product's age and its longevity. This introduces measurement error in relative prices, which can attenuate the estimated relationship between relative price and pass-through or bias the measured relationship if product age or longevity itself affects pass-through.

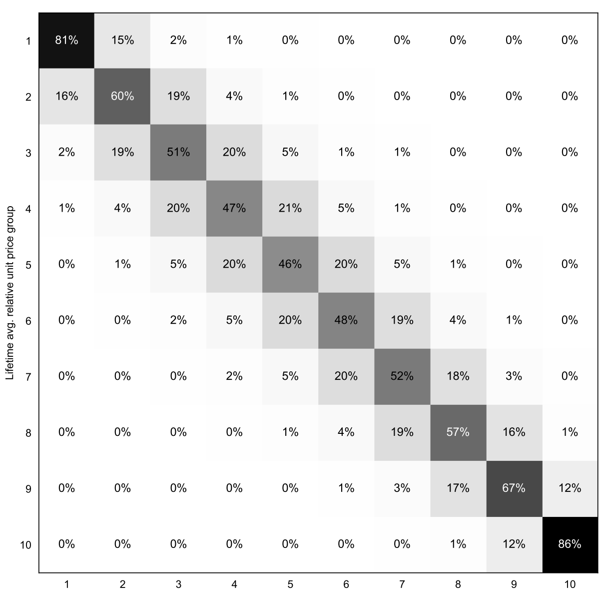
Figure C3: Comparison of baseline unit-price-group assignments to alternate rankings.



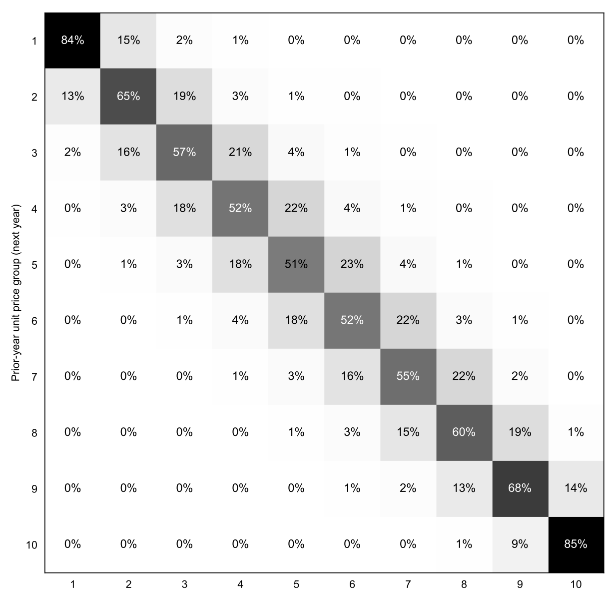
(a) Average unit price over last three years.



(b) Average unit price over last five years.



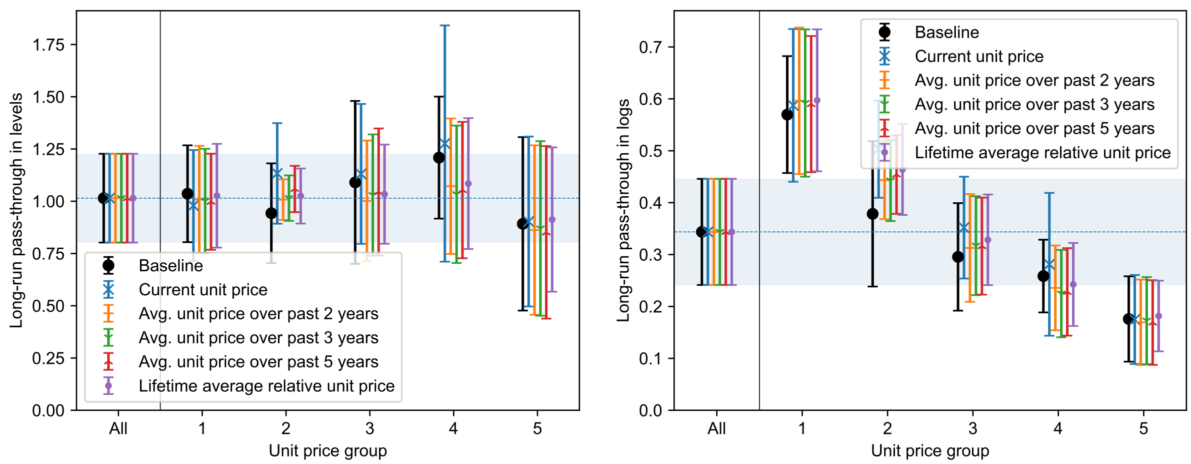
(c) Lifetime average relative unit price.



(d) Average unit price over next year.

Note: In each panel, the horizontal axis denotes varieties sorted into unit price groups using average unit price over the prior year, and the vertical axis denotes varieties sorted into unit price groups using the alternative measure. The percentage indicates the fraction of varieties in each column categorized to the according row. See the text of Appendix C.2 for definitions of the alternative unit price groupings.

Figure C4: Pass-through of coffee commodity costs to retail prices with alternate rankings.



(a) Pass-through in levels.

(b) Pass-through in logs.

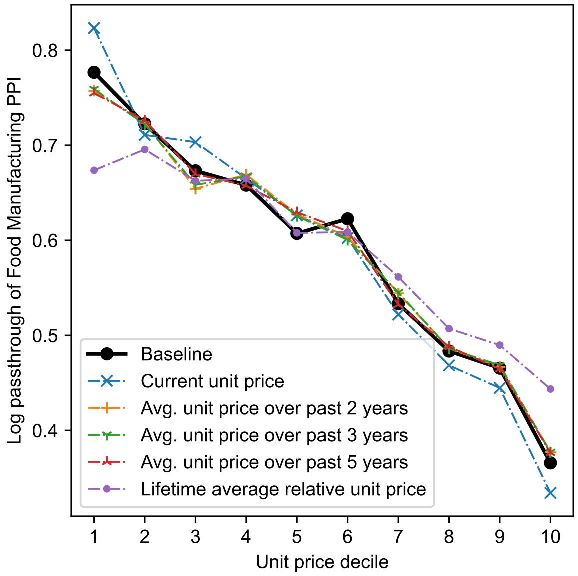
Note: This figure replicates Figure 2 using alternate definitions of unit price groups. See the text of Appendix C.2 for definitions of the alternative unit price groupings.

similar across groupings: in each case, the pass-through in levels across unit price groups is indistinguishable from one, and the pass-through in logs declines with unit price.

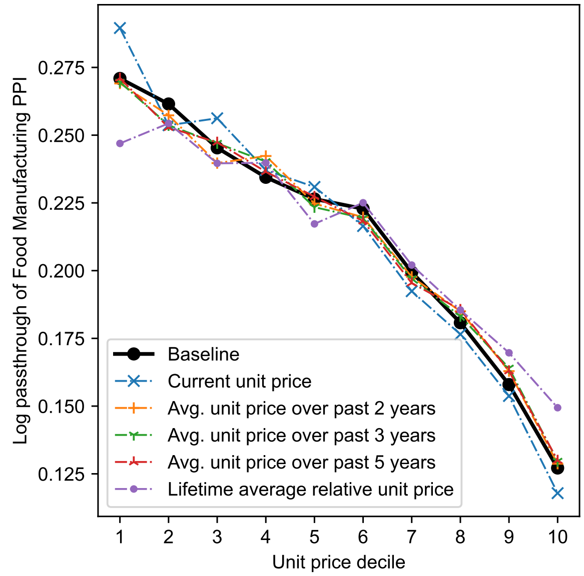
Similarly, Figure C5 panels (a) and (b) plot the pass-through of the Food Manufacturing and Farm Products PPIs to all food-at-home prices split by unit price decile. Using current-quarter prices to sort products slightly steepens the relationship between pass-through and unit prices, and using lifetime average relative unit prices slightly flattens the relationship, though the differences are not particularly large. Note that the flatter relationship between pass-through and unit price found using lifetime average relative unit prices could also reflect attenuation from measurement error, since lifetime average relative unit prices can reflect factors like product age and longevity (see Footnote 29).

Consistent with the idea that secular inflation differences are more sensitive to mean reversion than pass-through, panels (c) and (d) show that trend inflation across unit price groups varies more with the choice of ranking. That said, while using current-quarter prices yields a much steeper relationship between trend inflation and unit price, using a variety's unit price over the prior year yields very similar results to using prices over the prior two, three, or five years. Thus, averaging over the prior year appears to correct for the bulk of the mean reversion caused by transient price differences. Using products' lifetime average relative unit prices actually flips the relationship between unit price and trend inflation; however, the positive relationship could in part reflect reverse causality (higher trend inflation leading to higher relative prices over a product's lifetime).

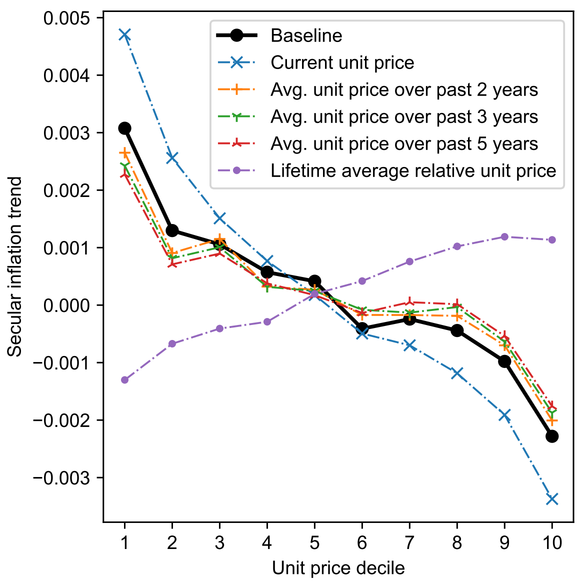
Figure C5: Log pass-through of upstream PPI and secular trends with alternate rankings.



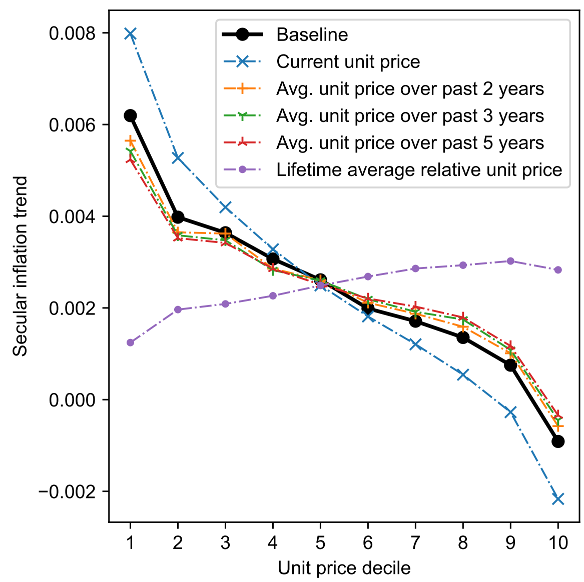
(a) Pass-through: Food Manufacturing PPI



(b) Pass-through: Farm Products PPI



(c) Secular trend: Food Manufacturing PPI



(d) Secular trend: Farm Products PPI

Note: Panels (a)–(b) replicate Figure 5 using alternate definitions of unit price groups. Panels (c)–(d) report $\{\alpha_n\}_{n=1}^{10}$ from the specification,

$$\Delta \log p_t^n = \alpha_n + \sum_{k=0}^{K} \beta_k^n \Delta \log \mathrm{PPI}_{t-k} + \varepsilon_{nt}.$$

This specification mirrors (5) but excludes the quarter-of-year fixed effects, so that the intercept terms α_n are identified. See the text of Appendix C.2 for definitions of the alternative unit price groupings.

C.3 External Validity in Consumer Panel Data

Appendix Table A2 reports that about 60 percent of expenditures reported in the Homescan data can be matched by UPC to the Retail Scanner data, and 25 percent of expenditures can be matched by both UPC and retail chain. The incomplete match rate raises an external validity question: do prices for UPCs and retailers not matched to the Retail Scanner data exhibit similar patterns to matched products?

In this appendix, we construct inflation rates for each product in the NielsenIQ Homescan data using purchase prices reported by panelists. This allows us to create measures of inflation for UPCs and retailers observed in the panelist data but not matched to the Retail Scanner data. For each UPC i at each retail chain r, we construct the inflation rate from quarter t to quarter $t+4$ using average effective prices paid by panelists p_{irt}^{cons},

$$\pi_{irt}^{\text{cons}} = \frac{p_{irt}^{\text{cons}}}{p_{irt-4}^{\text{cons}}} - 1, \qquad \text{where} \qquad p_{irt}^{\text{cons}} = \frac{\text{TotalPanelistExpenditures}_{irt}}{\text{TotalPanelistUnitsPurchased}_{irt}}.$$

Compared to the inflation rates in the Retail Scanner data, these measures of inflation are prone to a greater degree of sampling error and may further reflect changes in effective prices due to changes in shopping behavior (such as coupon use). Nevertheless, we can use π_{irt}^{cons} to test whether inflation dynamics for UPCs and retailers not matched to the scanner data exhibit cheapflation cycles. Proposition 1 predicts the inflation of a variety given the average inflation of all varieties in the same product category π^{all}, the wage inflation rate π^{w}, and the unit price of the variety relative to all varieties in the category, p_{irt}/p^{all},

$$\text{PredictedInflation}_{irt+4} = \pi_{t+4}^{\text{all}} + \left(p_t^{\text{all}}/p_{irt} - 1\right)\left(\pi_{t+4}^{\text{all}} - \pi_{t+4}^{w}\right). \tag{15}$$

We test whether UPCs and retailers that are not matched to the Retail Scanner data adhere to these predictions using the specification,

$$\begin{aligned}\text{RealizedInflation}_{irt+4} = {} & \beta\text{PredictedInflation}_{irt+4} \\ & + \delta\left(\text{PredictedInflation}_{irt+4} \times \text{Unmatched}_{irt}\right) + \gamma\text{Unmatched}_{irt} + \varepsilon_{irt},\end{aligned} \tag{16}$$

where $\text{PredictedInflation}_{irt+4}$ is the inflation rate from quarter t to $t+4$ predicted by (15), $\text{RealizedInflation}_{irt+4} = \pi_{irt+4}^{\text{cons}}$ measured in the Homescan data, and Unmatched_{irt} is an indicator for whether UPC i or retailer r is not matched to the Retail Scanner data. The coefficient β captures the degree to which realized inflation rates follow the predictions of (15), and the coefficient of interest is δ, which measures whether unmatched products

Table C2: External validity for UPCs and retailers not covered in scanner data.

	Realized inflation (in effective prices)					
	(1)	(2)	(3)	(4)	(5)	(6)
Predicted inflation	0.933**	0.946**	0.964**	0.459**	0.451**	0.479**
	(0.020)	(0.017)	(0.018)	(0.044)	(0.045)	(0.049)
Predicted infl. × Unmatched UPC		−0.033			0.017	
		(0.025)			(0.040)	
Predicted infl. × Unmatched Retailer			−0.052			−0.029
			(0.039)			(0.045)
Product module × date FEs				Yes	Yes	Yes
N (millions)	56.1	56.1	56.1	56.1	56.1	56.1
R^2	0.16	0.16	0.16	0.17	0.17	0.17

Note: This table reports β and δ from specification (16). Standard errors two-way clustered by retailer and quarter. * indicates significance at 10%, ** at 5%.

adhere more or less to the predictions of (15) than matched products.[30]

Column 1 of Table C2 shows that (15) strongly predicts realized inflation rates constructed from average purchase prices. Columns 2–3 find that unmatched UPCs and unmatched retail chains have no systematic differences in the sensitivity of realized inflation to predicted inflation ($\delta \approx 0$). In other words, Proposition 1 continues to describe inflation rates for varieties not matched to the Retail Scanner data. Columns 4–6 repeat the analysis after adding product module–date fixed effects, thereby isolating how the inflation rates of individual varieties compare to average inflation in their product category. While the overall sensitivity of realized inflation rates to predicted inflation rates falls from about 0.95 to 0.45, we again find no systematic differences in inflation patterns for UPCs or retailers not matched to the Retail Scanner data.

While Table C2 suggests that inflation dynamics are not substantially different for products not matched to the Retail Scanner data, another potential concern is the degree to which high-income households buy more expensive varieties than low-income households may be overstated in the sample of products matched to the Retail Scanner data. Again, we can use the purchase prices reported in the Homescan data to test this concern. Table C3 reports results from a regression of log unit prices on income quintile dummies and product module–time fixed effects, separating apart matched and unmatched UPCs and retailers. We find that the degree to which high-income households buy more ex-

[30]We allow for different trend inflation across matched and unmatched products, estimated by γ. Different trend inflation rates could arise if, for example, unmatched products tend to be more or less expensive than matched products, and there are secular differences in inflation across cheap and expensive varieties.

Table C3: Effective unit prices relative to lowest income quintile.

	Log (effective) unit price paid		
	All purchases (1)	Separating unmatched UPCs (2)	Separating unmatched retailers (3)
Income quintile 2	0.044**	0.037**	0.030**
	(0.005)	(0.006)	(0.006)
Income quintile 3	0.085**	0.067**	0.057**
	(0.008)	(0.009)	(0.007)
Income quintile 4	0.134**	0.106**	0.091**
	(0.010)	(0.011)	(0.009)
Income quintile 5	0.199**	0.154**	0.143**
	(0.012)	(0.012)	(0.012)
Income quintile 2 × Unmatched		0.016**	0.022**
		(0.006)	(0.010)
Income quintile 3 × Unmatched		0.040**	0.042**
		(0.009)	(0.015)
Income quintile 4 × Unmatched		0.062**	0.066**
		(0.014)	(0.017)
Income quintile 5 × Unmatched		0.099**	0.088**
		(0.018)	(0.019)
Unmatched		−0.240**	−0.021
		(0.048)	(0.038)
Product Module × Time FEs	Yes	Yes	Yes
N (millions)	311.6	311.6	311.6
R^2	0.84	0.84	0.84

Note: This table reports results from a regression of log effective unit prices on income quintile dummies and product module–time fixed effects. Each observation is a product–income quintile pair. Standard errors two-way clustered by retailer and quarter. * indicates significance at 10%, ** at 5%.

pensive varieties than low-income households is, if anything, understated in the set of matched products. In the full sample, effective unit prices reported by the top quintile are 20 percent higher than the bottom quintile, while they are only 15 percent and 14 percent higher than the bottom quintile in the samples of matched UPCs and matched retail chains, respectively.[31] Thus, our results on inflation differences across high- and low-income groups in the matched sample are likely conservative relative to true differences in the broader sample.

[31]These differences are larger than Table 3 Panel A, because Table 3 does not reflect differences in unit prices due to choice of retailer (inflation data is merged at the UPC level) or shopping behavior.

Appendix D Pass-Through Across the Price Distribution in Nakamura and Zerom (2010)

Consistent with our results on pass-through in levels in Section 3, Nakamura and Zerom (2010) estimate that changes in coffee bean commodity prices are passed through completely in levels to wholesale and retail coffee prices. However, they do not estimate pass-through across the price distribution. In this appendix, we show that the demand system estimated by Nakamura and Zerom (2010) predicts systematic variation in pass-through across products with different prices, at odds with our evidence of uniform pass-through in levels across the price distribution.

Simulating the Nakamura and Zerom (2010) demand system. Nakamura and Zerom (2010) assume that the utility to individual i from purchasing product j takes the form,

$$U_{ijmt} = \alpha_i^0 + \alpha_i^p(y_i - p_{jmt}) + \xi_{jmt} + \epsilon_{ijmt},$$

where y_i is the individual's income; p_{jmt} is the price of product j in market m at time t; ξ_{jmt} is an unobserved demand shifter that varies across products, markets, and time periods; ϵ_{ijmt} is a type-1 extreme value taste shock; and the parameters α_i^0 and α_i^p, which determine an individual's mean utility of purchasing coffee and her price sensitivity, are allowed to vary with income $\tilde{y}_i$ (normalized to have mean zero and variance of one across all markets),

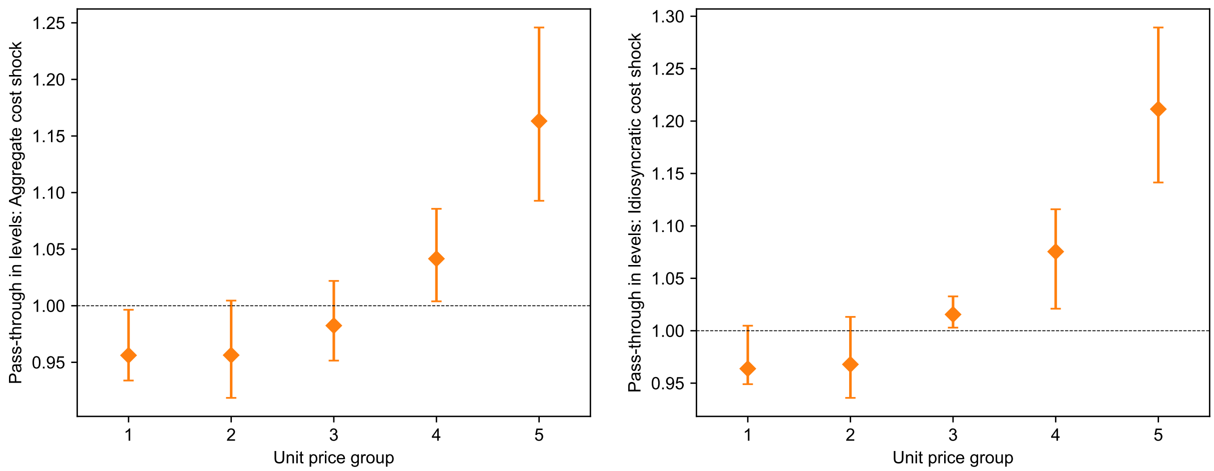
$$\alpha_i^0 = \alpha^0 + \Pi_{y0}\tilde{y}_i, \qquad \text{and} \qquad \alpha_i^p = \alpha^p + \Pi_{yp}\tilde{y}_i.$$

Nakamura and Zerom (2010) assume that each individual chooses the product j that maximizes her utility U_{ijmt}, or chooses to forego purchasing any product if $U_{ijmt} < 0$ for all j (i.e., the utility of the "outside option" is normalized to zero).

Nakamura and Zerom (2010) report estimates for the parameters α^0, α^p, Π_{y0}, and Π_{yp}. However, in order to estimate the pass-through of aggregate cost shocks, we also need data on products' prices p_{jmt} and the unobserved demand shifters ξ_{jmt}. A challenge is that the underlying market data used by Nakamura and Zerom (2010) are from 2000–2004 and are not available through current agreements with NielsenIQ.

To recover estimates of pass-through for coffee products, we thus use data on prices and sales shares of coffee products in NielsenIQ Retail Scanner data from 2006–2020 and assume that the demand parameters estimated by Nakamura and Zerom (2010) apply to this period. As we will show, their model implies meaningful differences in pass-through across the price distribution, and we can trace these differences in pass-through to the

Figure D1: Pass-through in levels across unit price groups in simulation of Nakamura and Zerom (2010) demand system.



(a) Aggregate cost shock.

(b) Idiosyncratic cost shock.

Note: Panels (a) and (b) show the pass-through of an aggregate cost shock (a) and idiosyncratic cost shock (b) across coffee products split by unit price. Scatter points indicate the average across all products, and error bars indicate the interquartile range. The dotted line indicates complete pass-through in levels.

underlying demand system.

Applying the demand system to data from 2006–2020 necessitates a few changes to the process used to assemble the data, which we now describe. We retain the set of products that Nakamura and Zerom (2010) include in their estimation, plus six additional brands that have substantial market shares in the later sample: Eight O' Clock, Millstone, Seattle's Best Coffee, Peets Coffee, New England, and Chase & Sanborn. All other products are grouped with the outside option. Ownership matrices are also updated to reflect subsequent acquisitions over 2006–2020. Rather than use demographic data from the CPS, we use demographic data from the Homescan consumer panel, aggregated using weights provided by NielsenIQ. Finally, we choose the relationship between the number of adults in a market and market size to match the median share of the outside option in Nakamura and Zerom (2010), which is 74%.

Given this demographic data and products' market shares and prices, along with the parameter estimates from Nakamura and Zerom (2010), we can invert the demand system in each market and each month to recover the unobserved demand shifters ξ_{jmt}. The model is then fully specified to simulate pass-through of idiosyncratic and aggregate cost shocks.

Pass-through along the price distribution. The left panel of Figure D1 plots the pass-through of an aggregate cost shock—defined as an infinitesimal increase in the marginal cost of producing all coffee products—across coffee products grouped by unit price in each month. While the pass-through of aggregate cost shocks on average is close to one, it systematically increases with unit price, averaging 0.96 for products in the lowest unit price quintile and 1.16 for products in the highest unit price quintile.

Why does the model imply pass-throughs of aggregate cost shocks different from one? Note that, when there is no outside option, mixed logit demand is shift-invariant with respect to products' prices, and firms exhibit complete pass-through in levels of aggregate cost shocks. However, the Nakamura and Zerom (2010) calibration includes an outside option that is not exposed to the aggregate cost shock. Thus, each product's pass-through of an aggregate cost shock depends on the curvature of its residual demand curve.

The influence of the firms' residual demand curves on the pass-through of aggregate cost shocks can be seen in the right panel of Figure D1, which shows that the model-implied pass-through of idiosyncratic cost shocks is also increasing in unit price. As we show below, the pass-through of idiosyncratic shocks is dictated by the shape of firms' residual demand curves. Figure D1 thus shows that while the pass-through of aggregate cost shocks is compressed toward one, it inherits its pattern from the pass-through of idiosyncratic shocks, since the presence of an outside option means that the shape of firms' residual demand curves also affects the pass-through of aggregate shocks.

To characterize the determinants of pass-through more systematically, let us define the elasticity and super-elasticity of the residual demand curve for product j in market m at time t as

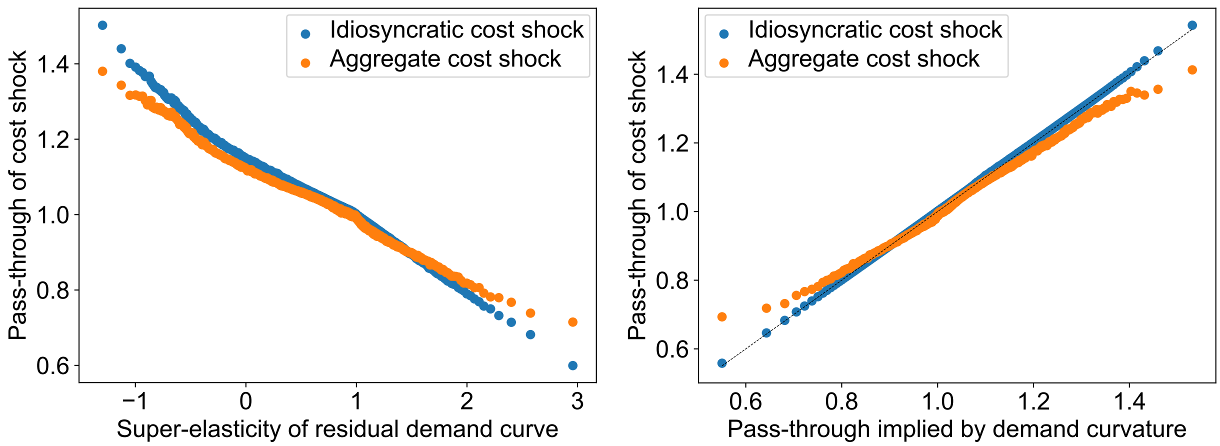
$$\sigma_{jmt} = -\frac{\partial \log q_{jmt}}{\partial \log p_{jmt}}, \qquad \text{and} \qquad \varepsilon_{jmt} = \frac{\partial \log \sigma_{jmt}}{\partial \log p_{jmt}}.$$

For single-product firms, profit-maximizing markups are given by the usual Lerner formula, $\sigma_{jmt}/(\sigma_{jmt} - 1)$. One can show that the pass-through of an idiosyncratic cost shock (in levels) is thus

$$\rho_{jmt}^{\text{idiosyncratic}} = \frac{\sigma_{jmt}}{\sigma_{jmt} + \varepsilon_{jmt} - 1}. \tag{17}$$

Figure D2 plots the pass-through of idiosyncratic and aggregate cost shocks against products' super-elasticities of demand and the pass-through implied by (17). The pass-through of idiosyncratic shocks is well-approximated by (17) (the presence of multi-product firms in the data means that the pass-through of idiosyncratic shocks could in practice differ from (17)). While the pass-through of aggregate cost shocks is more condensed than the pass-through of idiosyncratic cost shocks, it nevertheless also varies considerably across products. The super-elasticity of a product's residual demand curve

Figure D2: Pass-through of aggregate and idiosyncratic cost shocks in simulations of Nakamura and Zerom (2010) demand system.



(a) Super-elasticity of demand. **(b)** Pass-through implied by demand curvature.

Note: Each plot shows a binscatter with 1,000 bins. The pass-through of idiosyncratic shocks is calculated as the response (in levels) of firms' optimal prices for a product to an infinitesimal change in the product's cost. The pass-through of aggregate shocks is calculated as the response of firms' optimal prices to an infinitesimal change in all products' costs within the market, excluding the outside option. The pass-through implied by the demand curvature in panel (b) is given by (17).

and the pass-through implied by the product's residual demand curve are systematic predictors for the pass-through of aggregate cost shocks: products with a super-elasticity of demand below one tend to pass through both idiosyncratic and aggregate cost shocks more than one-for-one, while products with a super-elasticity above one tend to pass through both types of shocks less than one-for-one. In other words, the Nakamura and Zerom (2010) model predicts systematic variation in the pass-through in levels of aggregate cost shocks.

Table D1 further illustrates the variation in pass-through by regressing products' pass-throughs of aggregate cost shocks on various product characteristics. Doubling price per ounce of coffee is associated with a 0.14 increase in the pass-through of aggregate cost shocks, and increasing the market share of a brand or firm is associated with a decline in the predicted pass-through. Likewise, the curvature of products' residual demand curves directly affects the extent of pass-through. These predictions contrast with the uniform pass-through in levels that we document in Figure 2.

One may ask whether the variation in pass-through that we find is due to the 2006–2020 data that we use to simulate the model. Some ancillary statistics reported by Nakamura and Zerom (2010) suggest that the degree of heterogeneity in pass-through is likely to be

Table D1: Determinants of pass-through of aggregate cost shocks in Nakamura and Zerom (2010) demand system.

	Pass-through in levels of aggregate cost shock				
	(1)	(2)	(3)	(4)	(5)
Log Unit Price	0.144**				
	(0.003)				
Log Brand Market Share		-0.033**			
		(0.001)			
Log Firm Market Share			-0.036**		
			(0.001)		
Super-elasticity				-0.153**	
				(0.001)	
Predicted pass-through eq. (17)					0.865**
					(0.009)
Intercept	1.154**	0.872**	0.876**	1.125**	0.115**
	(0.004)	(0.004)	(0.004)	(0.002)	(0.009)
N	374607	374607	374607	374607	374607
R^2	0.39	0.29	0.30	0.80	0.80

Note: Standard errors clustered by market. ** indicates significance at 5%.

as large in their sample. First, Nakamura and Zerom (2010) report that the median super-elasticity of demand in their sample is 4.64. Figure D2a shows that products with super-elasticities above 3 tend to have pass-throughs quite far from one. Second, Nakamura and Zerom (2010) describe the market in 2000–2004 as a near duopoly between Maxwell House and Folgers. In the NielsenIQ data in 2006, the market is less concentrated due to increased penetration by higher-end brands like Starbucks and Peets Coffee. As shown in column 2 of Table D1, variation in market share tends to be associated with significant variation in pass-through across products.

Appendix E Cheapflation and Inflation Inequality in Automobiles

In this appendix, we use microdata on vehicle prices and consumer vehicle purchases to show that the same fluctuations in cheapflation and inflation inequality that we observed in the main text appear in the market for automobile purchases.

E.1 Data

We combine datasets on vehicle prices and characteristics from Ward's Automotive with data on vehicle purchases by households from the Consumer Expenditure Survey.

Vehicle prices and characteristics. We use data on vehicle sales, manufacturers' suggested retail prices (MSRPs), and characteristics from Ward's Automotive, as collected by Grieco, Murry, and Yurukoglu (2024). Vehicle brands—called "makes"—each produce several models, and models in turn are typically available in several different varieties, or "trims." Since sales are only available at the make-model level, the data are aggregated to the make-model level in each year, using the median for the MSRP and other product characteristics across trims.

Consumer purchases. The Consumer Expenditure Survey (CEX) includes detailed interview questions on households' vehicle purchases. We use these data to construct measures of the income of each make's buyers. We use data from 2006 to 2018, during which the makes of purchased vehicles are consistently coded (variable `MAKE`). We exclude vehicles that are purchased for business (i.e., `VEHBSNZ` $\neq 0$) and exclude vehicles that are purchased for someone outside the household or that are received as gifts (i.e., `VEHGFTC` $\neq 1$). For each vehicle make in each year, we calculate the average log before-tax income (`FINCBTAX`) and income quintile (calculated from `INC_RANK`) of households that purchase that vehicle make.

Summary statistics. Table E1 reports summary statistics for each vehicle make-model-year observation in our dataset covering 2006–2018. The first section of the table reports vehicle sales, prices (MSRPs), and characteristics such as horsepower and dimensions. The second section of the table reports changes in model prices and characteristics from one year to the next. The average change in each of these characteristics is small on average (less than 1 percent in all cases), and for each of the non-price characteristics, the median

Table E1: Summary statistics in vehicle make-model-year dataset, from 2006–2018.

Variable	N	Mean	Standard deviation	Min	Max
Sales	3986	49351.05	81315.60	10.00	844448.00
MSRP	3986	39.25	17.23	13.54	99.99
Miles per gallon	3986	21.32	7.85	10.00	50.00
Horsepower	3986	242.77	84.74	66.00	645.00
Height	3986	63.25	8.04	46.60	107.50
Width	3986	73.44	4.00	60.90	89.00
Length	3986	188.24	17.89	106.10	273.80
Curbweight	3986	3934.64	890.58	1808.00	7230.00
Δ Log MSRP ($\times$100)	3381	0.18	6.04	-59.74	57.26
Δ Log Miles per gallon ($\times$100)	3381	0.20	6.89	-117.12	65.23
Δ Log Horsepower ($\times$100)	3381	0.82	8.10	-69.31	58.94
Δ Log Height ($\times$100)	3381	0.01	1.02	-11.15	12.79
Δ Log Width ($\times$100)	3381	0.09	1.46	-18.37	19.62
Δ Log Length ($\times$100)	3381	0.12	1.22	-22.30	22.30
Δ Log Curbweight ($\times$100)	3381	0.28	2.99	-37.01	37.01
Average Log Income of Buyers	3957	10.89	0.32	9.89	12.47
Average Income Quintile of Buyers	3957	3.53	0.35	2.25	4.64

Note: Each observation is a make-model-year. Summary statistics are unweighted. MSRPs are in thousands of 2015 dollars, deflated using core CPI. Vehicle height, length, and width are in inches, and vehicle curbweight is in pounds.

change in model's characteristics is zero. When considering the year-over-year growth in prices (MSRPs) for a vehicle model, our robustness exercises will also control for other changes in vehicle characteristics. The final section of the table reports the average of log income for CEX buyers of the make to which a model belongs and the average income quintile of make buyers. Makes exhibit considerable variation in the average income of buyers; for a vehicle model one standard deviation above average, the average income of its buyers is 32 percent higher than average.

E.2 Cheapflation and Inflation Inequality

We begin by assessing whether vehicle prices exhibit cheapflation—i.e., the prices of relatively inexpensive models are more sensitive to overall growth in prices for a make. To do so, we estimate the specification,

$$\begin{aligned}\text{MSRPGrowth}_{imt+1} = \beta \left(\text{AvgMSRPGrowth}_{mt+1} \times \text{LogMSRP}_{imt}\right) \\ + \gamma \text{LogMSRP}_{imt} + \delta' X_{imt} + \phi_{mt} + \varepsilon_{imt}, \quad (18)\end{aligned}$$

where $\text{MSRPGrowth}_{imt+1}$ is the MSRP growth from year t to year $t+1$ for model i of make m, $\text{AvgMSRPGrowth}_{mt+1}$ is the (sales-weighted) average growth in prices for all models with make m, LogMSRP_{imt} is the log price of model i of make m in year t, X_{imt} is a vector of controls, and ϕ_{mt} are make-year fixed effects.

Columns 1–3 of Table E2 report the results from estimating (18). Each column includes successively more controls in X_{imt}: column 2 includes fixed effects for each model i, and column 3 includes both model FEs and controls for the changes in each vehicle characteristic (miles per gallon, horsepower, height, width, length, and curbweight). In each case, we find that $\beta < 0$: the growth in prices for relatively cheaper models is more sensitive to make inflation rates. In other words, the same systematic fluctuations in cheapflation appear in the vehicle context, with lower-priced varieties exhibiting more inflation (deflation) when prices on average are rising (falling).

Next, we test whether this leads to disproportionate sensitivity of inflation for low-income households. We do so by estimating an analogous specification that instead estimates how the sensitivity of price growth for each vehicle model to overall vehicle price changes varies with the average income of its make's buyers. We estimate the specification,

$$\begin{aligned}\text{MSRPGrowth}_{imt+1} = \beta\Big(&\text{AvgMSRPGrowth}_{t+1} \times \text{LogBuyerIncome}_{mt}\Big) \\ &+ \gamma\text{LogBuyerIncome}_{mt} + \delta' X_{imt} + \phi_t + \varepsilon_{imt}, \quad (19)\end{aligned}$$

where $\text{MSRPGrowth}_{imt+1}$ is the MSRP growth from year t to year $t+1$ for model i of make m, $\text{AvgMSRPGrowth}_{t+1}$ is the (sales-weighted) average growth in prices for all vehicles over the same window, $\text{LogBuyerIncome}_{mt}$ is the average of log income for all households purchasing make m in year t, and ϕ_t are year fixed effects. Note that, unlike specification (18), this specification exploits variation in buyer incomes across vehicle makes, since that is the finest level of disaggregation at which we observe buyer choices in the CEX.

Columns 4–6 of Table E2 report the results from (19). Regardless of the set of controls, we find that price growth for vehicles with a higher-income customer base is less sensitive to overall vehicle inflation rates. That is, the same systematic fluctuations in inflation inequality arise when considering vehicle purchases.

As in the cases discussed in the main text, the heterogeneity in inflation across vehicle makes purchased by low- and high-income households is masked in publicly available statistics. The BLS differentiates between inflation rates for new car and truck purchases (ELI code TA011) and used cars and trucks (ELI code TA021), but does not capture further differences in product-level inflation within these categories.

Table E2: Cheapflation and inflation inequality for automobiles.

	MSRP Growth for Make					
	(1)	(2)	(3)	(4)	(5)	(6)
Avg. MSRP Growth for Make						
× Log MSRP	−0.699**	−0.531**	−0.392**			
	(0.192)	(0.175)	(0.163)			
Avg. MSRP Growth for All Vehicles						
× Log Buyer Income				−0.894**	−0.860**	−0.911**
				(0.270)	(0.255)	(0.243)
Make-Year FEs	Yes	Yes	Yes			
Year FEs				Yes	Yes	Yes
Model FEs		Yes	Yes		Yes	Yes
Controls for other attribute changes			Yes			Yes
N	3381	3381	3381	3358	3358	3358
R^2	0.32	0.54	0.59	0.15	0.21	0.31

Note: Columns 1–3 report results from (18), and columns 4–6 report results from (19). In columns 3 and 6, controls for attribute changes include the change in log horsepower, miles per gallon, height, length, width, and curbweight. Standard errors two-way clustered by year and model. * indicates significance at 10%, ** at 5%.

Appendix F Differential Inflation Across Other Units

While the main text focuses on how pass-through in levels generates different inflation rates across income groups, the same mechanism applies to inflation across other units with heterogeneous expenditure patterns across varieties. In this appendix, we provide empirical evidence that input cost shocks lead to systematic differences in inflation across cities and import price inflation across countries. In both cases, we use coffee products as an example to illustrate these effects.

F.1 Inflation Across Cities

Data. We use data on prices of coffee from the C2ER Cost of Living Index. These data report the price of a "11 to 11.5 ounce can of Maxwell House, Hills Brothers, or Folgers" across U.S. urban areas on a quarterly basis from 1990Q1 to 2010Q1. The set of urban areas for which coffee prices are collected varies slightly from year to year, ranging from 203 to 260 urban areas per observation period.

For each urban area c in each quarter t, we calculate the $\text{RelativePrice}_{ct}$ of coffee relative to other urban areas as the log difference between the price in c and the national average, and the inflation rate Inflation_{ct} as the log change in the price of coffee in c from quarter $t-4$ to quarter t,

$$\text{RelativePrice}_{ct} = \log\left(\frac{p_{ct}}{\frac{1}{|C|}\sum_{c'} p_{c't}}\right), \qquad \text{and} \qquad \text{Inflation}_{ct} = \log\left(\frac{p_{ct}}{p_{ct-4}}\right).$$

We winsorize both relative prices and inflation rates at the 1 percent level. The national inflation rate, $\text{NationalInflation}_t$, is the average across urban areas from quarter $t-4$ to t.

We supplement these data with estimates of income in each urban area using the Bureau of Economic Analysis (BEA) annual estimates of per capita income by metropolitan and micropolitan statistical area (CAINC1).

Specifications and results. Pass-through in levels predicts that cities with lower coffee prices will be more sensitive to fluctuations in input prices. We estimate the specification,

$$\text{Inflation}_{ct+4} = \beta\,(\text{NationalInflation}_{t+4} \times \text{RelativePrice}_{ct}) + \delta_c + \alpha_t + \epsilon_{ct}, \tag{20}$$

where δ_c are urban area fixed effects that absorb secular trends in coffee inflation across urban areas and α_t are quarter fixed effects that absorb the average inflation rate in each quarter as well as any other shifters in coffee inflation across time periods.

Table F1: Heterogeneous incidence of cost shocks on inflation across cities: Coffee.

	Coffee Inflation in City			
	OLS	IV	OLS	IV
	(1)	(2)	(3)	(4)
Average Coffee Inflation × Relative Price	−1.003**	−1.054**		
	(0.253)	(0.285)		
Average Coffee Inflation × Log Income			−0.129**	−0.119**
			(0.061)	(0.061)
City FEs	Yes	Yes	Yes	Yes
Quarter FEs	Yes	Yes	Yes	Yes
N	19 431	19 431	16 990	16 990
R^2	0.67	0.67	0.68	0.68

Note: Columns 1–2 report the estimated coefficient β from (20), and columns 3–4 report the estimated coefficient γ from (21). In columns 2 and 4, the average coffee inflation from quarter t to quarter $t + 4$ is instrumented with the annual inflation in coffee commodity prices and two lags. Standard errors two-way clustered by city and date. * indicates significance at 10%, ** at 5%.

Column 1 of Table F1 reports that urban areas where coffee is relatively more expensive indeed have muted responses of inflation to national fluctuations in coffee inflation. Increasing the relative price of coffee by 10 percent reduces the sensitivity of local inflation to national coffee inflation by 10 percent. Column 2 finds similar results when we isolate fluctuations in coffee prices induced by input costs, using current and lagged inflation in green coffee bean commodity prices to instrument for national coffee inflation rates.

Prices of coffee may differ across cities because of variation in distribution margins or because of differences in consumer preferences over coffee varieties (though our data from the cost of living index largely minimizes the second component by focusing on the prices of roughly equivalent varieties like Maxwell House, Folgers, and Hills Brothers). An important determinant of both sources of variation is local income, which may be correlated with higher retail wages and thus higher costs of distribution services as well as with preferences for more expensive varieties.[32] To isolate how pass-through in levels implies different inflation sensitivity across urban areas with different incomes, we estimate the specification,

$$\text{Inflation}_{ct+4} = \gamma \left(\text{NationalInflation}_{t+4} \times \text{LogIncome}_{ct}\right) + \delta_c + \alpha_t + \epsilon_{ct}. \tag{21}$$

Column 3 reports that coffee inflation in higher-income cities is likewise less sensitive to nationwide fluctuations in coffee inflation. A 10 percent increase in per capita income

[32]Sangani (2022) estimates that retail markups increase with income across cities with an elasticity of 0.11, consistent with the magnitude of the results in Table F1.

is associated with a 1.3 percent lower sensitivity of local coffee inflation to national inflation. We find similar results when instrumenting for national inflation with coffee bean commodity cost fluctuations in column 4.

F.2 Import Price Inflation Across Countries

Data. We use data on the value and quantity of coffee imports across countries from the World Integrated Trade Solution (WITS). We define coffee imports as imports under HS-6 code 090121 ("Roasted coffee, not decaffeinated"). These data report the total trade value in USD and the quantity of imports in kilograms for up to 172 countries over the period 1992–2023. For each country c in each year t, we calculate the unit value of coffee imports as the total trade value divided by the quantity of imports,

$$\text{UnitValue}_{ct} = \frac{\text{TradeValue}_{ct}}{\text{Quantity}_{ct}}.$$

We define WorldUnitValue_t analogously as the total trade value across all countries divided by the total quantity of imports across all countries. The relative price of imported coffee, $\text{RelativeImportPrice}_{ct}$, is defined as the log difference between import unit values for country c relative to the world,

$$\text{RelativeImportPrice}_{ct} = \log\left(\frac{\text{UnitValue}_{ct}}{\text{WorldUnitValue}_t}\right).$$

The import price inflation for country c in year t is the log growth in unit values from $t-1$ to t,

$$\text{ImportPriceInflation}_{ct} = \log\left(\frac{\text{UnitValue}_{ct}}{\text{UnitValue}_{ct-1}}\right).$$

We winsorize import price inflation at the 5 percent level. The world import price inflation is defined analogously as the log growth in the world unit value from year $t-1$ to t.

Specification and results. Pass-through in levels predicts that countries that import relatively more expensive coffee varieties will see lower import price inflation in response to global increases in coffee prices. We test this prediction using the specification,

$$\begin{aligned}\text{ImportPriceInflation}_{ct+1} = \beta\left(\text{WorldImportPriceInflation}_{t+1} \times \text{RelativeImportPrice}_{ct}\right)\\ + \gamma\, \text{WorldImportPriceInflation}_{t+1} + \delta_c + \epsilon_{ct}. \quad (22)\end{aligned}$$

Table F2: Heterogeneous incidence of global cost shocks on import price inflation: Coffee.

	Country Import Price Inflation			
	OLS	IV	OLS	IV
	(1)	(2)	(3)	(4)
World Import Price Inflation × Relative Import Price	−0.221**	−0.285**	−0.225**	−0.167
	(0.075)	(0.101)	(0.093)	(0.106)
Country FEs	Yes	Yes	Yes	Yes
Year FEs			Yes	Yes
N	4480	4480	4480	4480
R^2	0.12	0.07	0.25	0.25

Note: The table reports the estimated coefficient β from (22). Columns 3–4 augment specification (22) with year fixed effects. In columns 2 and 4, the log change in the coffee commodity price from year t to year $t+1$ is used as an instrument for world import price inflation. Standard errors two-way clustered by country and year. * indicates significance at 10%, ** at 5%.

where δ_c are country fixed effects that absorb secular differences in import price inflation across countries.

Table F2 reports the results from estimating specification (22). Column 1 finds that fluctuations in world coffee prices are indeed associated with relatively less import price inflation for countries that have relatively more expensive coffee imports: a 10 percent increase in relative import prices is associated with 2.2 percent lower sensitivity of import price inflation to world coffee prices. Isolating commodity cost-driven fluctuations in world coffee import prices using coffee bean commodity prices as an instrument produces similar results in column 2. Columns 3–4 also find similar quantitative results, though statistical significance is attenuated, when we augment specification (22) with time fixed effects that absorb any arbitrary variation in coffee import price inflation across time. Thus, these results suggest that countries that import more expensive varieties within a HS-6 code experience muted responses of import price inflation in response to global input cost shocks.